BACHELOR OF ARTS IN FINANCIAL SERVICES

BFS 322: TREASURY MANAGEMENT

MODULE

BACHELOR OF ARTS IN FINANCIAL SERVICES

THIRD YEAR

Semester 2

Author: Cynthia Mwansa

© Copyright

ALL RIGHTS RESERVED

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic or mechanical, including photocopying, recording or otherwise without the permission of the *Zambia Centre for Accountancy Studies*.

TABLE OF CONTENTS

ALL RIGHTS RESERVED……i
1.0 INTRODUCTION .....
1 1.1 MODULE AIM .....
1 1.2 OBJECTIVES .....
1 1.3 ASSESSMENT DETAILS .....
2 1.4 READINGS .....
2 Prescribed Reading
..... 2 Recommended
Reading ..... 2 1.5 TIME
FRAME……..3 1.6
STUDY SKILLS 3
1.7 NEED HELP? .....
3
ACKNOWLEDGMENTS……...4

UNIT 1 INTRODUCTION TO TREASURY MANAGEMENT……...5
1.1 INTRODUCTION…….5
1.2 AIM……...5
1.3 OBJECTIVES……...5
1.4 TIME REQUIRED……5
1.5 REFLECTION……..5
1.6 Introduction to Treasury Management…….5
1.7 ACTIVITY……..12

1.8 SUMMARY……....12

UNIT 2 WORKING CAPITAL/LIQUIDITY MANAGEMENT…….....14

2.1 INTRODUCTION……..14

2.2 AIM…….14

2.3 OBJECTIVES…….14

2.4 TIME REQUIRED……..14

2.5 REFLECTION……....15

2.6 Defining Working Capital……...15

2.7 Working Capital Policy……...16

2.8 The Operating Cycle……17

2.9 Debtor Management……....19

2.10 Stocking……25

2.11 Creditor Management……...27

2.12 Cash Management……27

2.13 Managing Cash Surpluses……30

2.14 Overdrafts……31

2.15 Activity……33

2.16 Summary……...33

UNIT 3 MANAGEMENT OF INTEREST RATE RISK……35

3.1 INTRODUCTION……..35

3.2 AIM……35

3.3 OBJECTIVES……35

3.4 TIME REQUIRED……35

3.5 REFLECTION……....35

3.6 Background……36

3.7 Interest Rate Risk……36

3.8 Interest Rate Risk Management……..42

3.9 ACTIVITY……..48

3.10 SUMMARY……..48

UNIT 4 CURRENCY RISKS……...49

4.1 INTRODUCTION……...49

4.2 AIM……49

4.3 OBJECTIVES……49

4.4 TIME REQUIRED……..49

4.5 REFLECTION……49

4.6 Currency Risk-Overview……50

4.7 Attitudes to Risk……..59

4.8 ACTIVITY……..60

4.9 SUMMARY……61

UNIT 5 CURRENCY RISK MANAGEMENT……...62

5.1 INTRODUCTION…….62

5.2 AIM……....62

5.3 OBJECTIVES……....62

5.4 TIME REQUIRED……62

5.5 REFLECTION……...62

5.6 Currency Risk Management-Overview…….63

5.7 ACTIVITY……....71

5.8 SUMMARY……..71

UNIT 6 CASH FORECASTING……....72

6.1 INTRODUCTION…….....72

6.2 AIM……...72

6.3 OBJECTIVES……...72

6.4 TIME REQUIRED……....72

6.5 REFLECTION……...73

6.6 Introduction: Cash Forecasting…….73

6.7 ACTIVITY……83

6.8 SUMMARY……..83

UNIT 7 CASH CONCENTRATION……..85

7.1 INTRODUCTION…….85

7.2 AIM……...85

7.3 OBJECTIVES……...85

7.4 TIME REQUIRED……85

7.5 REFLECTION……...86

7.6 Introduction: Cash Concentration……..86

7.7 ACTIVITY……97

7.8 SUMMARY……..98

UNIT 8 DEBT MANAGEMENT……...99

8.1 INTRODUCTION…….99

8.2 AIM……...99

8.3 OBJECTIVES……99

8.4 TIME REQUIRED…….....99

8.5 REFLECTION……..99

8.6 Types of Debt……..100

8.7 Credit Rating Agencies…….....109

8.8 Debt Related Controls…….....112

8.9 Debt Related Policies……..115

8.10 Debt Related Procedures…….....116

8.11 ACTIVITY……116

8.12 SUMMARY……..116

UNIT 9 EQUITY MANAGEMENT……118

9.1 INTRODUCTION…….....118

9.2 AIM…….118

9.3 OBJECTIVES…….118

9.4 TIME REQUIRED……..118

9.5 REFLECTION…….118

9.6.1 Defining Equity…….119

9.6.2 Stock Registration…….119

9.6.3 Accounting for Stock Sales…….....122

9.6.4 Equity Related Controls……124

9.7 ACTIVITY…….....124

9.8 SUMMARY……125

UNIT 10 INVESTMENT MANAGEMENT……...126

10.1 INTRODUCTION……126

10.2 AIM……..126

10.3 OBJECTIVES……..126

10.4 TIME REQUIRED……...126

10.5 REFLECTION……126

10.6.1 Investment Criteria……...127

10.6.2 Investment Options……127

10.6.3 Investment Strategies……129

10.6.4 Outsources Investment Management……131

10.6.5 Risk Management Strategies…….....131

10.6.6 Accounting For Investments…….....131

10.6.7 Investment Reporting……134

10.6.8 Investment Management Procedures……136

10.6.9 Investment Management Controls……138

10.7 ACTIVITY…….....142

10.8 SUMMARY…….....142

UNIT 11 CLEARING AND SETTLEMENT SYSTEMS……..144

11.1 INTRODUCTION……...144

11.2 AIM…….....144

11.3 OBJECTIVES…….....144

11.4 TIME REQUIRED……144

11.5 REFLECTION……144

11.6 Overview of the Clearing & Settlement Process……145

11.7 ACTIVITY…….....158

11.8 SUMMARY……...158

UNIT 12 TREASURY SYSTEMS……....159

12.1 INTRODUCTION…….159

12.2 AIM……159

12.3 OBJECTIVES……159

12.4 TIME REQUIRED……159

12.5 REFLECTION……..159

12.6 Treasury Systems-An Introduction……...160

12.7 ACTIVITY……...165

12.8 SUMMARY……165

UNIT 13 CASH TRANSFER METHODS……..166

13.1 INTRODUCTION……166

13.2 AIM…….....166

13.3 OBJECTIVES…….....166

13.4 TIME REQUIRED……166

13.5 REFLECTION…….....166

13.6 Cash Transfer Methods-An Introduction…….....167

13.7 ACTIVITY……179

13.8 SUMMARY……..179

UNIT 14 BANK OF ZAMBIA-TREASURY ROLE……...180

14.1 INTRODUCTION…….....180

14.2 AIM……180

14.3 OBJECTIVES……180

14.4 TIME REQUIRED…….180

14.5 REFLECTION……...180

14.6 Bank of Zambia-An Overview……...180

14.7 ACTIVITY…….....189

14.8 SUMMARY……189

UNIT 15 MINISTRY OF FINANCE AND NATIONAL PLANNING……190

15.1 INTRODUCTION……..190

15.2 AIM…….....190

15.3 OBJECTIVES…….....190

15.4 TIME REQUIRED……..190

15.5 REFLECTION…….....190

15.6 Introduction-MOF……191

15.7 ACTIVITY……...194

15.8 SUMMARY……194

15.9 MODULE SUMMARY……...195

1.0 INTRODUCTION

Welcome to the study of Treasury Management. This module has been designed to enable you acquire knowledge of Treasury Management.

1.1 MODULE AIM

The aim of this module to help you have insights into issues of treasury management such as the role of the treasury department at the Ministry of Finance, BOZ's treasury functions, the evolution of cash and treasury management, working capital management, interest rate risk management, foreign exchange risk management and other topics that relate to treasury management.

1.2 OBJECTIVES

By the end of the course, you should be able to:

1) critically explain the role of treasury management;
2) state the importance of working capital in treasury management;
3) discuss cash forecasting, cash concentration using models and strategies;
4) discuss debt, equity and investment management;
5) critically evaluate the role of foreign exchange risk management as well as interest rate risk management;
6) discuss the role of the Ministry of Finance and National Planning's treasury department;
7) discuss Bank of Zambia's treasury role in Zambia;
8) outline the various clearing and settlement systems provided by the central bank.

1.3 ASSESSMENT DETAILS

Continuous Assessment

2 Tests of equal weights	30%
2 Assignments of equal weights	20%
	50%
Final examination	50%
Total	**100%**

1.4 READINGS

Prescribed Reading

1. Bragg, M.S. (2010) *Treasury Management.* John Wiley & Sons, Inc., Hoboken, New Jersey.

Recommended Reading

1. Masson, D.J., Weber, P. and Wikoff, D.A. (2010) *Essentials of Treasury Management,* 3rd Edition. Association for Financial Professionals. Pitman Publishers, London.
2. Ogilvie, J (1999) *Treasury Management-Tools and Techniques for Countering Financial Risks.* Kogan Page Publishers, London.
3. Parkinson, K.L. (2005) *Cash Flow Management,* 1st Edition. Pearson Publishers, London.
4. Webster, M.K. (2014) *Essentials of Treasury Management,* 4th Edition. Association For Financial Professionals, Bethesda, Maryland.

1.5 TIME FRAME

You are advised to spend at least a total 45 hours studying, writing notes and attempting all activities in this module.

1.6 STUDY SKILLS

For you to be able successfully complete this module, you need do the following:

- exercise commitment and self-discipline;
- plan your study times and commit yourself to it, as studying at a distance can be challenging;
- draw up a study time-table and ensure you stick to it; study daily and attempt all activities in this module.

1.7 NEED HELP?

If you need help on the module, please use the following contacts:

Course Tutor

Email: *information@zcas.edu.zm*

Zambia Centre for Accountancy Studies (ZCAS)

Dedan Kimathi Road, P O Box 35243, Lusaka, Zambia

Tel: +260 1 232093/5, Fax: +260 1 222542

ACKNOWLEDGMENTS

I would like to thank God Almighty and the Lord Jesus Christ through the wisdom of the Holy Spirit for this rare opportunity to write this module.

Special thanks go to Mr Rabbie Mukuma my reviewer whose major contributions to this module cannot go unacknowledged. His valuable comments, suggestions and inclusions have made this module a better piece of work.

I would also like to thank the ZCAS management for the confidence they had in me, by granting me this opportunity to write this module.

Lastly, I would also like to thank my family, friends and colleagues from ZCAS for their encouragement and support. God bless you all.

UNIT 1 INTRODUCTION TO TREASURY MANAGEMENT

1.1 INTRODUCTION

Welcome to unit 1 of the module: BFS 322- Treasury Management!

1.2 AIM

The aim of this unit is to enable you acquire knowledge of treasury management that you need to pass this course as well as knowledge you may need in the place of work.

1.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) define Treasury Management;
2) discuss the key areas of financial management decisions;
3) critically evaluate the key decisions of the finance function;
4) explain the key tasks of the treasurer;
5) list the advantages and disadvantages of operating treasury as a profit centre rather than a cost centre.

1.4 TIME REQUIRED

You are advised to spend at least 3 (three) hours studying this unit and attempting all questions at the end of the unit.

1.5 REFLECTION

Take time to think about treasury management and try defining it in your own words. If you work in a financial institution, write down some functions that your treasury department does.

……
……
……
……
……
……

1.6 INTRODUCTION TO TREASURY MANAGEMENT

The structure and role of the finance function within an organization will vary according to such factors as its size, the nature of its business, and the complexity of its organizational structure. However, within an organization, the finance function will be responsible for the provision and management of financial resources.

The establishment of a specialist treasury function within the finance department can be traced back to the late 1960s. (Ogilvie, 1999). Developments in technology, the breakdown of exchange controls, increasing volatility in interest and exchange rates, combined with the increasing globalization of business, have all contributed to greater opportunities and risks for businesses over the last four decades. To survive in today's complex financial environment, businesses need to be able to actively manage both their ability to undertake these opportunities, and their exposure to risks.

Businesses have needed to become more aware of the expanding range of hybrid capital instruments (e.g. convertible preference shares issued in the name of a subsidiary registered in the Dutch Antilles) and the financial instruments (forward markets and the various derivatives markets) and to be able to select from these the ones which are appropriate to the businesses' needs in the prevailing circumstances. A separate treasury function is more likely to develop the appropriate skills, and it should also be easier to achieve economies of scale; for instance, in achieving lower borrowing rates, or netting-off balances.

1.6.1 Key Decision Areas of the Finance Function

The activities of the finance function may be grouped together into three key decision areas to reflect the responsibilities of providing financial resources and managing those resources. These key decision areas are; investment, financing and dividend.

a) ***Investment decisions*** are those which determine how scarce resources in terms of funds available are committed to projects, which can range from acquiring a piece of plant to the acquisition of another entity. Investing in fixed assets usually carries the need for supporting investment in working capital, e.g. stocks and debtors, fewer creditors, an aspect often not properly taken into account by management. Investment to enhance internal growth is often called 'internal investment' as compared with acquisitions, which represent 'external investment'.
   The other side of the investment coin is 'disinvestment', which means the preparedness to withdraw from unsuccessful projects, and the disposal of parts of an entity which no longer fit with the parent entity's strategy. Such decisions usually involve one special element-the right timing for the action to be taken. Disinvestment decisions can also be involved in reconstructions whereby an entity has to alter its capital structure, possibly to survive as a result of grievous losses.

b) ***Financing decisions*** relate to acquiring the optimum finance to meet financial objectives and seeing that fixed and working capital are effectively managed. The financial manager needs to possess a good knowledge of the sources of available funds and their respective costs, and needs to ensure that the entity has a sound capital structure, i.e. a proper balance between equity capital and debt. Such managers also need to have a very clear understanding as to the difference between profit and cash flow, bearing in mind that profit is of little avail unless the entity is adequately supported by cash to pay for assets and sustain the working capital cycle. Financing decisions also call for a good knowledge of evaluation of risk, e.g. excessive debt carries high risk for an entity's equity because of the priority rights of the lenders. A major area for risk-related decisions is in overseas trading where an entity is vulnerable to currency fluctuations, and the manager must be well aware of the various protective procedures such as 'hedging' (minimizing the risk of loss) available.

c) ***Dividend decisions*** relate to the determination as to how much and how frequently cash can be paid out of the profits of an entity as income for its proprietors. The

owner of any profit-making organization looks for reward for investment in two ways, the growth of the capital invested and the cash paid out as income; for a sole trader this income would be termed 'drawings' and for a limited liability company, the term is 'dividends'.

The dividend decision thus has two elements: the amount to be paid out and the amount to be retained to support the growth of the entity often referred to as 'retained earnings'. The level and regular growth of dividends represent a significant factor in determining a profit-making company's market value, i.e. the value placed on its shares by the stock market.

All three types of decisions are interrelated, and cannot be considered in isolation. However, as an organization increases in size, the responsibilities of the finance function are likely to be subdivided, broadly in line with these decision areas, into two main functions or departments, namely treasury and financial control.

Treasury management is therefore the raising of finance, management of liquid resources, management of currency, management of interest rate risks, and the management of relationships with banks.

1.6.2 The Role of the Treasurer

The treasury function exists in every business, though in small businesses it may from part of a department covering other functions, such as accounting or company secretarial work. In a larger company, it is likely to be a separate department reporting to the chief financial officer, but communications with the rest of the organization need to be in good order if an effective service is to be provided.

Treasury is concerned with the relationship between the entity and its financial stakeholders, which include shareholders, fund lenders and taxation authorities, while **financial control** provides the relationship with other stakeholders such as customers, suppliers and employees.

In larger companies and groups, treasury will usually be centralized at Head Office, providing a service to all the various units of the entity and thereby achieving economies of scale, e.g.by obtaining better borrowing rates, whereas financial control is now frequently delegated to individual units, where it can more closely impact on customers and suppliers and relate more specifically to the competition which those units have to face. As a result, treasury and financial control may often tend to be separated by location as well as by responsibilities.

The main functions of the treasurer can be classified as:

- **Banking:** The treasurer will be responsible for managing relationships with banks.
- **Liquidity management:** This will involve working capital and money management. The treasurer will need to ensure that the business has the liquid funds it needs, and invest surplus funds.
- **Funding management:** This is concerned with identifying suitable sources of funds which require knowledge of the sources available, the cost of those sources, whether any security is required, and management of interest rate risks.
- **Currency management:** The treasurer would be responsible for providing the business with forecasts of exchange rate movements which, in turn will determine the procedures adopted to manage exchange rate risks. Dealing in the foreign exchange markets and day-to-day management of foreign exchange risks becomes a key function for the treasurer.

The treasurer's key tasks can also be categorized according to the three levels of management:

1. *Strategic:* e.g. matters concerning the capital structure of the business and distribution/retention policies, the actual raising of capital, including share issues, the assessment of the likely return from each source and the appropriate proportions of funds from each source, the decision as to the level of dividends, and consideration of alternative forms of finance.
2. *Tactical:* e.g. the management of cash/investments and decisions as to the hedging of currency or interest rate risk.
3. *Operational:* e.g. the transmission of cash, placing of 'surplus' cash and other dealings with banks.

Treasurers require specialist skills to be able to handle effectively an ever-growing range of capital instruments, e.g. convertible preference shares issued in the name of an offshore subsidiary, and to determine the most suitable way to protect their company from foreign exchange risk, which demands a good knowledge of forward markets and an ability to select the most appropriate methods of hedging and foreign exchange cover. They also need knowledge of taxation in all areas in which the group operates and, deriving from that, the ability to advise effectively on policies such as transfer pricing in permissible ways to minimize overall tax liability, and to be able to liaise competently with the group taxation department.

The capacity to make large gains or losses is enormous: a treasurer can wipe out, in a few hours, all the profit made from making and selling things over several months. It is important, therefore, that authority and responsibility associated with the treasury function are carefully defined and monitored. This becomes even more important as the range of derivatives increases. Senior managers need to be aware of which risks are being carried, which ones are laid off, and where appropriate, taken on. There is also growing pressure for companies to disclose, in their annual reports, more information about their treasury policies, and their 'positions' as at the balance sheet date.

Both the treasurer and the financial controller will usually be responsible to the finance director, whose concern will be to harmonize their activities in an effective manner.

Examples comparing their relationships are:

a) The treasurer is best able to assess cost of capital and quantify the entity's aversion to risk, while the financial controller relates these factors to group strategy;
b) The financial controller identifies the entity's currency risks, while the treasurer advises on the best means to hedge the risks.

1.6.3 Cost centre or profit centre

An area for debate is whether the treasury activities should be accounted for simply as a cost centre, or as a business in its own right, seeking to make a profit out of its activities, e.g. by charging other businesses in the enterprise for its services (and giving those businesses the choice of whether they use it or a bank). In Zambia for example, most banks consider the treasury department as a profit centre.

As indicated above, the case for a separation of treasury- not only from accounting but also from financial control – is a strong one. The question remains, however, as to how its performance/progress should be measured/ assessed.

Some people would argue for a ‘profit centre’ approach. This usually means that the treasury charges individual business units a market rate for the service it provides. If it writes currency options, for example, it charges the business unit a premium in line with that charged by the banks. (Currency options are options that give the holder the right but not the obligation to buy/sell a particular currency at a predetermined rate at a given point in time). It then has a task of managing that option (by buying and selling in the forward market, or by using derivatives) for a cost which leaves it with a profit. The main argument for this is that the treasurer is then motivated to do what is best for the company as a whole, i.e. to minimize the cost of the operation.

There are obviously some administrative costs involved, but the main drawback in practice has been that some treasurers have interpreted the profit concept as encouraging them to speculate. If it can make a profit from writing options, why not write them for other companies? If it understands the foreign currency markets, why not speculate, e.g. swap funds from currencies expected to depreciate into those expected to appreciate? Often spectacular gains can be made, but the record shows that spectacular losses are at least as likely. A sound internal control system is a necessity in any treasury function, but especially so when speculation is encouraged.

Consequently, there are many who advocate a ‘cost centre’ approach. This usually means that the costs of running the department are collected (and, no doubt, compared with budget) but that the substance of the transactions which they manage is reflected in the business unit’s books, e.g. the ‘profit’ made on writing options is credited to the business for which it was written. Alternatively the profits/losses are allowed to lie where they fall: e.g. the business unit carries the profit or loss on an overseas sale (according to the spot rate when the cash is received) and the treasury carries the offsetting currency loss/gain respectively (the difference between the forward rate obtained and the eventual spot rate). This approach collects the total cost/benefit of hedging. It discourages speculation but may, by the same token, discourage initiative.

Confrontation can occur when the business unit bears a loss, caused – as it sees it – by the treasury.

In summary, the main advantages of operating treasury as a profit centre rather than a cost centre are:

- Individual business units if the entity can be charged a market rate for the service provided, thereby making their operating costs more realistic;
- The treasurer is motivated to provide services as effectively and economically as possible to ensure that a profit is made at the market rate, e.g. in managing hedging activities for a subsidiary, thereby benefiting the group as a whole.

The main disadvantages are:

- The profit concept is a temptation to speculate, e.g. by swapping funds from currencies expected to depreciate into those expected to appreciate;
- Management time is unduly spent in arguments with business units over charges for services, even though market rates may have been impartially checked (say, by the internal audit department);
- Additional administrative costs may be excessive.

The decision as to whether to operate treasury as a profit centre may well depend on a particular 'style' of the company and the extent of centralization or decentralization of its activities.

1.7 ACTIVITY

1. Discuss what you understand by Treasury Management.
2. What are the key decision areas of the finance function?
3. What are the functions of the treasurer? 4. What are the key tasks of the treasurer?
5. What is meant by the treasury function as being 'a profit centre' or a 'cost centre'?
6. Critically evaluate the main advantages and disadvantages of operating treasury as a profit centre rather than as a cost centre.

1.8 SUMMARY

In this Unit, we have discussed the three groups into which financial management decisions can be classified, namely; investment, financing and dividends. We have identified the objectives of treasury management as being:

- Acquiring the optimum finance for the business to meet its financial objectives;
- Management of interest and exchange rate risks;
- Providing advice to management on capital structure and dividend policy issues;
- Management of the organization's liquid assets; Liaison with the organization's banks.

In the remaining units, we will consider the activities of treasury management to see how the treasury management to see how the treasury manager may achieve the above objectives.

Unit 2 is going to be a build- up Unit 1 and it will discuss working capital management.

UNIT 2 WORKING CAPITAL/LIQUIDITY MANAGEMENT

2.1 INTRODUCTION

Welcome to Unit 2. In this unit, we will look first at the overall management of working capital and then progress to look more closely at the individual components of debtors, inventory, creditors and cash.

2.2 AIM

The aim of this Unit is to help you acquire knowledge of working capital.

2.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss an organization's working capital policy;
2) Calculate the operating cycle;
3) Describe debtor management;
4) Critically evaluate inventory management;
5) Discuss creditor management;
6) Discuss cash management;
7) List the characteristics of overdraft facilities.

2.4 TIME REQUIRED

You are advised to spend at least three (3) hours reading through this unit and attempting questions at the end of the unit.

2.5 REFLECTION

In your own words, define working capital.

.....

.....

.....

.....

.....

.....

If you have studied some accounting somewhere, you probably should have met the word 'working capital'. However in treasury management we are more concerned with how working capital should be managed in different organizations and not its definition. But just in case you have forgotten your accounting, we will define working capital for you.

2.6 DEFINING WORKING CAPITAL

Working capital is generally defined as 'the capital available for conducting the day-to-day operations of an organization. (Ogilvie, 1999).

In accounting terms, this is a static statement of financial position concept, referring to the excess- at a particular moment in time-of permanent capital plus long-term liabilities over the fixed assets of the business. As such, it depends on accounting rules, such as what is capital and what is revenue, what constitutes a retained profit, the cut-off between long term and short term (12 months from the statement of financial position date for published accounts) and when revenue should be recognized.

If working capital, thus defined, exceeds net current operating assets (inventory plus debtors less creditors) the company has a cash surplus (usually represented by bank deposits and investments); otherwise it has a deficit (usually represented by a bank loan and/ or overdraft).

A business must be able to generate sufficient cash to be able to meet its immediate obligations and, therefore, continue trading. Unprofitable businesses can survive for quite some time if they have access to sufficient liquid resources, but even the most profitable business will quickly go under if it does not have adequate liquid resources.

Working capital is therefore essential to the company's long term success and development, and the greater the degree to which the current assets cover the current liabilities, the more solvent the company.

The efficient management of working capital is important from the point of view of both liquidity and profitability. Poor management of working capital means that funds are unnecessarily tied up in idle assets, hence reducing liquidity and also reducing the ability to invest in productive assets such as plant and machinery, so affecting profitability.

2.7 WORKING CAPITAL POLICY

A company's working capital policy is a function of two decisions;

a) The appropriate level of investment in and mix of current assets to be decided upon, for a level of activity. This is the *investment decision*.
b) The methods of financing this investment-the *financing decision*.

2.7.1 The investment decision

In conditions of uncertainty, firms must hold some minimal level of cash and inventories based on expected sales, plus additional safety stocks. With an *aggressive working capital policy*, a firm would hold minimal safety stock. Such a policy would minimize costs, but it could lower sales because the firm may not be able to respond rapidly to increases in demand.

Conversely, *a conservative working capital policy* would call for large safety stocks. Generally, the expected return is lower under a conservative than under an aggressive policy, but the risks are greater under the aggressive policy.

A Moderate policy falls somewhere between the two extremes in terms of risk and returns.

2.7.2 The financing decision

Working capital financing decisions involve the determination of the mix of long-term versus short-term debt.

With an aggressive financing policy, the firm finances part of its permanent asset base with short-term debt. This policy generally provides the highest expected return (because shortterm debt costs are typically less than long-term costs) but it is very risky.
Under a conservative financing policy, the firm would have permanent financing (long-term debt plus equity) which exceeds its permanent base of assets. The conservative policy is the least risky but also results in the lowest expected return. The maturity matching policy falls between the two extremes.

There is a basic difference between cash and inventories, on the one hand, and the receivables on the other. In the case of cash and inventory, higher levels mean safety stock, hence a more conservative position. There is no such thing as a 'safety stock of receivables', and a higher level of receivables in relation to sales would generally mean that the firm was extending credit on more liberal terms. If we characterize aggressive as being risky, then lowering inventories and cash would be aggressive but raising receivables would also be aggressive.

The financing of working capital depends upon how current and non-current asset funding is divided between long-term and short-term sources of funding.

There are three (3) possible policies that exist when it comes to working capital decisions:

a) **A conservative policy** is where all of the permanent assets-both non- current assets and the permanent part of the current assets (i.e. the core level of investment in inventory and debtors etc.) is financed by long-term funding, as well as part of the fluctuating current assets. Short-term financing is only used for part of the fluctuating current assets.
b) **An aggressive policy** for financing working capital uses short-term financing to fund all the fluctuating current assets as well as some of the permanent part of the current assets. This policy carries the greatest risk of illiquidity, as well as the greatest returns.
c) **A moderate policy** matches the short-term finance to the fluctuating current assets, and the long-term finance to the permanent part of current assets plus non-current assets.

2.8 THE OPERATING CYCLE

The operating cycle is the length of time between the company's outlay on raw materials, wages and other expenditures, and the inflow of cash from the sale of the goods. In a manufacturing business, this is the average time that raw materials remain in inventory, less the period of credit taken from suppliers, plus the time taken for producing the goods, plus the time taken by customers to pay for the goods. On some occasions, this cycle is referred to as the **cash cycle.**

This is an important concept for the management of cash or working capital because the longer the operating cycle the more financial resources the company needs. Management needs to watch that this cycle does not become too long. Allowances should be made for any significant

changes in the level of inventories taking place over the period. If, for example the company is deliberately building up its level of stocks, this will lengthen the operating cycle.

Table 1

Raw Materials		**Days**
Period of turnover of raw materials stock	Average value of raw material stock / Purchase of raw materials per day	X
Less:		
Period of credit granted by suppliers	Average level of creditors / Purchase of raw materials per day	X
Period of production	Average value of work in progress / Average cost of goods sold per day	X
Period of turnover of finished goods stock	Average value of stock of finished goods / Average cost of goods sold per day	X
Period of credit taken by customers	Average value of debtors / Average value of sales per day	X
Total operating cycle		X

Some companies advocate computation of an annual operating cycle for each quarter, since with a seasonal business the cycle would vary over different periods. The numerators in the equations can be found by taking the arithmetic mean of the opening and closing balances for inventory, creditors and debtors. If a quarterly statement is being prepared, the opening and closing balances for the quarter would be used.

N.B: Owing to the nature of the simplified information provided, end-of-year values-rather than average values- have been used for inventory, debtors and creditors.

A number of steps could be taken to shorten the operating cycle. The volume of debtor balances could be cut by a quicker collection of debt; finished goods could be turned over more rapidly;

the level of raw material inventory could be reduced or the production period could be shortened. The operating cycle is only the *time-span* between production costs and cash returns; it says nothing in itself about the amount of working capital that will be needed over this period. In fact, less will be required at the beginning than at the end.

2.9 DEBTOR MANAGEMENT

In order to reduce the debtor days to a more respectable figure, companies will offer customers inducements, in the form of *cash discounts*. These discounts may well speed up collection but reduce the amount from each sale when collected.

Credit management involves balancing the benefits to be gained from extending credit to customers against the costs of doing so, and finding the optimum level of credit and discounts which will maximize the company's profits. It also involves such things as assessing the credit risk of customers wanting credit, collecting debts which are overdue, assessing what effect changing credit terms will have on the occurrence of bad debts, and setting individual credit limits for customers.

2.9.1 The cost of credit (estimating the real cost of discounts)

The cost of offering a cash discount in order to generate better cash flow is sometimes overlooked. The savings made by the company from a lower debtor balance and shorter average collecting period should be compared with the cost of the discount to see if the reduced period of credit could be financed by alternative means- e.g. a bank overdraft.

Example: simple interest approach

Zoona (Z) Ltd normally offers its customers 50-day payment terms, but to improve its cash flow is considering a 2% discount for payment within 10 days. Advise the company on the cost of this proposed action. Assume a 365-day year and an invoice for K100. **Solution**

$$\frac{2\%}{98\%} \times \frac{365}{50-10} = 18.6\%$$

The company would receive K98 on day 10 instead of K100 on day 50. Zoona (Z) Ltd would then be able to invest the K98 for 40 days (50 – 10).

The 18.6% would then be compared with the company's cost of capital to establish if this is an efficient method of financing the shorter credit period.

Note: a more accurate method can be used which deals with the compound interest effect.

The compound interest formula states that the value V attained by a single sum X , after n periods at r % is:

$V = X (1 + r)^n$

Using this formula to quantify the cost of offering the cash discount:

$100 = 98 (1 + r)^{365/40}$

Rearranging this formula gives:

$1 + r = 9.125 \sqrt{100/98}$ $1 + r = 1.202$

The cost of offering cash discounts is then 20.2%.

2.9.2 Credit control

As stated above, to have good credit management you should assess the credit risk of your customer base. This would involve giving consideration to your credit control procedures. Detailed below are key elements of what you need to take into account in setting your *credit control policy:*

a) The terms of trade, notably the period of credit to be granted, and any discounts to be allowed for early settlement. This will largely be determined by practice within the industry but there is usually some scope for differentiation- from competitors, and as between customers (the riskier prospects being put on a shorter-period, higherdiscount arrangement). It is important to record all the terms agreed.
b) On a customer-by-customer basis, it is necessary to assess creditworthiness and to establish limits in terms of amount and time. Late payment is seen as a major problem in Zambia.

Consideration should be given to assessing a customer's creditworthiness, particularly for new customers. In addition, consider proceedings to review existing customers from time to time, especially if they request that their credit limit should be raised.

Information about a customer's credit standing can be obtained from a variety of sources:

- *Bank references*- Most banks in Zambia are able to offer references for their clients.

- *Trade references*- Suppliers already giving credit to the customer can give useful information about how good the customer is in paying bills on time.
- *Published information*- The customer's annual accounts and reports will give some idea of the general financial position of the company and its liquidity.
- *Credit agencies*- Agencies such as the Credit Reference Bureau here in Zambia keeps information of individual and company borrowings from different banks.
- *Company's own sales record*- For an existing customer, the sales ledger will show how prompt a payer the company is, although it cannot show how able the customer is to pay.

Assessing customers' creditworthiness can be expensive in terms of both time and money, and as with all credit management the costs and benefits have to be considered together. The more detailed and expensive investigations should be reserved for potentially important or especially risky customers.

2.9.3 Chasing overdue accounts

- There need to be well-defined procedures for following up-allowing in the first instance for the possibility of a genuine query-keeping notes which can be referred to later. If the worst comes to the worst, there will be a need to understand the law relating to contracts, insolvency, winding up and liquidation.
- You can also sue for debt or institute winding-up proceedings. The courts of law can also pass judgment about debt actions on a case by case basis. For example, here in Zambia, a court called '**the small claims court'** deals with non-payment of amounts owed by individuals and small firms.

2.9.4 Assessing the effectiveness of credit control

An outsider looking in, and wishing to assess the effectiveness of a credit control function, is generally limited to using static information and ratios. See the example below:

	Year 1	Year 2

K million:		
a) Sales for the year	100.00	120.00
b) Debtors at the end of the year	16.00	20.00
Times per annum:		
Asset velocity (a)/(b)	6.30	6.00
Days:		
Average collection period 365 (b)/(a)	58.40	60.80

In the example above, this would be interpreted as an apparent deterioration in performance, but such an approach inevitably ignores a number of possible explanations:

- Changes in the pattern of sales across the year, e.g. more towards the end;
- Changes in the mix of sales as between customers, e.g. more to those granted longer credit;
- Changes in terms, e.g. less attractive discounts to some or all customers;
- Different degrees of window-dressing, e.g. sales pulled forward from the subsequent year.

Internally, of course, these problems do not arise, thanks to the availability of management information, including daily sales and receipts. As sales are entered into the ledger, for example, it is possible to identify when they should be paid, thus providing an appropriate yardstick against which to measure performance.

Enterprises which seek to apply the principles of value assurance to activities characterized as giving rise to 'indirect costs' could do worse than start with credit control. Value in this context is the cost of the next best alternative, e.g. handing over the job to the professional factors.

This amounts to regarding the function as a business unit. It is credited with its actual receipts having been debited with:

- The price at which the factor would buy the debts, e.g. 98% of their face value;

- The cost of finance, say 0.03% per day on the aggregate balance; The costs of administration, etc.

If the net result is positive, the function is adding value to the enterprise.

2.9.5 Factoring

One business process with which financial managers have a natural affinity, but which may nevertheless be contracted out, is the management of receivables. In many industries, the capital required to fund the granting of credit to customers represents a substantial proportion of total and tends to vary directly with the level of activity: the greater the volume, the greater the investment in this particular component of current assets.

Rather than obtaining traditional finance (e.g. equity and overdrafts), some enterprises choose to define their businesses in such a way as to exclude this particular process. Effectively, they say that their business virtually ends at the point where goods are delivered. Specialist finance companies (again usually subsidiaries of banks) offer a factoring arrangement under which they:

- Provide finance by advancing, say, 80% of invoice value immediately, the remainder being settled when the client's customer settles the debt (but net of a charge for interest, typically 3% per annum above base rate);
- Take responsibility for the operation of the client's sales ledger, including assessment of creditworthiness and dealing with customers for an additional service charge, typically 2% of turnover. They may even, for an additional fee, offer **non-recourse finance**, i.e. guarantee settlement even if they are not paid by the customer.

In order to do this economically, they have developed their expertise in credit control, in terms of market intelligence (including credit scoring), information management (sophisticated databases, processing and decision support systems) and the skills required for dealing with customers-especially those who are in no hurry to pay!

Alternatively, they may offer a confidential **invoice discounting facility** under which they provide the finance as above but do not get involved with the operation of the sales ledger or hence become known to the customers. It is cheaper, and avoids creating a barrier between the business and its customers. It is less attractive to the providers of finance, however, being in the nature of supplying a commodity rather than adding value through expertise.

These financiers are usually subsidiaries of banks. Such banks are reluctant to lend just when the company needs it, e.g. ahead of a growth phase. A sales-based package is a logical, flexible alternative.

Having syphoned off the debtors in this way, however the returns from a business are going to be more uncertain, making it difficult to raise more traditional forms of finance except at high interest rates. It is also worth noting that factoring is associated in many people's minds with financial difficulties or at best with small businesses, which may have an impact on the image of the business in the eyes of its creditors.

The financial implications of a factoring agreement may need to be evaluated with regard to alternative financing methods.

Example

Mutengo PLC has been set up for the purpose of importing commodities which will be sold to a small number of reliable customers. Sales invoicing is forecast at K300,000 per month. The average credit period for this type of business is two and a half months.

The company is considering factoring its accounts receivables under a full factoring agreement without recourse. Under the agreement the factor will charge a fee of 2.5% on total invoicing. He will give an advance of 85% of invoiced amounts at an interest rate of 13% per annum. The agreement should enable Mutengo PLC to avoid spending K95,000 on administration costs.

Required

1. Calculate the annual net cost of factoring;
2. Consider the financial benefits of such an agreement, having regard to current interest rate on bank overdrafts of 12.5%. **Solution**

1.

Annual sales: K300,000 x 12	K3,600,000
Annual net cost of factoring	
Fee: 2.5% of K3,600,000	K90,000
Annual interest* (85% x 2.5/12 x 3.6m x 13%)	K82,875

Total annual cost	K172,875
Less: administration costs	K95,000
Net cost	K77,875

*Assuming the agreement is based on existing invoices and does not phase in.

2. The borrowing of K637,500 i.e. (2.5/12 x 3.6m x 85%) from the bank would cost K79,687.50). Therefore, factoring offers a saving of around K2,000 as well as providing certain advantages:

- **Flexibility:** As sales increase with the corresponding demand for finance, so finance from this source increases.
- **Security:** It allows the firm to pledge other assets as security for the finance.
- **Last resort:** It may be the most cost-effective lender to a firm which has no assets to offer as security.
- **Responsibility:** Relieves management of the responsibility for the sales ledger and can probably perform credit-checking better than the firm. Management must balance the disruption from cutting back its administrative costs with the financial and other advantages of factoring. Before reaching a decision, management should consider the possibility that the financial advantages may change and that re-establishing a sales ledger function may be costly.

2.10 STOCK MANAGEMENT

Stocks, like debtors, involve the commitment of a large amount of a firm's resources. Their efficient management is of great concern to the financial manager. Stocks include raw materials, bought-in parts finished goods awaiting sale and also work in progress, although the latter element will not be discussed here. Stocks should not be viewed as an idle asset; rather, they are an essential part of a firm's investment and operations. The optimum holding of stocks will maximize the benefits less costs involved.

Holding higher levels of finished goods stock will enable the company to be more flexible in supplying customers. More customers would receive immediate delivery rather than waiting for new items to be produced and they might obtain a greater choice of types of product.

There would be a smaller chance of sales being disrupted through interruptions in production. These benefits would have to be balanced against the storage costs incurred, the capital costs of financing the stock and the cost of stock becoming obsolete.

2.10.1 Just in time (JIT) purchasing

Developed by the Japanese, it is claimed that the implementation of JIT systems has been one of the major factors contributing to their success. It involves a continuous commitment to the pursuit of excellence in all phases of manufacturing. Its aims are to produce the required items of the required quality in the required quantity at the required time.

There are two aspects to JIT; JIT purchasing, and JIT production. With JIT purchasing, it means that materials or components are received from the supplier just in time to use them. They are in effect delivered to the factory floor.

The objectives of JIT purchasing may be stated as:

- Raw material stock reduction;
- Frequent deliveries with smaller orders from a smaller number of suppliers;
- Long-term contracts with suppliers;
- Quality assurance, with the supplier becoming responsible for the inspection of goods supplied.

The objectives of JIT production are to obtain low-cost, high quality, on time production, to order, by minimizing stock levels between successive processes and, therefore, minimizing idle equipment, facilities and workers.

The introduction of JIT may bring the following benefits:

- Reduced inventory;
- Savings in storage space required;
- Increased customer satisfaction-elimination of waste will lead to a better-quality product;
- Weaknesses and problems may be identified-problems such as bottlenecks, supplier reliability, and inadequate documentation may be revealed; Flexibility and the ability to supply small batches.

JIT will not, however, be appropriate in all situations. A full work study will be required to look at the production methods. A large amount of capital is needed to operate a JIT system, and flexibility is lost due to the nature of the contracts with suppliers.

2.11 CREDITOR MANAGEMENT

Trade credit is normally viewed as being a free source of finance. The policy adopted regarding trade creditors often then tends to be to maximize this resource, paying suppliers as late as possible. This policy may lead to cash-flow problems for key suppliers, which could adversely affect the viability of both organizations. Creditor management will broadly reflect debtor management, as one organization's debtor will be another organization's creditor.

2.12 CASH MANAGEMENT

The management of cash resources holds a central position in the area of short-term financing decisions. Results of investment decisions are estimated in cash terms and the value of a company to shareholders lies in its ability to add to their command over resources over time, which means to add to shareholders' command over cash.

Holding cash carries with it a cost-the *opportunity cost* of the profits which could be made if the cash was either used in the company or invested elsewhere. Therefore, a company has to balance the advantages of liquidity against profitability: cash should be held until the marginal value of the liquidity it gives is equal to the value of the interest lost.

Cash management is, therefore, concerned with optimizing the amount of cash available to the company, and maximizing the interest on any spare funds not required immediately by the company.

2.12.1 The time value of money

A recurring theme of financial management is the time value of money-or the money value of time, depending on your point of view. The need to offer the prospect of an adequate return is an important element in the setting of business objectives. That need is quantified as 'the cost of capital' for use as a criterion against which to make decisions and monitor there is no one generally applicable, constant price of money. Interest rates vary, according to the length of time of the investment, as portrayed in the familiar yield curve itself varies over time. Interest rates are different according to whether you are lending to, or borrowing from, the

banking system, and according to the flexibility of the arrangement, e.g. how much notice is required to move funds into or out of an account.

Other things being equal, in what amounts to a reflection of their expected costs, the banks pay higher interest on accounts subject to longer notices of withdrawal than they do on current accounts and charge higher interest on fluctuating overdrafts than they do on term loans.

A key task for the treasury function in any enterprise, therefore, is the management of the various accounts. Money is switched between them, so as to minimize aggregate costs (or maximize net income, as the case may be), recognizing both transaction costs and interest rate differentials.

Early theories in this area (generally credited to W J Baumol in the early 1950s) borrowed from the techniques which had been developed for controlling stocks. Assume, for example, that:

- Outgoings are K300,000 per annum, spread evenly throughout the year;
- Money on deposit earns 10% per annum more than money in a current account;
- Switching costs K20 per transaction.

According to Baumol, the optimum amount to be transferred (in K) each time is stated as:

$$\frac{\sqrt{2\ X\ transaction\ costs\ X\ outgoings}}{deposit\ rate}$$

That is:

$$\frac{\sqrt{2\ X\ 20\ X\ K300{,}000}}{0.1}$$

I.e. approximately K11,000. At this point, the number of transactions would be around 27 per annum, the average balance in the short-notice account would be K5,500, for an aggregate cost of K540 + K550, i.e. K1, 090 per annum.

Such a model was a gross oversimplification, of course, but did draw attention to the directional impact of the various factors, e.g. were the interest rate differential to be increased, then the size of the optimum transfer would be reduced.

One of the most serious weaknesses was seen to be the assumption that the net cash outflow from the short-notice account was steady and, therefore, predictable. In the real world, there are bound to be fluctuations, the exact timing of which may be difficult to predict. So, in the late 1950s, a more elaborate approach was developed, M H Miller and D Orr being credited with its origination.

Their idea was to transfer money into or out of the account to return the balance to a predetermined 'normal point' whenever the actual balance went outside a lower or upper limit (respectively, assuming the account is always in credit). The lower limit would be set by management, and the upper limit and return points by way of formulae which assume that cash inflows and outflows are random, their dispersion usually being assumed to repeat a pattern exhibited in the past.

The formulae are:

- Spread between upper and lower limits=

$$3x\left(\frac{3}{4}\frac{transaction\ cost\ x\ variance\ of\ cash\ flows}{interest\ rate}^{1/3}x\right)$$

- Return point = lower limit plus one-third of spread.

Assume for example, that:

- A lower control limit of K1,000 is decided upon;
- The interest rate is 0.025% per day;
- The standard deviation of the daily cash flows has been measured as K500, i.e. a variance of K250,000;
- Switching costs K20 per transaction.

Then the spread would be:

$$3x\left(\frac{3}{4}\frac{K20\ x\ K250{,}000}{0.00025}^{1/3}x\right)$$

This means that approximately K7,400. Hence the upper limit would be K8, 400 and the return point K3,467.

Again, the directional pointers provide useful reminders, e.g. if the variability of cash flows increases, so does the spread. Variations on the theme can be developed to take into account of seasonality, numerous different accounts, the lead time between breaking the limit and cash actually being transferred, etc., but these are likely to be of only specialist appeal.

Cash flows are not completely random and are not totally unpredictable and are not independently variable. Treasurers will be forewarned of many of the payments to be made, e.g. wages, dividends, and tax. They will also have some flexibility as regards the timing of many others, e.g. to hold a payment to a creditor until after receipt from a debtor.

Consequently, as in so many fields, decisions are rarely made on the basis of models which require you to believe that cash flows behave like a game of chance, constrained by a pattern derived from an analysis of the facts of the past.

Rather, they are made on the basis of a synthesis of judgments about the future. Spreadsheets provide a useful form in which to prepare and express these, and offer the facility of asking the 'what if?' type of question.

2.13 MANAGING CASH SURPLUSES

Cash surpluses arise for many reasons and last for varying time periods. In principle, where there is no foreseeable use for the surpluses, the cash should be returned to shareholders, or used to repay debt. Usually, though cash will be retained to protect against unexpected losses or to fund unexpected investment opportunities. It should be remembered that idle cash does not generate any income unless it is invested, so keeping to much of it may not be beneficial to a company.

The treasurer will need to consider opportunities for short-term investment in order to put any cash surpluses to work. The following considerations should be made in assessing how to invest short-term cash surpluses:

- Length of time for which the funds are available;
- Amount of funds available;
- Return offered on the investment in relation to the amount involved;
- Risks associated with calling in the investment early (e.g. the need to give three months' notice to obtain the interest); Ease of realization.

The aim of the treasurer is to maximize the post-tax return from the investment, but also to minimize the risk to the original capital invested. Examples of short-term investment opportunities that the treasurer may consider include:

- **Treasury bills:** These are short-term debt instruments issued by the central bank on behalf of government as a way of raising finance and control money supply in the economy. Treasury bills are normally issued at a discount to their face value. They offer good liquidity for investors that buy them. They have a fixed maturity period.
- **Bank deposits:** A wide range of financial instruments are available from banks such as:
  - ***a) Call deposits-*** These are current accounts that pay interest. Large corporate customers can open such accounts and keep excess cash in them. Other features of a current account remain the same (i.e. cheque book, overdraft, unlimited transactions etc.). Banks such as Stanbic Bank here in Zambia do offer such accounts.
  - ***b) Term deposits-*** These accounts allow the customer to keep a fixed amount of money for a fixed period of time say, 30 days, 60 days, or 90 days and being paid a fixed interest which is attractive.
  - ***c) Certificates of deposit-*** These are issued by commercial banks at a fixed interest realized on the discount market at any time.
  - ***d) Money market accounts-*** Most major financial institutions offer schemes for investment in the money market at variable rates of interest.

It is unlikely that short-term cash surpluses will be invested in equities because of the risks associated with achieving a return over a short period.

2.14 OVERDRAFTS

One of the most important external sources of short-term finance, particularly for small firms, is the overdraft.

An overdraft is a short-term advance granted to a customer with a current account. It is loan repayable within twelve months. The bank allows the customer to overdraw their account up to an agreed limit. Debit interest is charged over the accrued amount of the loan.

Overdrafts taken by companies must be used for working capital purposes, as for individuals; they are used to meet expected salary payments and to buy consumer goods.

Many businesses require their bank to provide financial assistance for normal trading operations, to bridge the gap in time between spending money to make or sell goods and services, and eventually receiving cash from the sales. The business, if it is a manufacturing business, it must purchase raw materials from suppliers, who must be paid for the purchase, so that is where the overdraft comes in handy.

Business must also pay wages or salaries to its workforce which converts the raw materials into finished goods. The finished goods will be warehoused on completion and eventually sold at a profit, perhaps on credit. The buyers of these finished goods will be debtors of the business until they eventually pay what they owe. When they do so, the cash cycle will have gone round one full turn from 'cash out to make the goods to cash in from the debtors'. A business can and should put up finance to cover the needs of the cash cycle i.e. the need to invest in stocks and debtors and to some extent, trade creditors will provide financial support by granting credit. However, a bank might also provide financial support by allowing the business an overdraft facility.

Advantages of an overdraft

- ***Flexibility:*** The bank will agree to a maximum overdraft limit, or facility. The borrower may not require the full facility immediately, but may draw funds up to the limit as and when required.
- ***Minimum documentation:*** Legal documentation is fairly minimal when arranging an overdraft. Key elements of the documentation will be to state the maximum overdraft limit, the interest rate payable, and the security required.
- ***Interest*** is only paid on the overdrawn amount rather than on the full facility. If the facility is secured, a lower rate of interest may be available because of the lower risk for the lender.

Disadvantages of overdrafts

- The facility is repayable on demand.
- Companies with few assets to offer as security will find it difficult to arrange further overdraft finance.
- The borrower is exposed to interest rate fluctuations in times when interest rates are volatile.

- The interest rate charged by the bank will vary depending on the perceived credit risk of the borrower.

2.15 ACTIVITY

1) Explain why an organization needs a good working capital policy.
2) How is the operating cycle calculated?
3) Discuss the various debtor management techniques that a company can use.
4) Discuss the advantages and disadvantages of a company holding higher levels of stock.
5) What is Just In Time (JIT)?
6) Outline one creditor management tool that a company can use as a way of managing its working capital.
7) Explain why cash management is important to an organization.
8) Discuss the features, advantages and disadvantages of an overdraft facility.
9) What is the 'time value of money'?
10) Discuss the techniques that companies can use to manage cash surpluses.

2.16 SUMMARY

This unit has been able to discuss the theory of working capital policies that companies use, the operating cycle, debtor management, stock management, creditor management, cash management and managing cash surpluses. The unit further went on to discuss overdraft facilities, their features, advantages and disadvantages.

The next unit will discuss interest rate risk management.

UNIT 3 MANAGEMENT OF INTEREST RATE RISK

3.1 INTRODUCTION

Welcome to unit 3! This unit will equip you with knowledge of tools of interest rate risk management.

3.2 AIM

The aim of this unit is discuss interest rate risks, and how these risks could be mitigated as part of treasury management.

3.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Define interest rate risk;
2) Explain the theory of the term structure of interest rates;
3) Outline the tools required in interest rate risk management.

3.4 TIME REQUIRED

You are advised to spend at least 3 (three) hours studying this unit.

3.5 REFLECTION

Attempt to define interest rate risk management. When can interest rate risk occur?

……
……
……
……
……
……

3.6 BACKGROUND

Large companies particularly multinationals, often rely heavily on the financial markets for finance. Interest and exchange rates change continually, often by significant amounts which make these markets extremely volatile. One of the treasurer's responsibilities is to manage an organization's exposure to interest and currency risks.

3.7 INTEREST RATE RISK

Businesses wish to reduce their exposure to risk in all its forms. Interest rate risk is the potential variability in a firm's profits and market value of equity due to changes in the level of market interest rates.

Interest rate risk arises because some of the firm's assets and liabilities are sensitive to changes in market interest rates, while others are not.

When a firm gets a loan from the bank, the bank may decide to charge a fixed rate of interest or a floating (variable) rate. A fixed rate of interest is where a firm will be charged the same amount of interest throughout the loan period, e.g. 20% p.a. for a loan to be repaid after say 5 years. A floating rate of interest on the other hand, is one that is likely to bring about interest rate risk as this rate is often adjusted according to market forces especially in times of high inflation levels. The firm may then be exposed to the volatility of interest rates or in short interest rate risk.

Firms are likely to reduce their exposure to risk by firstly identifying the different elements of risk (interest rate, foreign exchange, political), through to its quantification (portfolio theory and capital asset pricing model). There remain, however, large areas of uncertainty. These uncertainties may include political and economic trends.

Interest rate risk therefore, deals with the possibility of loss arising from a change in the level of interest rates. One important aspect of the treasury function is management of debt in such a way as to reduce the company's exposure to risk resulting from interest rate movements.

3.7.1Term structure of interest rates

The term structure of interest rates is the spread of interest rates that are paid on the same type of assets with different terms to maturity.

One of the primary considerations in evaluating debt is the likely movement in interest rates. This will affect the relative costs of long-and short-term debt, as well as increasing or decreasing the preference for fixed interest rates. In practice, long-term rates will normally be higher than short-term rates, owing to the additional risk borne by the lender. Hence, an interest premium is required to attract investors to longer-term securities.

This effect may be magnified or reversed by investors' expectations of future rates, an anticipated rate rise producing higher longer-term rates. This difference between long-and short-term rates is known as term structure. The term structure of interest rate is often plotted on a yield curve.

3.7.2 The yield curve

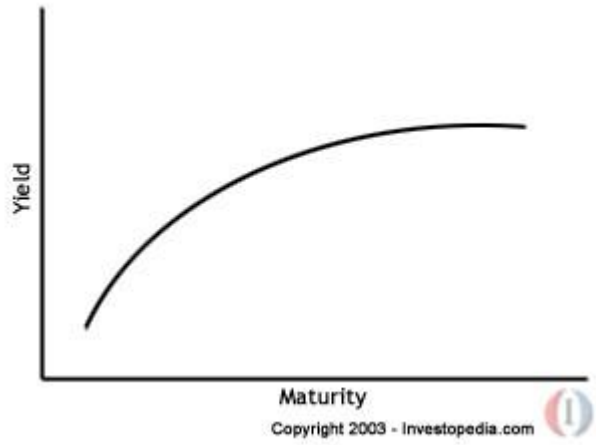

Figure 1: Normal yield curve

A normal yield curve slopes up-wards as depicted in Figure 1 above. It indicates that borrowers who wish to borrow for long periods of time pay a higher rate of interest than medium-term borrowers, who in turn pay a higher rate of interest than short term borrowers.

The characteristic up-ward slope of the normal yield curve is attributable to **liquidity preference and to risk.**

With regard to **liquidity preference,** any lending involves a loss of immediate **purchasing power.** When lending for a short term, this loss of immediate purchasing power may be relatively slight and so the lender may not require much compensation. When lending longterm, the lender would generally require a higher level of compensation. This compensation is reflected in the interest rate and this is why interest rates on long-dated loans will be higher than on short-dated loans.

The overall risks associated with lending increase over time, because other risks increase. These are the risks of:

a) **Default,** or either interest or principal, which will generally increase with the length of the loan;

b) **Real capital loss** due to inflation, which increases with the length of the loan;

c) **Capital loss on marketable securities** which, if they have to be sold prior to maturity, increases with the length of the loan;

d) **Changes in the lender's circumstances** which, making a particular loan within a portfolio inappropriate to new circumstances will increase with the length of the loan.

The yield on an asset includes not only the interest income from that asset, but also any capital appreciation or capital depreciation. The return that includes any capital appreciation or depreciation is termed the **redemption yield,** while the return that only takes into account of the interest income is termed the **interest yield.**

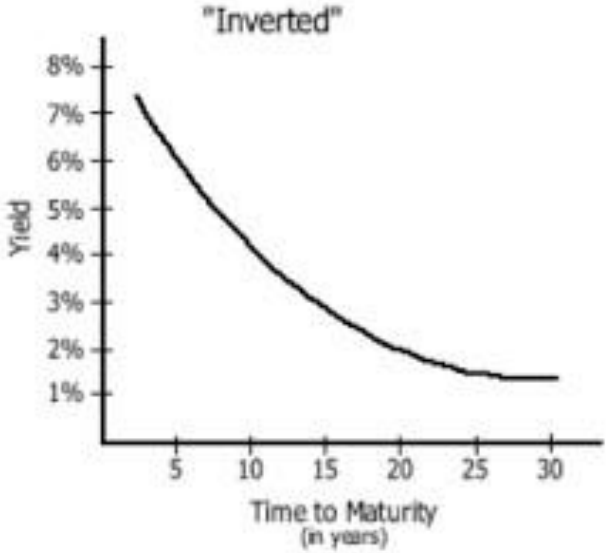

Figure 2: Negatively sloping curve (inverted curve)

A negatively- sloped yield curve indicates that borrowers who wish to borrow in the short term pay a higher rate of interest than medium-term borrowers, who in turn pay a higher rate of interest than long-term borrowers.

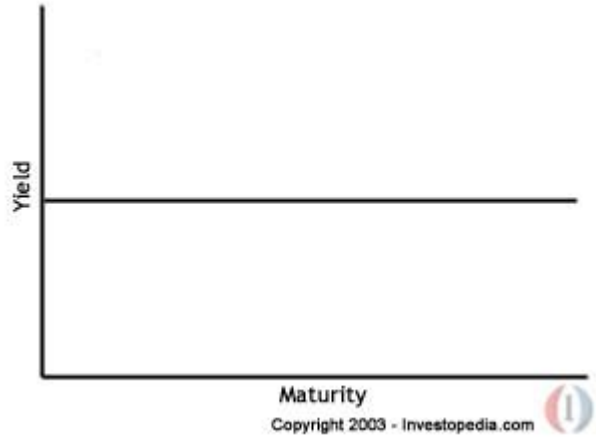

Figure 3: Flat Yield curve

A flat yield curve shows that short-term borrowers, medium term borrowers and long term borrowers all pay the same rate of interest.

In the real world there is no single interest rate charged to borrowers but a variety of interest rates depending upon numerous factors such as the length of the loan, risk assessment of the borrower and the expected rate of inflation.

Interest rates vary from yields on treasury bills and bonds to yields on deposit rates and interbank interest rates.

Yield curves can be quite volatile over time. Positively- sloped yield curves may become **inverted yield curves,** and yield curves can shift both upwards and downwards.

If interest rates are expected to fall, long-term interest rates will be depressed relative to short-term rates. Lenders will prefer to make longer-term loans now at the higher rate. On the other hand, borrowers wish to borrow short term in order to be able to rollover their loans.

When a rise in interest rates is generally expected, the impact will be for long-term rates to be increased relative to short-term rates. Borrowers here seek to borrow long-term in order to lock in at the current low rates, will seek to avoid borrowing short-term. Lenders on the other hand will be unwilling to lend long-term.

Yield curves are influenced by interest rate changes.

3.7.3 Factors that influence term structure

In general terms, an increasing term structure results from two factors:

1) increased risk of longer debt;
2) anticipated general interest rate rises.

Below are listed formal theories as to why interest rates increase with time.

a) Expectations theory

This states that the forward interest rate is due solely to expectations of interest rate movements. If an individual wishes to borrow for two years, two obvious possibilities present themselves: borrow for two years at an agreed rate; or borrow for one year and refinance for the second year (i.e. pay off the first loan by taking out a second).

In option (1), the interest paid on the loan will be based on the current interest rate and the forward rate for one year; and in option.

(2) the individual will consider the current interest rate and the expected interest rate for year two. Thus the choice between the options hinges on whether the forward rate for year two is higher or lower than the expected rate.

From the lender's point of view, if the expected rate was higher, they would only lend short, preferring to renegotiate at the end of one year and take advantage of the anticipated rate rise. A similar argument could be made if the expected rate was lower than the forward rate. Thus, for long and short-dated debt to co-exist, expected future rates and forward rates must be equal. Thus the term structure of interest rates is due purely to investor expectations.

b) Liquidity preference theory

According to the liquidity preference theory, bond holders for example are generally risk-averse in that they will wish to be compensated for the higher risk involved in holding long-term bonds by an appropriate risk premium on those bonds. According to the liquidity preference theory, the yield on longer-term bonds will therefore reflect not only market expectations, as outlined in the expectations theory, but also a liquidity premium.

The liquidity premium theory therefore predicts a generally higher set of interest rates than the pure expectations theory.

The problem with the expectations theory therefore is that it ignores risk-if the expected rate for year two is the same as the forward rate, then an individual needing to borrow for two years would choose a two-year loan since this eliminates the uncertainty of the actual interest rate to be paid in year two. Thus borrowers will aim to borrow for the period for which they need the funds. If lenders wish only to lend for one year, there will be a shortage of long funds and an excess of short funds. This will lead to a premium on forward rates, i.e. lenders will get a bonus for lending for two years and borrowers will have to pay extra if they insist on a two-year loan. In this case, the term structure of interest rates would again be up-ward-slopping but now it would be due to the liquidity preference of lenders and borrowers.

c) Market segmentation

The market segmentation theory was proposed by Culbertson (1957), and also assumes that investors have preferred habitats for their investments. It assumes that there are barriers to switching between short, medium and long-term investments. These barriers may be due to the need to meet regulatory requirements, or they may be self-imposed regulations or even transaction costs. According to this theory, the

shape of the yield curve is determined by separate supply and demand forces in each particular maturity segment.

The theory presumes that changes in interest rates in a particular segment of the market will have relatively little influence on other segments of the market because investors do not shift funds between the various sectors.

It has been argued that demand for capital funds in practice can be segmented particularly on a time basis. Thus, for example, companies tend to finance stocks with short-term funds and equipment with long-term funds. This leads to different factors affecting long-and short-term and a lack of a clear trend in the yield curve, characterized by irregularities such as humps and dips.

d) Preferred habitat theory

The preferred habitat theory was first formulated by Modigliani and Sutch (1966). The theory argues that the term structure reflects both the expectations of the future path of interest rates and a liquidity premium. However, in the preferred habitat theory, the liquidity premium does not have to rise uniformly with the maturity of the bond as assumed by the liquidity preference theory.

According to the preferred habitat theory, investors have specific objectives for their investments and this usually leads to specific time horizons (preferred habitats). For example someone wanting to save and buy a car in 5 years- time will tend to invest in an investment that will mature in 5 years- time, e.g. 5 year bonds, otherwise the investor faces the risk that a fall in interest rates will mean that he has insufficient funds at the end of five years to meet the investment objective.

According to the preferred habitat theory, risk premiums that exist to clear the bond market may be positive at certain time horizons in order to induce market participants to invest at different time horizons than their preferred habitat. For example, if most investors have a preferred habitat of five –year investments, then short and longerterm interest rates may both be higher to induce investors to accept investments at these time horizons. Conversely, if both short-and long-term investments are preferred habitats, then medium-term interest rates may be higher to induce investors to undertake medium-term investments.

3.8 INTEREST RATE RISK MANAGEMENT

The traditional ways of managing interest rate risk has been fixed-rate borrowing in the form of loans. It is simple, and companies know how much they will need each year to service the debt. However, it is not always possible to obtain a loan at the rates, or for the amounts required.

An enterprise may wish to take precautions against interest rates moving adversely in the future, or may wish to change the existing structure of its funding or deposits, for instance from a fixed rate of interest to a floating rate. With the development of the financial markets and, in particular, the financial futures markets, a number of instruments have arisen which allow the treasurer to hedge interest rate risk.

3.8.1 Interest rate swaps

An interest rate swap is an exchange of interest rate commitments, such that a fixed-rate commitment is exchanged for a floating-rate commitment. The parties to a swap retain their obligations to the original lenders, which means that the swap parties must accept counterparty risk.

Example

Zambia Sugar PLC has a high credit rating. It can borrow at a fixed rate of 10% or at a variable interest rate of (London Interbank Borrow and Offer Rate) LIBOR + 0.3%. It would like to borrow at a variable rate.

Lafarge Cement PLC has a lower credit rating. It can borrow at a fixed rate of 11% or at a variable rate of LIBOR + 0.5%. It would like to borrow at a fixed rate.

Using the principle of comparative advantage, both parties could benefit from a swap arrangement, whereby:

- Zambia Sugar PLC borrows at a fixed rate of 10%;
- Lafarge Cement PLC borrows at a variable rate of LUBOR + 0.5%;
- The parties agree a rate for swapping their interest commitments with, perhaps: Lafarge Cement PLC paying a fixed rate of 10.1% to Zambia Sugar PLC, and Zambia Sugar PLC paying a variable rate of LIBOR to Lafarge Cement PLC.

The outcome would be:

Zambia Sugar PLC

Borrows at	10%
Receives from Lafarge PLC	(10.1%)
Pays Lafarge PLC	LIBOR-------
	LIBOR-0.1%
	(a saving of 0.4%)

Lafarge PLC

Borrows at	LIBOR + 0.5%
Receives from Zambia Sugar PLC	(LIBOR)
Pays to Zambia Sugar PLC	10.1%
Net interest cost	10.6%
	(a saving of 0.4%)

In the above example, both companies benefit from lower interest costs.

Interest rate swaps are used for purposes other than obtaining a cheaper financing rate. They could, for example be used to change future cash flows, or to enhance returns.

Interest rate swaps, are off balance sheet items, as the principal amount of the contract is not paid, and it is just an agreement to swap future cash flows. However, the existence of the swap should be mentioned in the notes to the financial statements. The interest payments and receipts should be accrued over the life of the swap on a straight line basis. Financial institutions which actively trade swaps revalue their positions to the current market value.

3.8.2 Forward-rate agreements (FRAs)

A forward rate agreement (FRA) is an agreement whereby an enterprise can lock in an interest rate today for period of time starting in the future. On the future date, the two counterparties in the FRA settle up and, depending on which way rates go, one will pay an amount of money to the other representing the difference between the FRA rate and the actual rate.

Example

XYZ PLC has a K1,000,000 loan outstanding on which the interest rate is reset every six months for the following six months, and the interest is payable at the end of that six-month period.

The next six-monthly reset period may now be just three months' away, but the treasurer of XYZ PLC thinks that interest rates are likely to rise between now and then. Current sixmonth rates are 8% and the treasurer can get a rate of 8.1% for a six month FRA starting in three months' time. By transacting an FRA, the treasurer can lock in a rate today of 8.1%. If interest rates as expected to say, 9%, XYZ PLC has reduced its interest rate charge as it will pay the current 9% rate on its loan but will receive from the FRA counterparty the difference between 9% and 8.1%.

If however, rates drop to 7%, XYZ PLC will still end up paying an effective rate of 8.1% because, although the interest rate on the loan is lower, the company will pay the FRA counterparty the difference between the 7% and 8.1%.

If rates are 9% in three months' time:

Interest payable on the loan 9% x K1m x 6/12	K45,000
Amount receivable on FRA (9% - 8.1%) x K1m x 6/12	(K4,500)
Net amount	K40,500)

The K40,500 is the net amount payable, giving an effective rate of 8.1%.

If rates are 7% in three months' time:

Interest payable on the loan 7% x K1m x 6/12	K35,000
Amount payable on FRA (8.1% - 7%) x K1m x 6/12	K5,500
Net amount	K40,500

The K40,500 is the net amount payable, again giving an effective rate of 8.1%.

3.8.3 Interest rate futures

Futures are standardized traded forms of FRAs, and the above FRA example could also have been used for an interest rate future.

FRAs are normally transacted with banks and other financial institutions, and because of this, they are tailor-made to suit the dates and amounts that each individual company requires. However, interest rate futures are exchange-traded and each contract is for a pre-specified amount and date. In the UK, interest rate futures are traded on the London International Financial Futures Exchange (LIFFE). In Zambia, interest rate futures are not common, although Development Bank of Zambia (DBZ), does sometimes deal with them. However, with the introduction of the second stock market, Bonds and Derivatives Exchange (BAdeX), we expect to see these transactions becoming common in Zambia.

If a company does not need a specific amount or set dates, then futures are useful. For most organizations, however, they are not margin payments on futures contracts which many organizations find burdensome. However, futures contracts are very liquid. It is easy to liquidate a contract. This is not so common with FRAs. In the above FRA example, the treasurer could not have sold on the reverse effect of the first. With a future, it is a very easy matter just to sell the future back. In practice, many organizations use FRAs, relatively few use futures.

The pricing of interest rate futures is based on taking the interest rate away from 100. Thus, if the interest rate was 7%, then the futures price would be 100-7, i.e. 93. If the interest rate were 12%, the futures price would be 100-12, i.e. 88. Futures contracts can be bought and sold. If the company were hedging against interest rate rising, for instance, where interest payments were due, a futures contract would be sold. Suppose that the interest rate was 7% and thus the futures price was 93. The company could sell the futures contract as 93. If the rates then move up to 8% instead of 7%, the futures price would be 92. The company would lose out on the loan interest as it would now have to pay 8% instead of 7%. However, the company would profit on the futures contract, by 93-92, i.e. 1%, which would offset the increase in loan interest. This is because, the company originally sold the futures contract at a price of 93 when rates were 7% and purchased later when rates had moved to 8% and the futures price was 92.If a company wished to hedge against interest rates decreasing, for instance, if it was to deposit money, then it would buy the contract initially, and the sell the contract at a later date. Both FRAs and futures ‘lock’ in a price today for a transaction which will occur at some time in the future.

3.8.4 Interest rate options

An option is a right but not the obligation, to carry out a transaction at a price set today, at some time in the future. Swaps, FRAs and futures are all contracts which two parties agree to transact and which must be carried out even if circumstances change. An option, however, gives the buyer the choice of whether to transact or not.

A company would generally buy an option from an option seller or option writer. An option is a form of insurance, and as such, a premium is paid at the time the option is taken out, for the period of the option. In the FRA example above, XYZ PLC paid a net amount K40,500 no matter what happened to interest rates.

Let us suppose that, instead of transacting the FRA above, an option was bought, entitling the buyer of the option to pay the same interest rate of 8.1%. This is known as the **strike price.** The period of the option is for three months, which is when the renewal period for the loan starts. Suppose that the option premium paid today is K1,000. In three months' time we could have the same scenarios.

If rates are 9% in three months' time:

	K
Interest payable on the loan 9% x K1m x 6/12	45,000
Exercise the option at a strike price of 8.1%, receive	(4,500)
Plus premium paid	1,000
Net amount	41,500

If rates are 7% in three months' time:

Interest payable on the loan 7% x K1m x 6/12	35,000
Plus premium paid	1,000
Net amount	36,000

Thus, if interest rates are 9% in three months' time, XYZ PLC will pay a net amount of K41,500 in interest over the six-month period, which is more than the FRA because of the option premium. However, if the rates fall to 7%, only K36,000 will be paid, as the company does not need to exercise the option. An option is, therefore, not quite so favourable when rates go as expected, but much better when rates move in the opposite direction. Thus, with an option, a company can take advantage of favourable interest rate movements.

Interest rate options can be purchased from banks, while standardized interest rate options are traded on LIFFE. A number of companies now use 'swap options', which are options on transacting a swap at a future date, and some companies may find these useful.

3.8.5 Caps, floors and collars

An interest rate cap is an option which sets a maximum interest rate on future borrowing for an agreed length of time.

A floor is an option that sets a minimum interest rate.

A collar arises when a borrower buys an interest rate cap and at the same time sells an interest rate floor. The premium received from selling the floor will reduce the cost of the premium paid for the cap.

Example

Zambian Breweries PLC can currently borrow at 11%, but is concerned that interest rates may rise in the near future to 14% or more. In this situation, the company could buy an interest rate cap from a bank which will fix the maximum rate for borrowing. The bank will reimburse the company if market interest rates rise above the cap rate.

As part of the arrangement, the company may also agree that it will pay a floor rate of, say, 10%. The bank will pay a premium to the company for agreeing to this floor rate.

3.9 ACTIVITY

1) Define interest rate risk.

2) Explain what the theory of 'term structure of interest rates' refer to.

3) What is a yield curve?

4) Outline the tools that can be used to manage interest rate risks.

5) Explain how futures are priced.

3.10 SUMMARY

Interest rate risk is a problem faced by all companies with gearing. Given the past volatility of interest rates, an awareness of the risks of interest rate movements is important. In this unit, we have considered some of the influences on interest rate movement, and identified a range of techniques that may be used to reduce exposure to interest rate risk, such as interest rate swaps, forward-rate agreements, futures and options.

In the next unit, you will learn about currency risks and currency risk management.

UNIT 4

CURRENCY RISKS

4.1 INTRODUCTION

Welcome to unit 4. The unit has been designed to equip you with knowledge on currency risks.

4.2 AIM

The aim of this unit is to highlight some of the currency exposures customer involved in importing and exporting could be faced with.

4.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Describe how currency risk arises;
2) Discuss how overseas operations can be financed;
3) Describe the foreign exchange terminology such as; spot, forward, discount and premium rates.
4) Critically evaluate the various overseas risks that a firm is likely to be exposed to; 5) Discuss the various attitudes firms have towards risk.

4.4 TIME REQUIRED

You are advised to spend at least 3 hours reading through this unit.

4.5 REFLECTION

Attempt to define currency risk in your own words.

……
……
……
……
……

4.6 CURRENCY RISK-OVERVIEW

Currency risk also known as exchange risk, arises when an obligation has to be settled in a currency other than your domestic currency that can be expected to be received for a given amount of foreign currency will move as a consequence.

Movements can be favourable, resulting in windfall gains but they can equally easily be unfavourable. Businesses that operate on tight profit margins can ill afford to have their expected income reduced, or their costs increased, through unfavourable exchange rate movements. Such businesses should either invoice in domestic currency when selling abroad, or insist in paying in domestic currency for imports-in other words, pass on the exchange risk to the buyer of goods when selling and/or to the seller of goods when buying. If this is not possible, fixing the domestic currency amounts to be received or paid though a hedging mechanism, such as a forward exchange contract, is advisable.

It is important for the treasury manager to be aware of the risks involved in a company trading in other countries, and the various components of currency risk. Currency risk may also be faced even if an enterprise has no overseas operations, or transactions with a currency component.

4.6.1 FINANCING OVERSEAS OPERATIONS-A GLOBAL STRATEGY

In times past, a business would, in its early days tend to be concentrated in one geographical area, and in the regulated environment which used to pertain, even quite large companies would define themselves in national terms, and raise capital and invest in facilities where their market was. If asked about competition, they would naturally think of other enterprises based in the same country.

We now live in a global village. Globalization has taken centre stage. Companies in one country will trade with companies in other countries. Technological innovation has contributed to this trend. The wide use of the internet has contributed to the trend of people buying things online from other countries.

From a business aspect point of view, it is very common nowadays to see enterprises raising capital in one country and going to invest in another country and produce goods which are to be marketed in a third.

Companies have developed global strategies, perhaps to exploit new markets or secure supplies of raw materials.

Some ways of expanding business in overseas markets include:

a) **Acquisitions**

   A company can acquire (buy) off another company in another country. This is a cheaper way of entering a new market in a new country.

b) **Retained earnings**

   A subsidiary could also rely upon its own internally generated funds. This would avoid many of the problems of financing overseas, but is unlikely to result in the necessary level of expansion to meet high-growth objectives, and is obviously not suitable for new overseas ventures.

c) **Funds from your country**

   Funds can be raised by the parent company locally and transferred overseas to subsidiary companies by way of a combination of equity and loans. A good example of this is the way the mobile telecommunication company MTN Group, who's Head Quarters is in South Africa, but operates as subsidiaries in Zambia, and other parts of Africa raises funds from South African banks, and expands its operations in the many African countries it operates in.

   The main advantage of this method is that the company will probably be familiar with the South African capital markets and, therefore, be capable of raising finance quickly and cost-effectively. However, if exchange controls exist, this method can become difficult and expensive.

   Another disadvantage of using local currency finance for an overseas expansion is that it would in no way reduce foreign exchange risk through matching receipts and

payments (a method of hedging currency risk explained in more detail in the next unit).

The company would be more exposed to political risks as well. If the investment is lost, perhaps through a war, like what has been happening in recent years in countries such as Libya, Egypt, Tunisia, Sudan, MTN's operations in such countries would be at risk of losing its investments and the foreign governments may not be able to compensate the firm. The other disadvantage is that, if a subsidiary fails, the holding company is left with the liability of loss.

d) Finance in the overseas country

The main advantage of this method is that it will result in reductions in risk. Foreign exchange risk will be reduced since any loses in the value of the overseas asset will be offset by gains on the liability and vice versa. The complete elimination of risk is, however, unlikely since it would require the exact matching of cash flows, on the asset and liability. Political risk can also be reduced since, if the investment is lost, the liability will be eliminated as well.

Difficulties may be experienced with this method of finance if the country concerned does not have a well-developed capital market. On the other hand, financing in the overseas country can make such investments more acceptable to that country since it is seen that not all the profits made are sent abroad.

Alternatively, it could also be argued that financing overseas limits the methods by which profits from the overseas investments can be repatriated-heavy reliance being placed upon the payment of dividends. When finance is raised in Zambia by investors for example, it is easy for this money to be remitted to other countries.

Most governments, like the Zambian government, take steps to encourage overseas investors since these are beneficial to the country's economy, creating employment and wealth. A good example is how the Zambian government sold most of its shares in the mines to foreign investors.

e) Financing from other capital markets

Finance can today be raised from a variety of capital markets in different countries. One possible incentive in raising money in other countries is that interest rates are likely to be lower. For example, in a country like Zambia, interest rates charged on loans are usually above 20% whereas in a country like the U.K. interest rates may be around 3%.

However, if the interest cost is lower, then it is likely that the currency borrowed is strong and will, therefore, appreciate with respect to the local currency and other currencies. The expected exchange loss on the borrowings would therefore, offset any benefit through a lower interest rate.

4.6.2 FOREIGN EXCHANGE TERMINOGY

a) Spot rate

This is the current exchange rate quoted by a bank for transactions which are due for immediate or very imminent completion. Mostly, it is used for transactions to be completed in two working days. The spot fluctuates frequently. Good examples of a spot rate are those rates that banks or bureaus display every day. E.g. K/$ 10.50-10.40

K10.50 in the above example is the rate at which a Zambian bank would sell dollars. The selling rate in short is a rate at which a bank sells foreign currency.

The bank's buying rate is the rate at which the bank buys foreign currency from its customers. In the example above, this would be K10.40.

The difference between the bank's buying rate and the bank's selling rate represents the bank's margin i.e. profit.

b) Forward rates

This is the rate which is quoted today for transactions that will be completed on a specified future date. The forward rate is calculated arithmetically using interest rate parity relationship. A Zambian bank might quote a one-month forward rate for US$ as follows:

	$/ZMK
Spot	10.50 – 10.40
One month forward	11.50 - 11.40

In the above example, a customer could agree to sell US$ to the bank in one month's time at an exchange rate of K11.40, or agree to buy US$ from the bank at a rate of K11.50 for a transaction scheduled to take place in a month's time.

i) **Discount:** This is added to the spot rate to arrive at the forward rate. This implies that the forward rate is cheaper than the spot rate.

ii) **Premium:** This is subtracted from the spot rate to arrive at the forward rate. This implies that the forward rate for a currency is more expensive than the spot rate.

Example

A bank is currently quoting the following rates for Kwacha spot and forward:

	ZMK/$
Spot	10.75 - 10.55
1 month forward	1.50 – 1.30 pm
2 months' forward	2.50 – 2.20 pm
3 months' forward	3.60 – 3.40 pm

a) The bank will buy spot dollars at a rate of K10.55: $1

b) The bank will buy dollars one month forward at a rate of K9.25: $1 i.e.

Spot rate	10.55
Less premium	1.30
	9.25

c) The bank will sell dollars two months' forward at a rate of $8.25: $1 i.e.

Spot rate	10.75
Less premium	2.50
	8.25

d) The bank will buy dollars three months' forward at a rate of K7.15 i.e.

Spot	10.55
Less premium	3.40
	7.15

4.6.3 OVERSEAS RISK

The ability of companies to successfully raise capital overseas, and thus expand their worldwide operations, has brought increased uncertainty. Not only do we have to be alert to

our international competitors' technological, marketing and managerial advances, but we have to think about the risks associated with currency volatility, interest rate changes and the political, social, economic climate. Most observers distinguish between two types of risk which affect the value of the entity as a whole (transactional and economic) and one which doesn't (translational). We are now going to look at each risk in the order mentioned above.

4.6.3.1 Transaction risks

Foreign exchange transactions for immediate effect (or to be precise, unless special arrangements are made, for completion in a couple of days) are known as 'spot' rates. These rates fluctuate according to demand and supply which could lead to an exchange loss being experienced by a customer. This has been the case in Zambia in recent months (August, 2015-September, 2015) where the Kwacha has depreciated so much against the dollar from around K6/$1 to around K10/$1. Importers buying foreign currency are likely to be affected negatively by such a trend as they would need more Kwacha to purchase few Dollars. On the other hand, the depreciation of the Kwacha can have a positive impact on the exporter as he is likely to obtain more local currency for a few Dollars.

Transaction risks are essentially those risks associated with the delays between transactions and settlements. They occur at the operational level of control, having a measurable effect on results, but need to be thoroughly understood before moving on to the higher levels.

4.6.3.2 Economic risks

These are essentially the risks which affect a business before a transaction actually takes place, and are not, therefore, measurable in an accounting sense. They relate to the tactical and strategic levels of control, and are sometime referred to as competitive risk.

Examples at the tactical level include such things as:

- Insisting in dealing in only home currency-kwacha for example, thereby avoiding transaction risk, but opening up the risk that suppliers and customers will prefer to deal with competitors;
- Deciding to increase the level of stock of an imported raw material, the world price of which (expressed in dollars) is expected to increase. This may well be borne out by events, but if Kwacha strengthens against the US $ in

the meantime, competitors who did not increase their stock would buy their material more cheaply;

- Entering into a competitive bidding process, in which it is necessary to quote a fixed price now, knowing that the buyer will not make s choice between suppliers for several weeks or even months;
- Investing in a marketing campaign in, say, South Africa, with a view to supplying goods or services in competition with local producers a few months hence. If Kwacha was to strengthen relative to Rand, the appropriate Rand price might not convert into enough Kwacha to make it worth- while.
- (equally important but often overlooked), investing in a marketing campaign in say, South Africa, with a view to supplying goods or services, in competition with Botswana producers a few months hence. In this case, if Kwacha were to strengthen relative to Pula, (even if both weakened against the Rand), a Rand price which appealed to the Botswana supplier might not convert into enough Kwacha to make it worth- while.

Examples at the strategic level include:

- Deciding to acquire resources, say equipment in China, with a view to supplying goods or services to the Zambian market. If Kwacha was to weaken against the US$ (China, usually quotes its exports in dollars), the costs of the operation could become uneconomic;
- Investing in mining equipment in Zambia so as to export more copper to overseas markets. If Kwacha appreciates against the currency of the overseas country, the Kwacha conversion costs could render the business uneconomic;
- (possibly devastatingly, but again widely appreciated) Investing in Zambia to supply the Zambian market, and not finding out until after Kwacha has appreciated against say the Rand, that the same goods could then be supplied more cheaply from South Africa.

It is perhaps because these items do not appear in the accounts (because the business is lost to competitors) that they are less well understood. The fact that, in today's conditions-as the last example showed-you do not even have to be importing or exporting to have an exposure to foreign currency risk. Identifying and quantifying

(with inevitable margins of error) economic risk calls for a thorough appreciation of the competitive position and prospects of the enterprise. A proactive financial controller should be in the best position to carry out this task.

These economic risks affect potential rather than performance. A monitoring system based on forward-looking values could accommodate them, but one based on backward-looking costs cannot. This brings us to the question of the third category of risk that associated with translation.

4.6.3.3 Translation risk

Translation risk does not affect the cash flows of the entity, but nevertheless attracts considerable treasury attention. It relates to the situation in which, for the purposes of preparing a balance sheet for publication, overseas assets and liabilities are translated at current rates into the currency of the country in which the entity is domiciled, e.g. into Kwacha for a Zambian registered company. If Kwacha has weakened, overseas assets and liabilities will be translated into higher Kwacha figures; if Kwacha strengthens, then they will be translated into smaller Kwacha figures.

Taking the first of these possibilities, let us imagine that a company borrows $10 million at a time when $1 is worth K10. At that time, the borrowings would appear on the company's statement of financial position at K100 million. Over the ensuing year, however, if the Kwacha continues to depreciate and becomes $1 = K12, the borrowings would now be K120 million. The company pays more by K20 million. If this were assumed to be a permanent change in the Kwacha-Dollar rate it might be argued that this was a genuine loss of value, in the sense that more Kwacha would be needed to pay the interest and repay the loan in due course. The problem is that it is unreasonable to assume that the change is a permanent one. It could be reversed, reduced or increased in the next and subsequent years.

Meanwhile, the change in parity will have affected reported profits (and hence earnings per share) total assets, borrowings, net worth (and hence gearing) but as already mentioned, it will not have affected the measured cash flow in the period being reported on. Academic theory argues that this risk of itself, need not concern financial managers, but in practice there are two strong forces at work:

- Although it does not affect the value of the entity as a whole, it can affect the attribution of that value as between the different stakeholders. Higher gearing may lead to higher interest rates being charged on bank loans, either directly in accordance with clauses in borrowing agreements or indirectly as a result of the company's credit rating being reduced. The banks benefit at the expense of the

  equity investors in the business. If the treasurer is pursuing an objective of maximizing the proportion of the value of the entity which accrues to the equity, he will want to manage this risk;
- If the accounts are being used 'beyond their design spec', e.g. as the basis for calculating bonus payments for directors and senior managers, then there is a temptation to protect the current year's figures, even though it is known that doing so has a long-term cost. This is comparable to pulling profit into the current year, knowing that it will both reduce next year's profit and result in tax being paid earlier than necessary.

The former is a good reason the latter is often the real reason.

4.6.3.4 Political risk

Political risk is the unwanted consequences of political activities that will have an effect on the value of the firm. We are really only concerned with the detrimental effects. Examples of actions that can have a detrimental effect on the value of the firm are:

- Discrimination against foreign businesses;
- Expropriation of assets: expropriation involves the government confiscating private property; this may be done by nationalizing local subsidiaries;
- Rules specifying use of local labour and materials;
- Price-setting constraints;
- Exchange controls; limitations on the extent to which a country's currency can be used to transfer funds or restrictions on the conversion of a currency into other currencies;
- Tax regulations: an increase in the tax rate or introduction of new taxes;
- Restrictions on access to local loans;
- Wars and political unrest e.g. in countries such as Libya, Egypt, Syria, etc.

- Change of government through elections e.g. in Zambia each time there is a new government, exchange rates are affected.

High political risk does not necessarily mean that a company should not invest in a particular country. It may be that the high political risk does not affect the firm's industry. Even if the project is affected by high political risk the level of returns available may be large enough to justify taking on the risk.

Liberalization of cross-border financial flows, coupled with technological developments which have speeded up and cheapened the costs of shifting funds around, have provided opportunities to protect against exchange rate and interest rate risks but, ironically, have opened up others which we may not have thought of before. The emergence of specialist managers (including those for hedge funds), and the growth in institutional investment abroad, have likewise added to liquidity in the market-but also added to instability.

The daily volume of trade in the foreign exchange markets is something like double the aggregate of national governments' reserves. The bulk represents speculative trading, driven by rumour, expectations, chartism, etc, and often national governments' are put under pressure. Politicians try to resist market pressures but, by doing so, merely play into the hands of the speculators. The spot market, incidentally, now represents only a minority of the trade, thanks to the explosive growth of derivatives such as options and futures.

4.7 ATTITUDES TO RISK

The examples quoted above tended to emphasize the downside risks, i.e. leading to an adverse effect on performance or potential. At this stage, we need to recognize the upsides, too: if Kwacha weakens after you have invoiced in dollars, for example, your Kwacha income increases. This prompts the recognition of a spectrum of attitudes to risk, including:

- Choosing to open up risks, i.e. to get into a position to benefit from exchange rrate volatility, e.g. buying currencies which are expected to strengthen, writing options for other businesses. Perhaps only a small number of organizations are prepared to do this, but the high stakes mean that those which do need financial managers of the highest skill;

- Accepting the risks which arise in the normal course of business as an inevitable element thereof, on a par with competitors discovering new ways of delivering the goods or services;
- Taking a view on currency movements and finding a way to cover some, (i.e. take some offsetting action) but not others. If you are confident that Kwacha can only weaken against the dollar, leave dollar receipts uncovered, but cover all payables, and vice versa. The basis for taking the view might be an econometric forecast, chartism or simply sentiment. As a general rule, confidence is misplaced at least as often as it is justified, as some embarrassing annual reports have shown;

- Cover all identified risks in certain categories, e.g. all liabilities, all transaction exposures, all economic exposures included in the current 12-month forecast;
- Hedge (i.e. cover a predetermined proportion of) identified exposures, e.g. 60% of transaction risks, 40% of economic risks included in the current 12-month forecast;
- Matching opposite risks, e.g. those associated with importing and exporting-actively seeking matches by leading and lagging, perhaps, and covering or hedging only the net balance.

In organizational terms, there have been the usual debates as to whether currency risk management should be centralized or decentralized, and if the former, whether a profit centre or a cost centre approach is appropriate. As time goes by, opinion seems to be hardening in favour of:

- Decentralizing the responsibility for identifying risk;
- Transferring identified risks to the centre at the market rate;
- Giving (within defined limits) the centre discretion as to whether they actually take cover in the market;
- Measuring their performance against the cost or benefit of covering everything at the market rate (i.e. they have to beat the market to be in credit).

Dealing with risk and individuals' attitudes to risk is vital to any decision-making process, and when multinationals look to evaluate investments overseas, they have to compare the risks with the rewards.

4.8 ACTIVITY

1. Define currency risk.
2. Describe some of the ways in which businesses can expand into overseas markets.
3. Distinguish between spot and forward rates.
4. State the four (4) key risks that relate to overseas transactions.

4.9 SUMMARY

In this unit, we have introduced some important foreign exchange terminology. We have also identified the main types of foreign exchange risks which may have an adverse impact firms' profitability and on the method of financing overseas operations. These risks include; transaction, translation, economic and political. The unit concluded by considering different attitudes to risk and the trade-off between risk and reward.

Now that currency risks have been discussed in this unit, the next unit will analyse currency risk management i.e. how the risks we have identified here can be hedged against or mitigated.

UNIT 5

CURRENCY RISK MANAGEMENT

5.1 INTRODUCTION

Welcome to unit 5!

5.2 AIM

The aim of this unit is to now discuss ways in which the currency risks we discussed in unit 4 can be mitigated.

5.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Describe the internal hedging techniques for transaction risk;
2) Discuss how hedging currency risk can be done using financial markets; 3) Critically discuss how to select an appropriate hedging method.

5.4 TIME REQUIRED

You are strongly advised to spend at least three (3) hours studying this unit.

5.5 REFLECTION

What is hedging?

.....

.....

.....

……..

5.6 CURRENCY RISK MANAGEMENT-OVERVIEW

In unit 4 we identified the types of risk associated with overseas trading. In this unit we look at methods suitable for hedging these risks. The techniques for hedging currency risk are many and varied. Most of them are aimed at hedging transaction risks, although some are also useful for hedging translation risks.

5.6.1 INTERNAL/TRADITIONAL HEDGING TECHNIQUES

5.6.1.1 Invoicing in local currency

Here a company simply invoices in its own local currency. The risk is not avoided here it is merely transferred to the customer. This technique may not always be possible, given that the company may well be competing with local industries invoicing in the local currency, and as such, the overseas quote may become uncompetitive.

5.6.1.2 Multilateral netting

This is a form of matching appropriate for multinational groups or companies with subsidiaries or branches in a number of overseas countries. The process is based on determining a base currency e.g. Kwacha or Dollar, so that the intra-group transactions are recorded only in that currency; each group company reports its obligations to other group companies to a central, say Zambian treasury department, which then informs each subsidiary of the net receipt or payment needed to settle their foreign exchange intra group positions.

While this procedure undoubtedly reduces transaction costs by reducing the number of transactions and also reduces exchange rate risk by reducing currency flows, the difficulties are that there are regulations in certain countries which severely limit or even prohibit netting, and there may also be cross-border legal taxation problems to overcome as well as the extra administrative costs of the centralized treasury operations.

The possible advantages of netting are that:

- There are lower transaction costs as a result of fewer transactions; Regular settlements may reduce intra-company exposure risks.

The possible disadvantages are that:

- If the base currency is generally weak against other currencies for a sustained period, subsidiary company results could be significantly distorted;
- If a particular subsidiary is suffering from cash-flow problems, the centre may have to arrange for it to have additional funds for settlements, thereby offsetting some or all of the transaction cost benefits and possibly incurring exchange losses;
- The central treasury may have difficulties in exercising the strict control that the procedure demands;
- Tax considerations may be adverse.

This is known as multilateral netting. If it involved only two companies, it would be referred to as bilateral netting.

5.6.1.3 Leading and lagging

This method involves changing the timing of payments in an attempt to take advantage of changes in the relative value of the currencies involved.

Leading is making payments in advance of the due date whereas *lagging* is delaying payments until after the due date. If a company feels that it can reasonably predict the direction of exchange-rate changes, then:

- If the company has a debtor, i.e. an asset, in a currency predicted to depreciate it should aim to dispose of this asset as soon as possible. This is the process of 'leading'.
- If, however, it has a creditor, i.e. a liability, in a currency predicted to depreciate then it should aim to delay payment for as long as possible. This is the process of 'lagging'.
- If the currency involved is strong or appreciating, the company should 'lag' its receipts and 'lead' its payments.

A 'lead' payment however, can involve a cost in terms of interest foregone.

5.6.1.4 Matching

This is the use of receipts in a particular currency to match payment obligations in the same currency.

A Zambian company may have a substantial trade with another country overseas, involving both debtor and creditor transactions, e.g. where goods are exported and invoiced in a foreign currency and the overseas sales transacted are also paid for in that currency. The Zambian company may well benefit from operating a foreign currency account, whereby its exchange risk is limited to any conversions of the net account balance into Kwacha. A further advantage is that balances on the account can be cleared, i.e. disposed of, if the exchange rate is favourable to the account-holder.

The foreign currency account can either be held with a domestic bank, e.g. in Zambia, or by using say, a bank deposit account in the overseas country. It is allowed in Zambia to hold accounts in different currencies like Sterling, Dollar, Rand etc, and many banks offer such a facility.

If there are likely to be significant balances on the account, or frequent remittances to Zambia, the account holder may well need exchange risk protection by taking out some form of hedging cover.

Matching is more commonly found in multinational group situations, where different subsidiary companies inter-trade. Matching is also a technique that may be used to hedge translation risks.

If a Zambian company has an investment in assets in say South Africa, exposure to movements in the exchange rate between Kwacha and the Rand can be reduced by financing the investment with a US dollar loan; if the exchange rate alters, the change in the Kwacha value of the asset will be matched by a corresponding change in the Kwacha value of the Rand loan. For example, a company may purchase a factory unit in South Africa for ZAR400,000 financing the purchase by means of a loan denominated in Rand.

5.6.2 HEDGING USING THE FINANCIAL MARKETS

5.6.2.1 Forward markets

A forward contract is one in which one party agrees to buy ‘something’, and another party agrees to sell that same ‘something’ at a designated date in the future. For example in the case of a forward exchange contract, one party agrees to deliver a specified amount of one currency for another at a specified exchange rate at a designated date in the future. The specified exchange rate is called the *forward rate*. The designated date at which the parties must transact is called the *settlement date or delivery date.*

When an investor takes a position in the market by buying a forward contract, the investor is said to be in a *long position.* If instead, the investor's opening position is the sale of a forward contract, the investor is said to be in a *short position.*

The foreign exchange forward market is an interbank market. Most forward contracts have a maturity of less than two years. For longer-dated forward contracts, the bid-ask spread increases; that is, the size of the spread for given currency increases with the maturity. Consequently, forward contracts become less attractive for hedging long-dated foreign currency exposure.

Forward contracts are not that common in Zambia. Banks would rather deal with spot transactions.

Forward exchange fixed contracts

A forward exchange contract is an agreement, entered into today, to purchase or sell a fixed quantity of a commodity, currency or other financial instrument on a fixed future date at a price fixed today. They are tailor-made to meet the exact requirements of the contract, and once entered into, the contract must be completed.

A major problem with a forward contract is the fixed settlement date. By using an option forward contract, this can be overcome.

Forward exchange option contract

A forward exchange option contract offers the same arrangement as a forward exchange fixed contract except that there is a choice of dates on which the user can exercise the contract. This is either on any date up to a specified date or at any time between two future dates. In either case, the forward rate that applies would be the forward rate, in the period in which the contract can be exercised, that is *least favourable* to the purchaser of the contract.

Example

ABC Mining (Z) Ltd exports copper to the USA and Europe. It is due to receive US$ 120,000 in six months' time. The rates are as follows:

Six- month US dollar forward rate = 12.456 – 13.456

The spot rate is 9.9200 – 10.1100

Calculate the Kwacha receipt if the company decides to hedge using a forward exchange contract.

Solution

$120,000 x 12.456 = K1, 494, 720=00

Problems with forward contract

A forward contract is binding. If, for example, importers discover eventually that they have contracted to buy more currency than they need to pay their suppliers, they are required to take up the full contractual amount at the agreed forward. They can then sell back any surplus to the bank at the spot rate on the day they settle. This is known as 'closing out' the contract. Similarly, exporters who have contracted to sell forward more currency than they actually receive from their customers will be required to buy extra amounts at spot amounts in order to close out their contracts.

5.6.2.2 Money market hedge

The use of the forward market as a 'hedge' against variations in exchange rates was illustrated above. Using the forward market, it was possible to ensure that the exchange rate applying at some future date was known with certainty. Another way of achieving this is to use the money market.

5.6.3 SELECTING A HEDGING METHOD

When a company has an outstanding foreign currency payable or receivable, it may choose to hedge against the currency risk by using the forward markets or the money markets. The method selected should be the one which leads to the smallest payment in Kwacha terms or the highest receivable in Kwacha terms.

5.6.4 FUTURES

Financial futures in foreign exchange rates are contracts to buy or sell an amount of foreign currency at a future date, and are traded on futures exchanges such as Chicago Mercantile Exchange, which has a London office and London International Financial Futures Exchange (LIFFE). The futures exchange quotes a price for each contract on every trading day, so that contracts are saleable before their delivery date. Although this makes them flexible, there are restrictions in that contracts have to be for specific amounts of currency.

An example of how futures contracts are calculated is shown below:

A Zambian company XYZ (Z) Ltd sells goods to Smith Inc. in the USA to the value of $2,650,000. The sale takes place in June with payment due in August. The finance director of XYZ (Z) Ltd is concerned that the Kwacha will strengthen against the US dollar which would reduce the Kwacha value of the contract. The spot rate in June is US$10.1100.

Assume that in June a September dollar/Kwacha futures contract is quoted at US$ 9.9200. This means that the futures market will buy or sell Kwacha for Dollars for future delivery, in September, at an exchange rate of US$9.9200/ZMK.

To hedge against the risk of the Kwacha strengthening against the dollar, XYZ (Z) Ltd would need to buy an appropriate number of Dollars/Kwacha futures contracts. This means that XYZ (Z) Ltd has agreed to buy Kwacha for Dollars on the future delivery date at the exchange rate specified by the price on the futures contract. The Kwacha futures are K62,500. The number of contracts that XYZ (Z) Ltd would need to buy is:

$2,650,000 x 9.9200 = K26,288,000 ÷ K62,500 = 420.27 contracts.

Only a whole number of contracts can be traded, so XYZ (Z) Ltd would need to buy either 420 contracts of 421 contracts. We shall assume 420 contracts though this will leave a small amount un-hedged.

In August, just before Smith Inc. is due to pay, the futures contract price has changed to US$8.2500 and the spot rate has also moved to US$8.2500/ZMK. XYZ (Z) Ltd will now close out its position in the futures contracts by selling the futures contracts.

	$
Profit on futures	
(9.9200 - 8.2500) x 62,500 =	
$104,375 per contract x 420 contract	43,837,500
Received from Smith Inc.	2,650,000
	46,487,500

The total receipts of $46,487,500 will be sold on the spot market to give a Kwacha receipt of 46,487,500 x 9.9200 = K461,156,000 and effective exchange rate for the US$2,650,000 received of K461,156,000 ÷ 46,487,500 = US$9.92

In practice, the profit on the futures contracts would be received as it arose during the period from June to August. Note also that if the Kwacha had weakened against the dollar, a loss would have been made on the futures contract, but this would have been offset by receiving more Kwacha in the spot market in August.

5.6.5 OPTIONS

Options give a customer the right but not the obligation to buy ('call') /sell ('put') a specific currency at a specific date. The banks take into account the costs of buying and selling currencies and the potential profits, and distil them all into the premium payable for the option. The level of premium depends on a number of factors:

1) **The strike price**. This might be:
    - 'at the money' i.e. the agreed price corresponds with that currently available (spot or forward);
    - 'in the money', i.e. the agreed price is more favourable to the client than is currently available (but then the premium would be higher);
    - 'out of the money' i.e. the price is less favourable to the client than is currently available (but then the premium would be low-possibly zero);
2) **The maturity:** the premium follows the line of a diminishing returns curve, e.g. a six-month option will not be twice as expensive as a three-month option;
3) **The volatility of the spot rate,** i.e. the greater the volatility, the greater the premium;
4) **Interest rate differentials,** which affect the banks' carrying costs;
5) **Liquidity in the market,** sentiment and other judgemental factors.

Example

A Zambian company is tendering for the sale of equipment to a South African company for 3 million Rand, settlement due in three months' time. The current spot rate is ZAR 0.70: K1. However, the company is worried about rand weakening against the kwacha, thus making the sales less profitable.

The company has been offered a three-month put option on rand at ZAR0.72:K1, costing

0.02 cents per kwacha. What is the company's position?

Solution

The kwacha amount received if the option is exercised:

ZAR 3,000,000 x 0.72 = K2,160,000

In addition, the cost of the option itself must be considered. This is quoted in cost per foreign currency unit:

0.02 x 2,160,000 = ZAR 43,200.

This is payable regardless of whether the option itself is exercised. Hence the net kwacha amount:

	K
To sell ZAR 3 million	2,160,000
Cost of option (at spot) 43,200 x 0.70	(30,240)
	2,129,760

Clearly, the advantage here is that if the spot rate moves in the company's favour (say to K0.50), then the option can be abandoned and the rand sold on the spot market. Secondly, if the tender is not won, the company has no binding obligation to deliver rand in three months' time.

Observation suggests that the option route is becoming increasingly popular. It sits comfortably alongside a situation in which the cash flows themselves are subject to considerable uncertainty, e.g. the tactical component of economic risk. If you cannot be sure the rand will be received, you do not want to be committed to delivering them to the bank at a fixed price.

There are rules and products to deal with delays in currency becoming available and, of course, the options have a (positive or negative) value at a point in time. This can be left unrealized, or could be realized by selling the option or taking up an opposing position in the market. The scope for switching profit between accounting periods is enormous.

There are various other products available on the futures markets, and there are various combinations of these products called derivatives. You can for example get a lower cost, or

even zero cost, option by agreeing to forgo some of the ‘upside’ potential or take some of the ‘downside’ risk. There are caps, floors, collars and swap agreements. These derivatives can be extremely important in financial management, but the above is probably sufficient to introduce the products and their underlying logic.

Selection and monitoring in this area has become a very specialized skill, given the need to be aware of developments, not only of new instruments, but also of the security of the company writing them.

Many banks seize the opportunity to diversify into these products as their credit intermediation businesses go through difficult times, and capital adequacy ratios are tightened. They and the regulators are not sure that they fully understand the products and are able to manage this level of complexity. Likewise, there are some concerns regarding the rapid spread of computerized trading systems and the increasing interdependence of market: problems in one area could easily spread to others. It does not help that the legal and taxation framework has not been able to keep pace.

5.7 ACTIVITY

1. What is hedging?
2. Describe the internal hedging techniques for transaction risk.
3. Discuss how currency risk can be hedged using financial markets.
4. Discuss the difference between futures contracts and options contracts.

5.8 SUMMARY

This unit has considered a range of potential techniques for hedging foreign currency risk. Most of these are aimed at hedging transaction risks, but the technique of matching may equally be used to hedge transaction or translation risk. Treasury managers need to be aware of the range of hedging techniques available to them and the relative attractions of each. Important considerations in the selection of an appropriate hedging technique will be cost, flexibility, ease of administration and attitude to risk. The next unit will discuss cash forecasting.

UNIT 6

CASH FORECASTING

6.1 INTRODUCTION

Welcome to Unit 6! This is the unit that will now discuss cash forecasting, which is a build-up of unit 2, working capital management.

6.2 AIM

The aim of this unit is to help you acquire knowledge of cash forecasting, which is a crucial aspect to the operation of every organization.

6.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss the cash forecasting model;
2) Discuss information sources for the cash forecast;
3) Know how to measure cash forecasts accurately;
4) Discuss cash forecasting automation;
5) Examine business cycle forecasting;
6) Discuss cash forecasting controls, policies and procedures.

6.4 TIME REQUIRED

You will be required to spend at least 3 hours grasping the content of this unit and attempting all questions.

6.5 REFLECTION

Before you read any further, attempt to define cash forecasting in your own words.

……
……
……
……..

6.6 INTRODUCTION: CASH FORECASTING

Larger companies with many subsidiaries, especially those with operations in multiple countries, maintain a significant number of bank accounts. This is an inefficient arrangement from the perspective of cash management, since the treasury staff must track all of the individual account balances.

Cash forecasting is an integral part of running an organization. It a company faces a cash shortage, payroll cannot be met, suppliers cannot be paid, scheduled loan repayments will not be met and other creditors will not be paid. In the end, investors will also not be able to receive a return on their investment in the form of dividends. A quality cash forecasting technique therefore will require the treasury manager to determine how much cash is available for short, medium and long-term investments, each having progressively higher returns. We need to remember here that long-term investments have less liquidity than short term investments which bring in cash in the shortest period of time. It is for this reason that the treasurer should be able to invest the correct amounts with confidence.

6.6.1 CASH FORECASTING MODEL

The core of any cash management system is the cash forecast. It is imperative for the management team to be fully appraised of any cash problems with a much lead time as possible. The cash model is basically based on the 'receipts' and 'disbursements' method. This is primarily based on a combination of actual and estimated receivables and payables.

The receipt and disbursements for example may be split into months, and weeks for easier monitoring and forecasting.

The cash forecast mostly will rely on anticipated cash receipts and disbursements for the first month or so and then gradually transition into a greater reliance on sales forecasts and expense trend lines. To extend the trend further into the future, the treasurer must rely to a greater extent

on the annual budget, as modified for the company's actual experience subsequent to finalization of the budget. Thus a long-duration cash forecast draws on input from three sources; receipt and disbursement, short-term projections and finally the budget. Since each of these sources of information is progressively less accurate, the forecasted results will be correspondingly less reliable.

6.6.2 INFORMATION SOURCES FOR THE CASH FORECAST

Good cash forecasts should include **scheduled items.** These are specific cash inflows and outflows that can be predicted with certainty. Examples may include payroll items, rent payable, and the like. If the treasurer wants cash forecasting which is accurate, then as much as possible this should comprise of scheduled items. It should however be pointed out here that it is not possible to include all of the scheduled items in the cash forecast if these items are too many. If this is the case, the best would be to summarize many of the smaller items using the **distribution method.** Under this method, the treasurer examines historical cash flows to create an estimate of the timing and amounts of future cash flows.

6.6.3 MEASURING CASH FORECAST ACCURACY

It is important that cash forecasts are as accurate as possible. There are a number of ways to improve the forecast, all involving the continuing comparison of past forecasts to actual results and correcting the system to ensure that better information is provided for future forecasts.

A key area in which the cash forecast can be wildly incorrect is in receipts from sales forecasts. A detailed review of this area will reveal that some salespeople do not want to forecast any sales because then they will be held accountable for their predictions. This problem requires constant feedback with the sales staff to correct, and may require reinforcement by including the sales forecasting function in the annual review and compensation plan for them.

Another problem is in the accounts payable area where actual cash outflows will typically exceed forecast cash outflows. This imbalance may be caused by a faulty accounts payable data entry process, where invoices are initially mailed by suppliers to people outside of the accounts payable department or because invoices are sent out for approval before they are logged into the accounting system, thereby resulting in their late appearance in the forecast, usually just before they need to be paid. These problems can be solved by asking suppliers to send invoices straight to the accounting department, and by entering all invoices into the accounting system before sending them out for approval. It is also possible to review open

purchase orders to see if there are any missing invoices that are supposed to be currently payable, thereby proactively starting a search for the missing invoices.

A major cash flow variance will arise if a fixed asset is suddenly purchased that was not included in the cash forecast. This problem is best resolved by giving the treasury staff complete access to the capital budgeting process, so that capital requests are in queue for approval, and when they are likely to require cash payments.

In short, the accuracy of the cash forecast requires great attention to processes that provide its source data. The treasury staff should regularly compare forecasted to actual results, and work its way back though the underlying systems to determine what issues caused the error and then correct them.

6.6.4 CASH FORECASTING AUTOMATION

When the cash forecast is generated on a regular basis, the required workload can be extraordinarily high. Automation can be used to avoid some of the most time-consuming steps. Many off-the-shelf accounting software packages contain standard reports that itemize the daily or weekly time buckets in which payments are scheduled to be made, based on each supplier invoice date and the number of days before they are due for payment, including any requirements for early payment in order to take advantage of early payment discounts. The cash flow information provided by such reports is quite reliable, but tends to be less accurate for the time period several weeks into the future, because of delays in the entry of supplier invoice information into the accounting system. This delay is caused by the divergence of incoming invoices to managers for approval. By first entering the invoice information and then sending the invoices out for approval, this time delay can be avoided, thereby improving the accuracy of the automated accounts payable payment timing report. The same report is available in many accounting software packages for accounts receivable itemizing the day or week buckets in which invoice dates and the number of days before customers are required to pay for them. However, this report tends to be much less accurate, for any overdue invoice payments are scheduled for immediate payment in the current period, when in fact there may be collection problems that will delay receipt for quite some time. Also, the report does not

account for the average delay in payments that varies by each customer, in accordance with each one's timeliness in making payments. Consequently, this report should be manually modified especially for the largest outstanding invoices, to reflect the collection staff's best estimates of when payments will actually be received.

In a few cases, software packages will also extend current payroll payments into the future, by assuming that the existing salaries for current employees will continue at the same rates, and that hourly employees will be paid for a regular workweek for all for all future reporting periods. This is not a viable option for those companies that outsource their payroll, since the in-house software will not have any way to predict cash flows if it does not contain any information about payroll.

There are so many variables, such as uncertain receipt dates for accounts receivable, changes in payroll levels, and the sudden purchase of fixed assets, that any automatically generated reports should be adjusted by the accounting staff's knowledge of special situations that will throw off the results of the reports. Also, the basis for automated reports is primarily very short-term accounts receivable and payable information that will rapidly become inaccurate for periods much greater than a month, so manual adjustments to the cash forecast that are derived from the short-term forecasts or the budget will become increasingly necessary for later time periods.

6.6.5 BULLWHIP EFFECT

It is usually possible to create reasonably accurate estimates of the amounts and timing of incoming payments from customers and outgoing ones to suppliers. However, the inventory component of the forecast can be down-right befuddling. Inventory levels may arise or fall so sporadically that it appears impossible to forecast accurate payables.

One element of inventory forecasting that causes so much heartburn is the **bullwhip effect.** This is when a company runs into a materials or capacity shortage and informs its customers that they are being put on an allocation basis. The customers immediately ramp up their order quantities so they can lock in a greater proportion of the company's output over a longer time horizon, which forces the company to increase its capacity to meet the unexpected demand; once the company starts meeting the larger orders and eliminates its shipment allocations, customers promptly shrink their panning horizons, find that they now have plenty of inventory for their immediate needs, and rescind most outstanding orders. The company has

just been the victim of the bullwhip effect. The cash forecast has also been bull-whipped, as the payables line item alternatively skyrockets and dives. So how can such a problem be avoided?

The treasurer should monitor the issuance of any order allocation notices to customers and see if there is a sudden order increase that occurs subsequent to the notice. If so, the treasurer should work with the materials manager to estimate "real" order volumes based on the historical order volumes of each customer, adjusted for any seasonal effects. Inventory planning should incorporate these historical values, rather than the incremental jump in orders, until such time as the allocation notice is rescinded and order volumes return to their normal levels. This kind of proactive planning will result in a much more accurate cash forecast, and will also keep a company from suffering through the inventory and production gyrations of the bullwhip effect.

6.6.6 BUSINESS CYCLE FORECASTING

The cash forecasting period is generally quite short ranging from one to six months. The level of accuracy of this forecast declines markedly if it projects multiple months into the future. However, it is possible to incorporate longer-range business cycle forecasting into the cash forecast in order to introduce a higher degree of accuracy into more distant parts of the cash forecast. Some possible actions to take when conducting business cycle forecasts include:

a) **Report on published forecasts:** In some countries, forecasts are published by major business magazines for the economy at large. The advantages of such forecasts are that they are usually accurate and free as they are conducted by expert forecasters. The problem on the other hand may be that each company operates in a smaller industry within the national economy, and as such is subject to "mini" business cycles that may not move in lockstep with that of the national economy. For this reason, the reported information may be only generally relevant to a company's specific situation.
b) **Subscribe to a forecasting service:** A company can pay a significant fee to a forecasting service for more specific reports that relate to the industry in which it operates. This is a good approach for those organizations that do not have the

resources to gather, summarize, and interpret economic data by themselves. However, the fee charged is too high in comparison to the value of the information received.

c) **Develop an in-house forecasting model:** In house forecasting can be useful if at all there are no forecasting service organizations externally available that can provide the information. This effort can range from a minimalist approach to a comprehensive one, with each level of effort yielding better results. The first step is to find the right kinds of data to accumulate, followed by implementing a data-gathering method that yields reliable data in a timely manner. Then arrive at a methodology for translating the underlying data into a forecast. This forecast should include the underlying assumptions and data used to arrive at the forecast, so that any changes in the assumptions are clearly laid out. Finally, there should be a methodology for comparing results against actual data and adjusting the forecasting methodology based on that information. Though this approach is a time-consuming approach, it can yield the best results if a carefully developed forecasting system is used.

6.6.7 CASH FORECASTING CONTROLS

The daily cash forecast usually is assembled quickly, using the preceding day's forecast as a template, and with only minor updates. A less frequent forecast may be assembled "from scratch", without attempting to roll forward the old forecast; this increases the risk of errors. Also, the person assigned to prepare cash forecasts on an infrequent basis is less likely to be familiar with the process, and so is more likely to make mistakes. These characteristics allow the treasurer to mandate a reduced set of controls for a daily cash forecast and a more comprehensive one for less frequent forecasts.

If a forecast is issued on a daily basis, then the treasurer should focus controls on the incremental daily changes in the forecast. This can be achieved with the following two controls:

a) **Investigate significant variances from the preceding day's forecast-** This is a side by side comparison of the current day's forecast and the immediately preceding day's forecast. If there are significant changes, the preparer should verify that the changes are reasonable.

b) **Obtain the approval of a knowledge person-**Another person should briefly review the forecast, initial it, and retain a copy. The reviewer may not necessarily be the preparer's supervisor; it may make more sense to have someone else with significant and recent cash forecasting expertise review it. Alternatively the reviewer could be someone with considerable knowledge of the information feeding into the forecast, such as the person responsible for collections, payables, payroll, or capital expenditures.

If a forecast is generated less frequently, the controls should include the preceding two controls (with some modifications), as well as additional controls. They include:

- *Match latest forecast against preceding forecasts.* It is useful to compare the new forecast against several preceding forecasts. In particular, compare the predicted ending cash balance for each time period to the same time period in the earlier forecasts. If there are significant differences in the ending balance for a specific time period as it moves closer to the present, it is likely that the forecast model needs to be changed for the periods farther in the future.
- *Match forecast against standard forecast checklist.* When a forecast is being prepared on an infrequent basis, there is an increased risk that some line items in the forecast will be inadvertently left out. It is extremely difficult to spot a missing item, so a reviewer should match the forecast against a standard checklist of forecast contents. This control is more useful when reviewing a forecast prepared by an inexperienced person who is not familiar with the company's business processes.
- *Obtain approvals.* The less frequently a forecast is prepared, the more approvals it needs, since the company is going to rely on it for a longer period of time. This may call for reviews by all of the people having provided input into the forecast, such as the managers of collections, payables, payroll and capital expenditures. At a minimum, at least one person with extensive knowledge of the company's cash flows should review and initial the forecast, and retain a copy.
- *Retain a copy.* The best way to investigate the accuracy of a cash forecast is to compare it to the forecasts from prior periods, so retain a copy of every forecast. If the forecast is compiled on a spreadsheet, the save each forecast on

a separate tab of the spreadsheet, and label each tab with the date of the forecast. It may also be useful to lock the spreadsheet, so that earlier versions are not inadvertently altered.

Finally, if the forecast is created using an electronic spreadsheet, then the departmental year-cash forecast. Any tinkering with the forecast model during the year could have caused a calculation error that no longer properly rolls up the cash forecast. This formula review should *not* be conducted by the person who prepares the cash forecast; instead, use someone who does not use the spreadsheet. This third party will be more likely to work through all of the formulas and how they work, whereas someone who is excessively familiar with the spreadsheet is more likely to assume that it works correctly.

6.6.8 CASH FORECASTING POLICIES

The treasurer should implement policies that will assist in the management of two aspects of cash forecasting, which are the issuance frequency and review frequency of the forecasts. Sample policies include the following:

a) **Cash forecasts shall be issued on a (daily/weekly/monthly) basis.** This policy ensures that cash forecasts are issued with sufficient frequency to match the periodic updating of a company's debt and investments. A larger company with significant transaction volume may need daily cash forecasts, while a smaller one may only require a monthly issuance.
b) **Cash forecasts shall be structurally updated at least once a (month/quarter/year)** A cash forecast may gradually become less accurate over time, due to changes in the businesses that are not reflected in the cash forecast model. This policy requires the treasurer to engage in a periodic updating that "fine-tunes" the forecast to make it more accurately reflect actual cash flows.

These policies should be integrated into the treasury's policies and procedures manual, and can also be integrated into the job descriptions of those responsible for the cash forecast.

6.6.9 CASH FORECASTING PROCEDURE

A formal cash forecasting procedure is least likely to be used if an experienced person prepares a forecast every day, but can be quite useful for more infrequent forecasting intervals or for training purposes, where there is less certainty about the various components

of the forecast. Exhibit 1 below shows a cash forecast procedure. It assumes the use of an electronic spreadsheet for preparation of the forecast.

EXHIBIT 1: CASH FORECASTING PROCEDURE

Procedure Statement Retrieval No: TREASURY-01 Subject:

Cash Forecast

1. PURPOSE AND SCOPE

This procedure is used by the treasury staff to create a periodic cash forecast.

2. PROCEDURES

2.1 Create Cash Forecast (Treasury Staff)

1. Create a new worksheet within the forecast spreadsheet.
2. Copy the forecast model forward from the most recent worksheet into the new worksheet.
3. Label the tab of the new version with the issuance date of the cash forecast.
4. Lock the most recent completed worksheet.
5. In the new worksheet, clear all numbers from the forecast model.
6. Enter the beginning cash balance from the daily bank reconciliation.
7. Obtain from the collections manager the best estimate of receipt dates for receivables exceeding K_____ . For smaller receivables, estimate the distribution of receipts based on the historical days' sales outstanding metric. Enter this information in the forecast.
8. Obtain from the payroll manager the estimated timing and amounts of scheduled wage and commission payments. Enter this information in the forecast.
9. Obtain from the payables system the timing and amounts of scheduled payables. Enter this information in the forecast.
10. Obtain from the financial analyst the timing and amounts of scheduled capital expenditures. Enter this information in the forecast.
11. Obtain from the sales manager the short-term sales forecast, and estimate cash receipts from these sales based on the historical days' sales outstanding.
12. Incorporate expenses into the short-term sales forecast period, based on recurring expenses and historical percentages of the cost of goods sold and administrative expenses.
13. Obtain from the financial analyst the receipts and expenditures based on the modified budget for the period beyond the short-term sales forecast. Enter this information in the forecast.

14. Review the preliminary cash forecast for reasonableness.
15. Forward the forecast to the assistant treasurer for review.

2.2 Review Cash Forecast (Assistant Treasurer)

1. Review and approve the cash forecast. This may include matching the timing and amounts of the projected collection, payables, and capital expenditures listed in the forecast to other reports, or asking for the additional reviews by the managers of those functions.
2. Print two copies of the cash forecast.
3. Initial both copies to indicate approval, return one to the forecast preparer, and retain the other copy in a binder, stored by date.

2.3 Issue Cash Forecast (Treasury Staff)

1. Upon receipt of the approved forecast, issue the forecast to the treasurer, assistant treasurer, controller, and chief financial officer.

6.7 ACTIVITY

1. Discuss how the cash forecasting model works.
2. What information sources are needed to carry out a cash forecast accurately.
3. Discuss what is meant by Cash Forecasting Automation.
4. What is meant by business cycle forecasting?
5. Discuss the cash forecasting controls, policies and procedures that a treasurer could use.
6. What is meant by "Bullwhip" effect?
7. Describe how information can accurately be measured in cash forecasts.

6.8 SUMMARY

We have seen in this unit that cash forecasting is an important task that deserves the utmost attention from the treasurer, since a cash shortfall can bring a company's operations to an abrupt halt in short order, or at least

require the use of expensive short-term debt. The cash forecasting process is based on multiple sources of information that can yield quite inaccurate results just a few months in the future. To improve forecasting accuracy, the treasury staff should regularly compare the forecast to actual results in order to locate and root out problems. A number of controls and a rigidly followed forecasting procedure can also help improve forecasting accuracy. Only by improving forecasting accuracy can a company take proper steps to mitigate its borrowing needs, as well as place excess funds in the proper types of investments.

Well done for reading this far. The next unit will introduce you to cash concentration. It will be a build- up of this unit you have just read.

UNIT 7

CASH CONCENTRATION

7.1 INTRODUCTION

Welcome to Unit 8! This unit is going to discuss cash concentration and its importance in treasury management.

7.2 AIM

The aim of this unit is to discuss cash concentration, the many cash concentration strategies and its importance in treasury management.

7.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Evaluate the various cash concentration strategies such as pooling concepts, notional pooling, physical sweeping and many more;
2) Discuss cash concentration controls, policies and procedures;
3) Compare account sweeping with notional pooling;
4) Discuss non-pooling situations;
5) Describe what is meant by bank overlay structure;
6) Outline the benefits of cash concentration.

7.4 TIME REQUIRED

You are strongly advised to spend at least 3 (three) hours studying this unit.

7.5 REFLECTION

Do you recall that unit 6 discussed cash forecasting? Am sure you have a good idea what it is. Attempt to define cash concentration in your own words before reading the description provided below.

……
……
……
……

7.6 INTRODUCTION: CASH CONCENTRATION

There are so many large organizations that maintain a lot of different bank accounts with different banks as a way of cash management. This method has often not proved to be effective because since the treasury staff must track all individual account balances, it becomes cumbersome. With such highly fragmented cash balances, it is extremely difficult to repurpose the funds for either centralized payments, debt pay down, or investments. An excellent solution to such a problem is **cash concentration.** This is where cash in multiple accounts is **pooled.** Pooling can be achieved either through **physical sweeping** or **notional pooling** (where funds are not actually transferred, but balance information is reported as though physical sweeping had occurred). For this unit will attempt to discuss these strategies.

7.6.1 CASH CONCENTRATION STRATEGIES

A company having multiple locations can pursue a variety of cash concentration strategies, which tend to bring larger benefits with greater centralization. The strategies include:

a) **Complete decentralization**

Every subsidiary or branch office with its own bank account manages its own cash position. So long as balances are small this is a good idea as it can lead to little synergy to be gained by concentrating cash in a single account. However, if large cash balances

are held, then a more centralized approach is needed. This strategy will involve centralizing accounts in the country of currency (Kwacha being kept in Zambia and Rand being kept in South Africa. This alternative is transactionally efficient, but does not sufficiently centralize cash for investment purposes.

b) **Centralized payments, decentralized liquidity management**

A company can implement a centralized payment factory that handles all payables for all company subsidiaries but issues payments from local accounts. This improves the overall planning for cash outflows but does not improve the management of excess cash balances, for which local managers are still responsible. Also, if investments are managed at the local level, there is a greater risk that the corporate investment policy will not be followed.

c) **Centralized liquidity management, decentralized payments**

The treasury staff centralized cash into a concentration account (either through sweeps or notional pooling) and has responsibility for investments. However, local managers are still responsible for disbursements.

d) **All functions centralized**

The treasury staff pools all cash into concentration account, invests it, and manages disbursements. This is an excellent structure for optimizing investment income and gives the treasurer considerable control over the accounts payable portion of the company's working capital. Larger companies usually follow this strategy, but may not carry it through to cross-border centralized. Instead, they may centrally manage key currencies, such as dollars and euros, while allowing regional control over other currencies.

In the cases where cash is invested from a central location and multiple currencies are involved, it may be necessary for the treasury staff to invest funds locally (i.e. in the home country of the currency), because of legal or foreign exchange restrictions. The final (and most centralized) of these strategies is the most efficient approach for cash concentration, but it may not be the most cost-effective. If a company has relatively few subsidiaries with low account balances, overhead than will be offset by increased interest income or reduced interest expense.

7.6.2 POOLING CONCEPTS

Cash concentration requires that a company create a **cash pool.** This comprises a cluster of subsidiary bank accounts and a concentration account. Funds physically flow from the subsidiary accounts into the concentration account under a **physical sweeping** method. Alternatively, cash balances in the subsidiary accounts can be concentrated in the master

account only within the bank's records, with the cash remaining in the subsidiary accounts. This later method is called **notional pooling.**

If a pooling arrangement includes accounts located in more than one country, this is known as a **cross-border cash pool.** A company may elect to pool cash within the country of each currency (e.g. Kwacha being pooled in Zambia), which is known as a **single currency centre** model. If a company pools all of its foreign currency accounts in a single location, this is a **multicurrency centre** arrangement. Multicurrency centres are generally easier to manage, but transactions are more expensive than under the single-currency-centre model.

7.6.3 PHYSICAL SWEEPING

When a company sets up a **zero-balance account,** its bank automatically moves cash from that account into a concentration account, usually within the same bank. The cash balance in the zero-balance account (as the name implies) is reduced to zero whenever a sweep occurs. If the account has a debit balance at the time of the sweep, then the money is shifted from the concentration account back into the account having the debit balance.

It is also possible to use **constant balancing** to maintain a predetermined minimum balance in a subsidiary account, which involves sweeping only those cash levels above the minimum balance, the reverse sweeping cash into the subsidiary account if the balance drops below the minimum balance.

Daily sweeping may not be necessary outside of a company's designated core currencies. This is especially likely when noncore currency account balances are relatively low. If so, it may be more cost-effective to sweep them less frequently, or to implement **trigger balances.** A trigger balance is an account balance level above which excess funds are swept out of the account.

Some concentration banks can also monitor a company's account balances at third- party banks using SWIFT (Society for Worldwide Inter-bank Financial Telecommunication) messages, and create transfer requests to move excess cash to the concentration bank. The key point with account sweeping is to fully automate it-the effort involved in manually tracking account balances and shifting funds on a daily basis is not only expensive but also likely to cause errors.

In most sweeping transactions, the sweeps occur on an **intraday** basis which means that balances are transferred to the concentration account before the end of the day.

Consequently, some cash may be left behind in subsidiary accounts, rather than being centralized. This occurs when cash arrives in an account after execution of the daily sweep. The cash will remain in the subsidiary account overnight and be included in the following day's sweep. If a company is not dealing with such a bank, then a proactive approach to depositing checks before cutoff times is the best way to avoid unused cash.

There may be a need to track the amounts of cash swept from each zero-balance account into the concentration account; if so, the company records an intercompany loan from the subsidiary to the corporate parent in the amount of cash transferred through the cash concentration process. There are several reasons for doing so:

a) **Subsidiary-level financial reporting requirements-**A subsidiary may have an outstanding loan, for which a bank requires the periodic production of a balance sheet. Since account sweeping shifts cash away from a subsidiary's balance sheet, detailed sweep tracking is needed to put the cash back on the subsidiary's balance sheet for reporting purposes. This can be done by recording an intercompany loan from the subsidiary to the corporate parent in exchange for any swept cash, which can then be reversed to place the cash back on the subsidiary's balance sheet.
b) **Interest income allocation-**A company may elect to allocate the interest earned at the concentration account level back to the subsidiaries whose accounts contributed cash to the concentration account in a low-tax jurisdiction, where the tax on interest income is minimized. Thus, the amounts of cash swept into and out of a subsidiary account must be tracked in order to properly allocate the correct proportion of interest income to that account.
c) **Interest expense allocation-**Some tax jurisdictions may require the parent company to record interest expense on intercompany loans associated with the transfer of cash in a physical sweeping arrangement. If so, the company must track the intercompany loan balances outstanding per day, which is then used as the principal for the calculation of interest expense. The interest rate used for these calculations should be the market rate; any other rate can be construed by local tax authorities to be transfer pricing designed to shift income into low-tax regions.
d) **Central bank reporting-**Some central banks require that they be sent reports on transfers between resident and non-resident accounts. This may be handled by the company's bank but still increase the administrative burden associated with the seep.

Some banks have the capability to track the amount of balance sweeps from each subsidiary account on an ongoing basis, which a company can use as its record of intercompany loans.

7.6.4 NOTIONAL POOLING

Notional pooling is a mechanism for calculating interest on the combined credit and debit balances of accounts that a corporate parent chooses to cluster together without actually transferring any funds. This approach allows each subsidiary company to take advantage of a single, centralized liquidity position, while still retaining daily cash management privileges. Also, since it avoids the use of cash transfers to a central pooling account, there is no need to create or monitor intercompany loans, nor are there any bank fees related to cash transfers (since there are no transfers). In addition, it largely eliminates the need to arrange overdraft lines with local banks. Further, interest earnings tend to be higher than if investments were made separately for the smaller individual accounts. It also offers a solution for partially owned subsidiaries whose other owners may balk at the prospect of physically transferring funds to an account controlled by another entity. And finally, the use of notional pooling is not a long-term commitment; on the contrary, it is relatively easy to back out of the arrangement.

Where global notional pooling is offered (usually where all participating accounts are held within a single bank), the pool offsets credit and debit balances on a multicurrency basis without the need to engage in any foreign exchange transactions. An additional benefit of global notional pooling lies in the area of intercompany cash flows; for example, if there are charges for administrative services, the transaction can be accomplished with no net movement of cash.

Once a company earns interest on the funds in a notional account, interest income is usually allocated back to each of the accounts making up the pool. For tax management reasons, it may be useful for the corporate parent to charge the subsidiaries participating in the pool for some cash concentration administration expenses related to management of the pool. This scenario works best if the corporate subsidiaries are located in the high-tax regions, where reduced reportable income will result in reduced taxes.

The main downside of notional pooling is that it is not allowed in some countries, especially in portions of Africa, Asia and Latin America (though it is very common in Europe). In these excluded areas, physical cash sweeping is the most common alternative. Also, the precise form

of the notional pooling arrangement will vary according to local laws, so that some countries allow cross-border pooling, while others do not.

In addition to the prohibition against notional pooling in some countries, it is difficult to find anything but a large multinational bank that offers cross-currency notional pooling. Instead, it is most common to have a separate notional cash pool for every currency area.

7.6.5 COMPARISON OF ACCOUNT SWEEPING AND NOTIONAL POOLING

When there is a choice between account sweeping and notional pooling notional pooling is usually the better alternative. Under notional pooling cash does not physically leave the bank accounts of each subsidiary, which greatly reduces the amount of intercompany loans that would otherwise have to be recorded. This eliminates the treasury overhead cost that would otherwise be associated with tracking and recording the intercompany loans.

Similarly, if the parent company allocates interest earned to its various subsidiaries based on their cash balances, this requires significant accounting resources to calculate the interest income allocation based on intercompany loans and deposits. In multicurrency situations, this also involves revaluing the interest income into different currencies. However, in a notional pooling environment, the bank may be able to automate the calculation of interest, with a physical transfer of funds at month-end to pay out interest income to each subsidiary.

Also, if the intent is to pool funds denominated in multiple currencies, account sweep require the use of foreign exchange transactions and possibly hedging. Notional pooling eliminates these activities, since funds are not shifted from one currency to another.

When account sweeping moves cash from one entity to another, this can have adverse tax consequences in some countries. Conversely, because notional pooling requires no physical transfer of funds between legal entities, no tax problems are triggered.

Notional pooling is more likely to be supported by the managers of local subsidiaries, since it means that they retain control over their cash balances. Indeed, they may not even realize that a notional pooling arrangement has been created.

However, a major disadvantage of notional pooling is that it is illegal in some countries. Thus, notional pooling should be taken advantage of where it is offered. When not available, physical sweeping is a very acceptable alternative to conducting no cash concentration activities at all.

7.6.6 NON POOLING SITUATIONS

A company may need to use a local bank that cannot be linked into its corporate account sweeping or notional pooling structures. If so, the company's account balances within the bank must be managed manually. A key focus of this activity is to prevent overdrafts, since their cost usually greatly exceeds the interest income to be earned on the same account; thuss, it is cost-effective to maintain excess balances in such accounts, rather than incurring overdraft fees.

If there are to be cash transfers out of such accounts, they tend to be infrequent, since the parent company will prefer to maintain excess cash balances in order to avoid overdrafts, and so will only siphon off cash when there is a clear excess balance. Also, because account management is manual, it is more cost-effective from a labor perspective to limit the number of transfers into or out of these accounts.

7.6.7 BANK OVERLAY STRUCTURE

Companies operating on an international scale frequently have trouble reconciling the need for efficient cash concentration operations with the use of local banking partners with whom they may have long-standing relationships and valuable business contracts. The solution is the **bank overlay structure.**

A bank overlay structure consists of two layers. The lower layer is comprised of all incountry banks that are used for local cash transaction requirements. The higher layer is a group of networked regional banks, or even a single global bank, that maintains a separate bank account for each country or legal entity of the corporate structure. Cash balances in the lower layer of banks are zero-balanced into the corresponding accounts in the higher layer of banks on a daily basis (where possible, subject to cash flow restrictions). These sweeps are accomplished either with manual transfers, SWIFT messages from the networked banks to the local banks, or with standing authorizations to the local banks. The approach allows funds to be consolidated on either a regional or global basis for centralized cash management.

7.6.8 CASH CONCENTRATION CONTROLS

There is a strong argument that having a physical sweep cash concentration system in place is a control, since it pushes all cash balances into a single location for easier monitoring.

Otherwise, the treasury staff would be faced with a large jumble of accounts, over which it might exercise little control.

The following controls can improve on the inherent control of a physical sweep system:

a) **Review target balances.** If a subsidiary account is allowed a target cash balance, then periodically review account usage to see if the target balance is appropriate.
b) **Review excluded accounts.** Some subsidiary accounts may have been excluded from the sweep on the grounds that there are local currency restrictions, or that minimal transaction activity does not justify a periodic sweep. These factors may change over time, so schedule an annual analysis of excluded accounts.
c) **Compare intercompany loan rates to market rates**
   If the company uses nonmarket interest rates when calculating intercompany charges, it may run afoul of transfer pricing rules in some tax jurisdictions. Thus, create a tracking procedure to periodically compare market rates to the intercompany loan rates.
d) **Verify the allocation of interest income to subsidiaries**
   Depending on the circumstances, it may be necessary to allocate interest earned back to the subsidiaries. If so, verify that the calculation is consistently applied across all subsidiaries.
e) **Verify the calculation of intercompany loan balances**
   A company using physical sweeping may need to record intercompany loan balances to reflect the shifts in cash to the corporate consolidation account. If so, have the internal audit staff periodically verify the calculation of these balances. This review can impact the amount of interest expense or income allocated to subsidiaries.

A notional pooling system does not create an inherent control over the physical cash balances, as was the case with the physical sweep, since cash is still sitting in what may be a large number of accounts in multiple locations. Thus, all controls normally exercised over a bank account would be in effect in this situation. The primary change with notional pooling is the presence of interest income distributions back to the originating bank accounts. This calls for a repetition of a preceding control, which is found in (d) above.

The receipt and disbursement of cash involve intricate controls.

7.6.9 CASH CONCENTRATION POLICIES

There is one key policy required for cash concentration, which is one that places responsibility for cash concentration on a specific group within the company. Otherwise, it can be extremely difficult to exert central control over far-flung accounts that are normally under the control of the managers of local subsidiaries. The policy follows:

- **All cash concentration activities shall be managed by the corporate treasury department.** This can be a surprisingly important policy in a larger firm where divisional managers may attempt to retain control over their bank accounts. The policy clearly places responsibility for cash concentration in the hands of the **corporate** treasury department.

A deliberate decentralization policy could revise the preceding policy to state that cash concentration activities are limited to a lower level within the company, such as at the division level.

It is also possible, though not necessary, to state the company's cash concentration strategy with a policy statement. Doing so clarifies the company's direction in this area, but also requires a subsequent policy change if the treasurer wants to turn in a different direction. Strategies for both small and large companies are noted in the following two policies:

1. **Small-company policy-**The company shall engage in cash concentration with a "physical sweeping" strategy. All subsidiary accounts shall be designated as zerobalance, and shall sweep into a concentration account, which in turn shall also be used as a central disbursement account. The treasurer may exclude accounts from this strategy if sweeping is not cost-effective. This policy works well for smaller entities using a single currency, in locations where notional pooling is not available.
2. **Large-company policy-**The company shall engage in cash concentration with a "cross-currency notional pooling" strategy. All subsidiaries are restricted to accounts with a bank designated by the treasurer. In areas where notional pooling is not available, the treasurer may opt for physical sweeping is not cost-effective. This policy assumes that notional pooling will be organized under a single global bank, and provides for the inevitable exceptions that will arise in some jurisdictions.

It is not necessary at the policy level to define any greater level of detail regarding cash concentration, such as the allowable methods of cash transfer for physical sweeping-these are tactical issues over which the treasury staff should have discretion.

7.6.10 CASH CONCENTRATION PROCEDURES

Once a cash concentration system is set up, the system should be fully automated, with cash balances either shifting to a concentration account with physical sweeping or appearing in a notional pooling arrangement. Thus, there is no need for a procedure to conduct the cash concentration. However, procedures are needed to track intercompany loans and related interest expense allocations (if physical sweeping is used), as well as to allocate interest income back to the subsidiaries.

Exhibit 1 Intercompany Loan Tracking Procedure

Procedure Statement Retrieval No: TREASURY-02

Subject: Tracking of Intercompany Loans and Allocation of Related Interest Expense Caused by Physical Sweeping

1. **PURPOSE AND SCOPE**
   This procedure is used by treasury staff to determine the amount of intercompany loans caused by the sweeping of cash from subsidiary accounts into the corporate cash concentration account as well as the amount of internal interest charges related to these loans.
2. **PROCEDURES**
   2.1 Record Intercompany Loans (Treasury Staff)
   1. At the end of each month, access the company's bank statements and calculate the amount of intercompany loans outstanding from each subsidiary to the corporate entity that was caused by physical sweeps into the corporate cash concentration account. Add to this amount the balance of any intercompany loans that were outstanding at the beginning of the month.
   2. Subtract from the revised intercompany loan balances the amount of any expenditures related to each subsidiary for disbursements from the corporate shared services centres for payroll and disbursements.
   3. Subtract from the remaining intercompany loan balances the amount of any cash flows from corporate to the subsidiaries for other purposes, such as debit balance replenishment.
   4. Calculate the incremental change in intercompany loan balances, and create a journal entry to reflect the change. If there is an increase in a subsidiary's loan to corporate, then this is a debit to their notes receivable account and a credit to corporate loans payable. If there is a decrease in a subsidiary's loan to corporate, then this entry is reversed.
   5. Also calculate the average loan balance outstanding during the month, which is calculated [*on a daily basis/by adding the beginning and ending balances*

and dividing by two], which shall be used for subsequent allocations of interest income or expense.

6. Retain a copy of the calculations for use during the next month's calculations.
7. Forward the intercompany loan calculations to the treasurer for approval.
8. Forward the approval calculations and journal entry to the general ledger accountant for entry into the accounting system.

2.2 Allocate Interest Expense (Treasury Staff)

1. Determine the market interest rate on debt, which is the monthly interest rate charged to the company on its primary line of credit.
2. Calculate the intercompany interest expense to be charged to corporate for use of the subsidiary's funds, and credited to the subsidiaries, based on the market interest rate multiplied by their average intercompany loans outstanding during the month.
3. Create a journal entry to reflect the interest expense, which is a debit to the corporate interest expense account, and a credit to the subsidiary's interest income accounts. If there is a net flow of cash to a subsidiary, then this entry is reversed.
4. Retain a copy of the calculations.
5. Forward the interest expense calculations and journal entry to the treasurer for approval.
6. Forward the approval calculation and journal entry to the general ledger accountant for entry into the accounting system.

Exhibit 2 Interest Income Allocation from Cash Concentration Account

Procedure Statement Retrieval No: TREASURY-03
Subject: Calculation and Recording of Interest Income from Cash Concentration Account

1. **PURPOSE AND SCOPE**
   This procedure is used by the treasury staff to calculate the interest income earned from the investment of cash concentration activities, less administrative charges.
2. **PROCEDURES**
   2.1 Calculate Allocable Interest Income (Treasury Staff)
   1. Summarize interest income earned for the month, as reported on investment statements.
   2. Record the interest income as corporate income, with a debit to cash and credit to the interest income account. Have the entry approved by the treasurer, and forward to the general ledger accountant for entry into the accounting system.
   3. Calculate the amount of administrative expense that is associated with the administration of cash concentration and subsequent investment activities. This should include bank pooling fees and loaded labour expenses. Compare this expense to the amount for previous months to establish the reasonableness of the amount.
   4. Subtract the administrative expense from the total interest earned.
   5. Allocate the remaining interest earned to the subsidiaries based on the proportional average amount of intercompany loans they had outstanding to the parent company during the month. This is a debit to the corporate interest income account, and a credit to the interest income accounts of the subsidiaries.
   6. Combine the interest allocation, administrative fee calculation, and copies of investment statements into a package, and retain a copy for backup purposes.
   7. Forward the originals of the package to the treasurer for approval.
   8. Forward the approved package to the general ledger accountant for entry into the accounting system.

7.7 ACTIVITY

1. Discuss the cash concentration strategies; pooling concepts, notional pooling and physical sweeping.
2. Distinguish between complete decentralization and centralized payments.
3. Compare account sweeping to notional pooling.
4. Discuss 5 (five) cash concentration control methods.
5. How are cash concentration policies and procedures determined?

7.8 SUMMARY

A company must determine if it makes sense to engage in cash concentration activities. This unit has been able to highlight the importance of cash concentration. We have also learnt that for cash concentration to be cost effective, the company should have large, on-going balances in multiple accounts that are not being efficiently invested. Further, the incremental reduction in overdraft fees and increase in interest income should outweigh the incremental increase in pooling fees and administrative costs. There is also the intangible factor of whether the managers of local subsidiaries will accept corporate-level interference in their bank accounts. If this analysis is favourable, then a company should engage in cash concentration activities. While notional pooling is theoretically to be preferred over physical sweeping, either method is significantly better than having no pooling at all. Also, manual sweeping is not recommended; involves a large volume of manually initiated transactions to transfer funds between accounts. The level of automation associated with these services mandates the use of a large bank with a broad range of automated services; a small local bank simply cannot assist a multi-location company with a cash concentration strategy.

The next unit will discuss debt management.

UNIT 8

DEBT MANAGEMENT

8.1 INTRODUCTION

Welcome to unit 8 of this module! This unit will discuss debt and its management.

8.2 AIM

The aim of this unit is to discuss the various debt options that could be available for a treasurer and how best he/she can manage such debt.

8.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss the types of debt available for a company; 2) Discuss the many debt-related controls; 3) Discuss debt-related policies.

8.4 TIME REQUIRED

You are advised to spend at least 3 (three) hours studying this unit.

8.5 REFLECTION

In your own words describe debt.

If you were a treasury manager of a large firm, where would you go if you needed to borrow for your company?

……
……
……
……
……

A treasurer is usually called upon to either manage a company's existing debt or procure new debt. In either case, this calls for knowledge of the broad variety of debt instruments available, as well as dealing with credit rating agencies. It may also be necessary to have a working knowledge of the accounting, controls, policies and procedures used to manage debt.

8.6 TYPES OF DEBT

8.6.1 Commercial Paper

Commercial paper is unsecured debt that is issued by a company and has a fixed maturity ranging from 1 to 270 days. A company uses commercial paper to meet its short-term working capital obligations. It is commonly sold at a discount from face value with the discount (and therefore the interest rate) being higher if the term is longer. A company can sell its commercial paper directly to investors, such as money market funds, or through a dealer in exchange for a small commission.

Because there is no collateral on the debt, commercial paper is an option only for large companies having high-level credit ratings from a recognized credit rating agency. For those companies capable of issuing it, the interest rate on commercial paper is extremely low. Commercial paper is not popular in Zambia, but in most developed countries, this is a big business.

8.6.2 Factoring

Under a factoring arrangement, a finance company agrees to take over a company's accounts receivable collections and keep the money from those collections in exchange for an immediate cash payment to the company. This process typically involves having customers mail their payments to a lock box that appears to be

operated by the company but is actually controlled by the finance company. Under a true factoring arrangement, the finance company takes over the risk of loss on any bad debts, though it will have the right to pick which types of receivables it will accept in order to reduce its risk of loss. A finance company is more interested in this type of deal when the size of each receivable is fairly large, since this reduces its pre-transaction cost of collection. If each receivable is quite small, the finance company may still be interested in a factoring arrangement, but it will charge the company extra for its increased processing work. The lender will charge an interest rate (at least 2% higher that the prime rate), as well as a transaction fee for processing each invoice as it is received. There may also be a minimum total fee charged, in order to cover the origination fee for the factoring arrangement in the event that few receivables are actually paid by the factor at once or can wait until the invoice due date before payment is sent. The latter arrangement reduces the interest expense that a company would have to pay the factor, but tends to go against the reason why the factoring arrangement was established, which is to get money back to the company as rapidly as possible. An added advantage is that no collections staff is required, since the lender handles this chore.

A similar arrangement is accounts receivable is accounts receivable financing under which a lender uses the accounts receivable as collateral for a loan and takes direct payments from the company. A lender will typically loan a maximum of only those accounts that are less than 90 days old. Also, if an invoice against which a loan has been made is not paid within the required 90-day time period, then the lender will require the company to pay back the loan associated with that invoice.

Though both variations on the factoring concept will accelerate a company's cash dramatically, it is an expensive financing option, and so is considered a viable long-term approach to funding a company's operations. It is better for short-term growth situations where money is in short supply to fund a sudden need for working capital.

8.6.2 Field Warehouse Financing

Here a finance company (usually one that specializes in this type of arrangement) will segregate a portion of a company's warehouse area with a fence. All inventory within it is collateral for a loan from the financing company to the company. The finance company will pay for more raw materials as they are needed, and is paid

back directly from accounts receivable as soon as customer payments are received. If a strict inventory control system is in place, the finance company will employ someone who will record all additions to and withdrawals from the secured warehouse. If not, then the company will be required to frequently count all items within the secure area and report this information back to the finance company. If the level of inventory drops below the amount of the loan, then the company must pay back the finance company the difference between the outstanding loan amount and the total inventory valuation. The company is also required under state lien laws to post signs around the secured area, stating that a lien is in place on its contents.

Field warehousing is highly transaction intensive, especially when the finance company employs an on-site warehouse clerk, and therefore is a very expensive way to obtain funds. This approach is recommended only for those companies that have exhausted all other less expensive forms of financing. However, lenders typically do not require any covenants in association with these loans, giving corporate management more control over company operations.

8.6.3 Floor Planning

Some lenders will directly pay for large assets that are being procured by a distributor or retailer (such as kitchen appliances or automobiles) and be paid back when the assets are sold to a customer. In order to protect itself, the lender may require that the price of all assets sold be no lower than the price the lender originally paid for it on behalf of the distributor or retailer. Since the lender's basis for lending is strictly on the underlying collateral (as opposed to its faith in a business plan or general corporate cash flows), it will undertake frequent recounts of the assets, and compare them to its list of assets originally purchased for the distributor or retailer. If there is a shortfall in the expected number of assets, the lender will require payment for the missing items. The lender may also require liquidation of the loan after a specific time period, especially if the underlying assets run the risk of becoming outdated in the near term.

This financing option is a good one for smaller or underfunded distributors or retailers, since the interest rate is not excessive (due to the presence of collateral).

8.6.4 Lease

A lease covers the purchase of a specific asset, which is paid for by the lease provider on the company's behalf. In exchange, the company pays a fixed rate, which includes interest and principal, to the leasing company. It may also be charged for personal property taxes on the asset purchased. The lease may be defined as an **operating lease**, under the terms of which the lessor carries the assets on its books and records a depreciation expense, while the lessee records the lease payments as an expense on its books. This type of lease typically does not cover the full life of the asset, nor does the buyer have a small-dollar buyout option at the end of the lease. The reverse of the situation arises for a **capital lease**, where the lessee records it as an asset and is entitled to record all related depreciation as an expense. In the latter case, the lessee payments are split into their interest and principal portions and recorded on the lessee's books as such.

The cost of e lease can be reduced by clumping together the purchases of multiple items under one lease, which greatly reduces the paperwork cost of the lender. If there are multiple leases currently in existence, they can be paid off and re-leased through a larger single lease, thereby obtaining a slightly lower-financing cost.
The leasing option is most useful for those companies that want to establish collateral agreements only for specific assets, thereby leaving their remaining assets available as a borrowing base for other loans. Leases can be arranged for all but the most financially shaky companies, since lenders can always use the underlying assets as collateral and rarely impose any other financing restrictions. Furthermore, future operating lease payments are not listed on the balance sheet as a liability instead future lease obligations are listed in a footnote.

However, unscrupulous lenders can hide or obscure the interest rate charged on leases, so that less financially knowledgeable companies will pay exorbitant rates. A company is obligated to make all payments through the end of a lease term, even if it no longer needs the equipment being leased.

8.6.5 Line of Credit

A line of credit is a commitment from a lender to pay a company whenever it needs cash, up to a preset maximum level. It is generally secured by company assets, and for that reason bears an interest rate not far above the prime rate. The bank will typically charge an annual maintenance fee, irrespective of the amount of funds

drawn down on the loan, on the grounds that it has invested in the completion of paperwork for the loan. The bank will also likely require an annual audit of key accounts and assets balances to verify that the company's financial situation is line with the bank's assumptions. One problem with a line of credit is that the bank can cancel the line or refuse to allow extra funds to be drawn down from it if the bank feels that the company is no longer a good credit risk. Another issue is that the bank may require a company to maintain a compensating balance in an account at the bank; this increases the effective interest rate on the line of credit, since the company earns little or no interest on the funds stored at the bank.

The line of credit is most useful for situations where there may be only short-term cash shortfalls or seasonal needs that result in the line's being drawn down to zero at some point during the year. If one's cash requirements are expected to be longer term, then a term note or bond is a more appropriate form of financing.

8.6.6 Loans

Asset-based loans: A loan that uses fixed assets or inventory as its collateral is a common form of financing by banks. Loans may also be issued that are based on other forms of collateral, such as the cash surrender value of life insurance, securities, or real estate. The bank will use the resale value of fixed assets (as determined through an annual appraisal) and/or inventory to determine the maximum amount of available funds for a loan. If inventory is used as the basis for the loan, a prudent lender typically will not lend more than 50% of the value of the raw materials and 80% of the value of the finished goods, on the grounds that it may not obtain full prices at the time of sale. Lenders will be much less likely to accept inventory as collateral if it has a short shelf life, is customized, is so seasonal that its value drops significantly at certain times of the year, or if it is subject to rapid obsolescence.

Given the presence of collateral, this type of loan tends to involve a lower interest rate. Lenders typically require minimal covenants in association with these loans, giving corporate management more control over company operations. However, the cost of an annual appraisal of fixed assets or annual audit by the bank (which

will be charged to the company) should be factored into the total cost of this form of financing. Lenders require frequent reports on the status of underlying assets.

8.6.7 Bonds

A bond is a long term debt instrument issued with a fixed interest rate. If the rate is excessively low in the current market, then investors will pay less for the face value of the bond, thereby driving up the net interest rate paid by the company. Similarly, if the rate is too high, then investors will pay extra for the bond, thereby driving down the net interest paid.

A bond at times maybe a **registered bond,** under which a company maintains a list of owners of each bond. The company then periodically sends interest payments, as well as the final principal payments, to the investor of record. It may also be a **coupon bond,** for which the company does not maintain a standard list of bondholders. Instead, each bond contains interest coupons that the bondholders send to the company on the dates when interest payments are due. The coupon bond is more easily transferable between investors, but the ease of transferability makes them more susceptible to loss. The most common types of bonds include:

a) **Collateral trust bond:** This is a bond that uses as collateral a company's security investments.
b) **Convertible bond:** This is a bond that can be converted into stock using a pre-determined conversion ratio. The presence of conversion rights typically reduces the interest cost of these bonds, since investors assign some value to the conversion privilege.
c) **Debenture:** This is a bond issued with no collateral. A subordinated debenture is one that specifies debt that is senior to it.
d) **Deferred interest bond:** This is a bond that provides for either reduced or no interest in the beginning years of the bond term, and compensates for it with increased interest later in the bond term. Since this type of bond is associated with firms having short-term cash flow problems, the full-term interest rate can be high.
e) **Floorless bond:** This is a bond whose terms allow purchasers to convert them to common stock, as well any accrued interest. Bond holders can convert some shares and sell them on the open market, thereby supposedly driving down the price and allowing them to buy more shares.

f) **Guarantee bond:** This is a bond whose payments are guaranteed by another party. Corporate parents will sometimes issue this guarantee for bonds issued by subsidiaries in order to obtain a lower effective interest rate.

g) **Income bond:** This is a bond that pays interest only if income has been earned. The income can be tied to total corporate earnings or to specific projects. If the bond term indicate that interest is cumulative, then interest will accumulate during non- payment periods and be paid at a later date when income is available for doing so.

h) **Mortgage bond:** This is a bond offering backed by any real estate owned by the company (called a real property mortgage bond), or by company-owned equipment (called an equipment bond), or by all assets (called a general mortgage bond).

i) **Serial bond:** It is a bond issuance where a portion of the total number of bonds is paid off each year, resulting in a gradual decline in the total amount of debt outstanding.

j) **Variable rate bond:** This is a bond whose stated interest rate varies as a percentage of a baseline indicator, such as the prime rate. Treasurers should be wary of this bond type because jumps in the baseline indicator can lead to substantial increases in interest costs.

k) **Zero coupon bond:** This is a bond with no stated interest rate. Investors purchase these bonds at a considerable discount to face value.

l) **Zero coupon convertible bond:** This is a bond that offers no interest on its face but allows investors to convert to stock if the stock price reaches a higher level of volatility in the stock price gives investors some hope of a profitable conversion to equity.

We need to remember that a bond is usually issued with a fixed interest rate. However, if the rate is excessively low in the current market, then investors will pay less for the face value of the bond, thereby driving up the net interest rate paid by the company.

There may be a bond indenture document that itemizes all features of the bond issue. It contains restrictions that the company is imposing on itself, such as limitations on capital expenditure or dividends, in order to make the bond issuance

as palatable as possible to investors. If the company does not follow these restrictions, the bonds will be in default.

Bonds are highly recommended for those organizations large enough to attract a group of investors willing to purchase them, since the bonds can be structured to precisely fit a company's financing needs. Bonds are also issued directly to investors, so there are no financial intermediaries to whom transactional fees must be paid. A company can issue long-maturity bonds at times of low interest rates, thereby locking in modest financing costs for a longer period than would normally be possible with other forms of financing. Consequently, bonds can be one of the lowest-cost forms of financing.

8.6.8 Bridge loans

A bridge loan is a form of short-term loan that is granted by a lending institution on the condition that the company will obtain longer-term financing shortly that will pay off the bridge loan. This option is commonly used when a company is seeking to replace a construction loan with a long-term note that it expects to gradually pay over the many years. This type of loan is usually secured by facilities or fixtures in order to obtain a lower interest rate.

8.6.9 Economic Development Authority Loans

Various agencies of state governments are empowered to guarantee bank loans to organizations that need funds in geographic areas where it is perceived that social improvement goals can be attained. In Zambia for example, Development Bank of Zambia (DBZ) and National Savings and Credit Bank (Natsave) are such institutions.

8.6.10 Long-Term Loans

There are several forms of long-term debt. One is a long-term loan issued by a lending institution. These loans tend to be made to smaller companies that do not have the means to issue bonds. To reduce the risk to the lender, these loans typically require the company to grant the lender senior status over all other creditors in the event of liquidation.

A long-term loan usually involves the use of fixed payments on a fixed repayment schedule, which will involve either the gradual repayment of principal or the regularly scheduled payment of interest, with the bulk of the principal being due at the end of the loan as a balloon payment.

Long-term debt is a highly desirable form of financing, since a company can lock in a favourable interest rate for a long time, and keeps it from having to repeatedly apply for shorter-term loans during the intervening years, when the business conditions may result in less favourable debt terms.

8.6.11 Receivables Securitization

A large company can consider securitizing its accounts receivable, thereby achieving one of the lowest interest rates available for debt. To do so, it creates a special purpose entity (SPE) and transfers a selection of its receivables into the SPE. The SPE then sells the receivables to a bank conduit, which in turn pools the receivables that it has bought from multiple companies, and uses the cash flows from the receivables to back the issuance of commercial paper to investors, who in turn are repaid with the cash flows from the receivables.

Receivables securitization is clearly a complex process to initially create; the primary benefit of doing so is that a company's receivables are isolated from its other risks, so that the SPE has a higher credit rating than the company, with an attendant decline in borrowing costs. These transactions are not yet common in Zambia but in developed countries in Europe as well as the United States. Actually, SPEs did contribute to the global financial crisis of 2007-2009.

The debt created through an SPE is not required to be recorded as debt on the original company's balance sheet, as this would have been transferred to the SPE.

Receivables securitization is available only to large companies having a broad customer base whose receivables experience minimal defaults. Furthermore, there must be adequate tracking systems in place to monitor the creditworthiness of those debtors whose receivables are included in the SPE, delinquency statistics, and customer concentrations as well as frequent reporting on receivable collections.

8.6.12 Sale and Leaseback

Under this arrangement, a company sells one of its assets to a lender and then immediately leases it back for a guaranteed minimum time period. By doing so, the company obtains cash from the sale of the asset that it may be able to use more profitably elsewhere, while the leasing company handling the deal obtains a guaranteed lessee for a time period that will allow it to turn a profit on the financing arrangement. A sale and leaseback is most commonly used for the sale of a corporate building, but can also be arranged for other large assets, such as production machinery.

A sale and leaseback is useful for companies in any type of financial condition, for a financially healthy organization can use the resulting cash to buy back shares and prop up its stock price, while a faltering organization can use cash to fund operations. It has the added advantage of not burdening a company's balance sheet with debt; furthermore, it puts cash back into the balance sheet, allowing a company to obtain additional debt. It is especially useful when market conditions make other forms of financing too expensive. Obviously, it is an option only for those organizations that have substantial assets available for sale.

8.7 CREDIT-RATING AGENCIES

These are agencies that giving ratings to companies. Good examples include Moody's, Standard & Poors and Fitch. These are the three top-tier international credit-rating agencies that the Securities and Exchange Commission (SEC) (global organization) allows to issue debt ratings. In Zambia, our stock exchange Lusaka Stock Exchange (LUSE), does give ratings to companies.

A debt rating results in a credit score that indicates the perceived risk of default on the underlying debt, which in turn impacts the price of the debt on the open market. Having a credit score is essentially mandatory in most developed countries like the U.S and the U.K. but not really so in Zambia. This is because in those countries, most funds are prohibited by their internal investment rules from buying debt that does not have a specific level of credit rating assigned to it.

It should also be pointed out here that if a company is wrongly rated it can be catastrophic, like what happened during the global financial crisis of 2007-2009, where some companies

received very high credit ratings, which counterparties relied on and only to discover that such companies ended up collapsing, e.g. Lehman Brothers in the U.S.

The rating scores used by the three credit rating agencies mentioned above (Moody's, Standard & Poor's, and Fitch are as follows:

Definition	**Fitch Rating**	**Moody's Rating**	**S&P Rating**
Prime	AAA	Aaa	AAA
High grade	AA+	Aa1	AA+
	AA	Aa2	AA
	AA-	Aa3	AA-
Upper medium grade	A+	A1	A+
	A	A2	A
	A- BBB+	A3	A- BBB+
Lower medium grade	BBB	Baa1	BBB
	BBB-	Baa2	BBB-
	BB+	Baa3	BB+
	BB	Ba1	BB
	BB-	Ba2	BB-
	B+	Ba3	B+
Highly speculative		B1	B
		B2	B-
	CCC	B3	CCC+
Substantial risk		Caa1	

A company should expect to deal with a credit-rating agency through a primary analyst who has considerable credit rating experience, and is usually ranked at the director level. The primary analyst is supported by a senior analyst having direct experience in the company's industry. The primary analyst is responsible for formulating a rating and for the ongoing monitoring of that rating.

The treasurer represents the company to the credit-rating agency. If there is a chief risk officer or similar position, then this person will also have discussions with the analyst team.
It is also entirely likely that the managers of the company's operating divisions will be asked to participate in some meetings with the analyst team or to assist them with tours of key company facilities.

In order to develop a credit rating, the analyst team uses the financial statements that a company has previously filed with the SEC, but also needs detailed information about its budgets, internal operating reports, risk management strategies, and financial and operating policies. To this end, it will ask the management team to complete an initial questionnaire that usually requires five years of historical financial data, five years of forecasted financial results, a summary of the business and its objectives, a comparison of its market share and growth prospects to those of its peer companies, and the biographies of the senior management team. The focus of this analysis is forward-looking, since the analyst team is most concerned with the company's future performance. Consequently, short-and mediumrange projections of a company's financial viability are considered more important than its historical performance.

We need to point out here that the analysts use their agency's standardized rating methodology to assign a credit rating to the specific company debt for which they are being hired. It is difficult to estimate in advance what the rating may be, since the relative weighting of factors in the methodology will vary based on individual circumstances.

If an agency issues a low credit rating or downgrades an existing rating, the best reaction by the company is not to publicly challenge it. The company has however, appeal rights, usually by presenting new information to the agency. Appeals are rare though, comprising only about one-half percent of all ratings changes.

If a company wants to improve its credit rating, then it must take specific steps to make its financial structure more conservative, such as by issuing more stock and using the proceeds to pay down debt. This requires the development of a plan to achieve the higher credit rating and communication of this information to the credit rating agency.

8.8 DEBT-RELATED CONTROLS

a) General Debt Transaction Controls

- *Require approval of the terms of all new borrowing agreements.* A senior corporate manager should be assigned the task of reviewing all prospective debt instruments to verify that their interest rate, collateral, and other requirements are not excessively onerous or conflict with the terms of existing debt agreements. It may also be useful from time to time to see if a lending institution has inappropriate ties to the company, such as partial or full ownership in its stock by the person responsible for obtaining debt agreements.
- *Require supervisory approval of all borrowings and repayments.* High-level supervisory approval is required for all debt instruments-except this time it is for final approval of each debt commitment. If the debt to be acquired is extremely large, it may be useful to have a policy requiring approval by the board of directors just to be sure that there is full agreement at all levels of the organization regarding the nature of the debt commitment. To be a more useful control, this signing requirement should be communicated to the lender, so that it does not inadvertently accept a debt agreement that has not been signed by the proper person.
- *Investigate the reasoning for revenue recognition related to attached rights that is not recognized ratably.* When a value is assigned to an attached right, the debit is to a discount account that will be ratably recognized as interest expense over the term of the debt; however, the credit will be to an unearned revenue account for which the potential exists to recognize revenue much sooner, thereby creating a split in the timing of revenue and expense recognition. Whenever revenue recognition related to this credit is not calculated on a ratable basis (which would create an approximate match between revenue and expense recognition), the calculation should first be approved by a manager.

b) Interest Rate Calculations

- *Require written and approved justification for the interest rate used to value debt.* When the stated interest rate on debt varies significantly from the market rate of interest, generally accepted accounting principles (GAAP) requires that the debt be valued using the market rate. However, the exact amount of this market rate is subject to interpretation, which has an impact on the amount of interest expense

recognized. Requiring justification for and approval of the rate used introduces some rigor to the process.

- *Include in the month –end closing procedure a task to record interest expense on any bonds for which interest payments do not correspond to the closing date.* The payment of interest to bondholders is a natural trigger for the recording of interest expense, but there is no such trigger when there is no payment. To enforce the proper recording not only of unpaid interest expense but also of any amortization on related bond discounts or premiums, a specific task should be included in the closing procedure, as well as a required signoff on the task.

c) Extinguishing of Debt

- *Include in the debt procedure a line item to charge unamortized discounts or premiums to expense proportionate to the amount of any extinguished debt.* The general ledger accountant may not remember to write off any unamortized discount or premium when debt is extinguished, so the debt extinguishment procedure should include a line item requiring that this task be addressed. Otherwise, expense recognition could potentially be delayed until the original payment date of the debt, which may be many years in the future.
- *Report to the board of directors the repayment status of all debt.* GAAP requires that all unamortized discounts and premiums be recognized in the current period if there is no reasonable chance that the debt will be repaid. Since the acceleration has a significant impact on reported earnings in the current period, there may be some unwillingness to classify debt as unable to be repaid. By requiring a standard report to the board of directors regarding the status of debt repayments at each of its meetings, the board can decide on its own when amortization must be accelerated, and can force management to do so.

d) Convertible Debt

- *Verify the market value of equity on conversion dates when the market value method is used.* If a company uses the market value method to record the conversion of debt to equity, it is possible to influence the gain or loss recorded, depending on fluctuations in the stock price from day to

day. Accordingly, the market price of the stock should be independently matched to the date on which the conversion took place. Also, it is possible to include in the conversion procedure a fill-in blank where the stock price can be noted, dated, and initialed. This approach makes it much easier to trace transactions, and also holds accountants responsible for their entries.

- *Verify the market value of equity on debt retirement dates when offsetting equity entries are being reverses.* When a convertible bond is issued with its equity conversion feature already in the money, the intrinsic value of the equity portion of the bond must be credited to the additional paid-in capital account. If the bond is later retired, the equity portion of the bond must be removed from the additional paid in capital. Any difference between the original and final intrinsic values is charged to either a gain or loss on the extinguishment makes it more likely for manipulation to occur in both the timing and calculation of the extinguishment transaction. One can match the date of the debt retirement to the equity valuation on that date to ensure that the proper equity value is used. Also, the correct retirement calculation can be included in the corporate accounting procedures manual to ensure that it is handled properly.
- *Include a review of accrued interest expense on all recently converted debt.* If the terms of a company's bond agreements state that bond holders must forfeit accrued interest on converted debt, then there will be a temptation to also avoid recording this accrual on the books as an expense, as is required by GAAP. Consequently, the formal procedure used to convert debt to equity should include a line item for the general ledger accountant to record this accrued interest expense, and also require a signature on the procedure to ensure its completion.
- *Verify expense calculations associated with any sweetened conversion offers.* GAAP requires the recognition of a debt conversion expense associated with any completed conversion from bonds to equity, in the incremental amount of the net increase in fair value of stock obtained through a sweetened conversion offer. Since this results in an added expense, there will be a tendency to simply process the total conversion

and not recognize the incremental expense, which could be substantial. Accordingly, a copy of the relevant portions of the original bond agreement should be attached to any journal entry that records a conversion price. When the calculation is verified by an internal auditor or senior accounting person, this provides documentation of the initial baseline conversion price.

8.9 DEBT-RELATED POLICIES

Debt related policies define the issuance and buyback of debt, control the timing of expense recognition, the setting of interest rates used for expense calculations, and similar issues. The intent of the bulk of these policies is to issue and buy back debt only when it is in the best business interest of the company to do so, as well as to ensure that debt-related transactions are recorded fairly.

a) General Debt Transaction Policies

- *All notes and bonds shall only be issued subsequent to approval by the board of directors.* This policy gives the board control over any new debt liabilities. In reality, anyone lending money to the company will require a board motion, so this policy is likely to be imposed by the lender even if it does not exist internally.
- *Debt sinking funds shall be fully funded on scheduled dates.* This policy is designed to force the treasury staff to plan for the timely accumulation of funds needed to pay off scheduled principal payments on debt, thereby avoiding any last-minute funding crises.
- *Recognition of unearned revenue for attached rights shall match offsetting discount amortization as closely as possible.* This policy is designed to avoid the manipulation of revenue recognition for attached rights.

b) Extinguishment of Debt

- *Debt shall not be extinguished early if the primary aim is to report a gain or loss on the extinguishment.* If a company buys back its bonds when the stated interest rate on the debt is lower than the current market interest rate, it will recognize a gain on the transaction, but must refinance the purchase with more expensive debt at current market rates. Thus, this policy is designed to keep

company management from creating transactions that appear to increase profits when the underlying results worsen the company's financial situation.

- *When interest rates allow, the company shall repurchase its debt with less expensive debt.* Though it sounds obvious, this policy is designed to force management to make the correct decision to always use less expensive debt, even though this will result in the recognition of a loss when the older, more expensive debt is eliminated from company records.

c) Convertible Debt

- *Debt conversions to equity shall always be recorded using the book value method.* This policy keeps the accounting staff from switching between the book value and market value method to recognize gains and the book value method to avoid losses.

8.10 DEBT-RELATED PROCEDURES

The detailed procedures used for borrowing from a line of credit, calculating the effective interest rate, extinguishing debt, converting debt to stock, and dealing with debt that has attached warrants are shown in this unit.

It is important that treasurers design their own debt related procedures for their individual companies as each company's debt needs will always be different from the other.

8.11 ACTIVITY

1. Outline 5 (five) types of debt available to a company.
2. Discuss 5 (five) types of bonds.
3. What is meant by "sale and lease back"?
4. Examine 2 (two) main debt-related controls.
5. Discuss 3 (three) common debt related policies available to a treasurer.
6. Discuss the role of Credit Rating Agencies.

8.12 SUMMARY

The treasurer is regularly called upon to obtain debt financing for a company. On a simplified level, this may involve obtaining a line of credit from the company's bank, or a factoring or

leasing arrangement. However, larger companies with excellent credit ratings have a number of additional tools available, such as bonds, commercial paper, and receivables securitization.

Of these types of debt, bond offerings in particular present a number of challenges from the perspective of accounting and control systems because there are ways to manipulate expense recognition in a company's favour. However, for the majority of companies that are restricted to basic lines of credit and loans, the accounting and control systems are relatively straight-forward, primarily involving the approval of new debt and the proper recognition of interest expense in the correct reporting periods.

Well-done for reading unit 8! Unit 9 will discuss Equity management.

UNIT 9

EQUITY MANAGEMENT

9.1 INTRODUCTION

Welcome to Unit 9! The unit has been designed to discuss equity management.

9.2 AIM

The aim of this unit is to discuss equity management.

9.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss stock registration; 2) Discuss how stock sales are accounted for; 3) Discuss equity related policies.

4) Discuss equity related controls.

9.4 TIME REQUIRED

You are advised to spend at least 3 (three) hours studying this unit.

9.5 REFLECTION

Attempt to define equity. Write your answer in the space provided below.

……

……...

9.6.1 DEFINING EQUITY

The term equity is often associated with stocks and shares. Equities are long term securities traded on the stock exchange market that enable an investor to raise capital. The stock exchange market where equities are traded, is sometimes called the ‘’equities market, capital market or securities market’’. To many investors the term ‘’investment’’ implies investment in the ordinary shares of a company meaning, in equities.

A key goal of the treasurer of a public company is to have its securities registered, so that it can more easily sell the securities, and so that its investors can freely trade them. The registration process is very time consuming and expensive, so companies attempt to circumvent it through a variety of exemptions.

9.6.2 STOCK REGISTRATION

If a treasurer wants to sell stock to investors that in turn can be immediately traded by the investors, then it is necessary to file a registration statement with the Securities and Exchange Commission (SEC). Compiling a registration statement and walking it through SEC review process is one of the most expensive and time-consuming tasks that a treasurer can engage in. Each country has its own pre-printed forms that must be filled in and filed with SEC. **a)**

Shelf registration

Shelf registration is the registration of a new issue of securities that can be filed with the SEC up to three years in advance of the actual distribution of such securities. This allows a company to obtain funds quickly when needed, rather than compiling a registration document and then waiting for the SEC to declare the registration effective. It is especially useful for debt offerings, since a public company can wait for interest rates to decline before issuing any securities.

A shelf registration must be declared effective by the SEC before any securities sales related to it can be initiated.

b) Declaring a Registration Statement Effective

A registration statement is reviewed by the SEC staff, and if they find that it conforms to SEC regulations and clearly states key information about the company, then they declare it ***effective.*** Once declared effective, either the company or those investors on whose behalf it is registering, the securities can initiate selling activities.

The problem is obtaining that "effective" status. The SEC's examiners delve into registration statements with great vigor, and it is a rare document indeed that is immediately granted "effective" status. Instead, the usual situation is for the SEC to spend one month reviewing the registration document and then sending back a comment letter detailing information about the review.

The key issues involving the effectiveness declaration are the time and expense involved. The company's attorneys and auditors are deeply involved in every question and answer iteration with the SEC, so the cost of their services builds over time.

c) **Regulatory requirements specified by SEC for corporate debt securities** Debt securities represent a debt by an organization. They do not give their owners, ownership of the organization. They are usually issued for a fixed term and the amount of capital represented by them has to be paid back to their owners at the end of the debt security's term. Debt securities usually carry an interest charge which is paid to their holders at specified periods during the debt security's term.

All securities issued by a public issue must be listed on the stock exchange. The issue of debt securities by companies including listed companies (i.e. which have any of their securities, either as equity or debt, offered through an offer document and listed on a recognized stock exchange like LUSE (Lusaka Stock Exchange) on a private placement basis must comply with the conditions prescribed by SEC from time to time, for getting them listed on the stock exchanges. Further, unlisted companies/statutory corporations/other entities in Zambia, if they so desire, may get their privately placed debt securities listed on the stock exchanges by complying with the relevant conditions. The debt securities shall carry a credit rating from a credit rating agency registered with SEC. Debt securities shall be issued and traded in demat form.

d) **Eligibility norms for making capital issues**

SEC Zambia has laid down eligibility norms for entities accessing the primary market through public issues. In the primary market, securities (shares/bonds/debentures) are offered to the public for the purpose of raising capital for the first time. There are no eligibility norms for a listed company that is making a rights issue (i.e. shares offered to already existing shares). There are also no eligibility norms for listed company making a preferential issue.

Any company making a public issue or a listed company making a rights issue is required to file a draft offer document with SEC for its observations. The validity period of SEC's observation letter is only three months, i.e. the company has to open its issue within a period of three months. (LUSE, 2015). There are no requirements of filing any offer document/notice to SEC in the case of preferential allotment and QIP. A QIP (Qualified Institutional Placement) is an institution such as a merchant banker that handles an issue and files a copy of the placement document with SEC postallotment, for record purposes.

Merchant banks are specialized intermediaries who are required to display due diligence and ensure that all the requirements of DIP (Disclosure and Investor Protection) are complied with, while submitting the draft offer document to SEC. Any non-compliance on their part attracts penalty action from SEC, in terms of SEC (Merchant Bankers) Regulations. The draft document filed by the merchant banker is also placed on the website for public comments. Officials of SEC, at various levels, examine the compliance with DIP guidelines and ensure that all necessary material information is disclosed in the draft offer documents.

Offer document means, a prospectus in case of a public issue, or offer for sale and Letter of Offer in case of a rights issue which are filed with the registrar of Companies (ROC) and stock exchanges. An offer document covers all the relevant information to help an investor to make his/her investment decision. A draft offer document means the offer document in a draft stage. The draft documents are filed with SEC, at least 21 days prior to the filing of the other documents with the Registrar of Companies (ROC). SEC may specify changes, if any, in the draft offer document and the issuer or the lead merchant banker (LM) shall carry out such changes in the draft offer document before filing the offer document with the ROC. The draft offer document will be available on the SEC website for public comments for a period of 21 days from the filing of the draft offer document with SEC.

e) Power of SEC to regulate or prohibit issue of prospectus, offer document or advertisement soliciting money for issue of securities

Without prejudice to the provisions of the Companies Act, SEC may, for the protection of investors,

(a) Specify, by regulations:

i) Matters relating to issue of capital, transfer of securities and other matters incidental thereto; and

ii) Manner in which such matters shall be disclosed by the companies; (b) By general or special orders:

(i) Prohibit any company from issuing a prospectus, any offer document or advertisement soliciting money from the public for the issue of securities;

(ii) Specify the conditions subject to which, the prospectus, such offer document or advertisement, if not prohibited, may be issued.

Without prejudice, SEC may specify the requirements for listing and transfer of securities and other matters incidental thereto.

9.6.3 ACCOUNTING FOR STOCK SALES

Most types of stock contain a par value, which is a minimum price below which the stock cannot be sold. The original intent for using par value was that a residual amount of funding was contributed to the company, and which could not be removed from it until dissolution of the corporate entity. In reality, most common stock now has a par value that is so low that its original intent no longer works. Thus, the accountant still tracks par value separately in the accounting records, it has little meaning.

If an investor purchases a share of stock at a price greater than its par value, the difference is credited to an additional paid-in capital account. For example, if an investor buys one share of common stock at a price of K82, and the stock's par value is K1, then the entry would be:

	Debit	**Credit**
Cash	K82	
Common stock-par value		K1
Common stock-additional paid-in capital		K81

When a company initially issues stock, there will be a number of costs associated with it, such as the printing of stock certificates, legal fees, investment banker fees, and security registration fees. These costs can be charged against the proceeds of the stock sale, rather than be recognized as expenses within the current period.

If the warrant attached to a debt instrument cannot be detached and sold separately from the debt, then it should not be separately accounted for. However, if it can be sold separately by the debt holder, then the fair market value of each item (the warrant and the debt instrument) should be determined, and then the accountant should apportions the price at which the combined items were sold amongst the two, based on their fair market values.

Example

If the fair market value of a warrant is K63.50, and the fair market value of a bond to which it was attached is K950, and the price at which the two items were sold was K1,005, then the entry should be made to an additional paid-in capital account for K62.97 to account for the warrants, while the remaining K94.03 is accounted for as debt. The apportionment of the actual sale price of K1,005 to warrants is calculated as follows:

Fair Market value of Warrant x price of combined instruments

Fair Market Value of Warrant

+ Fair Market Value of Bond

Or,

K63.50 x K1,005=K62.97

(K63.50 + K950)

If a warrant expires, then the funds are shifted from the outstanding warrants account to an additional paid-in capital account. To continue with the last example, this would require the following entry:

	Debit	**Credit**
Additional Paid-In Capital-Warrants	K62.97	
Additional Paid-in Capital-Expired Warrants		K62.97

If a warrant is subsequently used to purchase a share of stock, then the value allocated to the warrant in the accounting records should be shifted to the common accounts. To use the preceding example, if the warrant valued at K62.97 is used to purchase a share of common stock at a price of K10, and the common stock at a price of K10, and the common stock has a par value account is credited with K25 (since it is mandatory that the par value be recorded), and the remainder of the funds are recorded in the additional paid-in capital account. The entry is as follows:

	Debit	Credit
Cash	K10.00	
Additional paid-in-capital-warrants	K62.97	
Common stock-par value		K25.00
Common stock-additional paid-in capital		K47.97

9.6.4 EQUITY-RELATED CONTROLS

There are a number of controls that should be used during the process of selling equity to the investment community. The following controls are listed in sequential order, from the start of an equity offering. Those applying primarily to a specific registration document or exemption are also noted.

9.6.4.1 General stock sale controls

1. *Verify available authorized shares:* There must be sufficient number of authorized and unused shares available for distribution, which can be determined by comparing the authorized number of shares listed in the articles of incorporation to the share total reported by the company's stock transfer agent.

2. *Verify board authorization:* The board minutes should reflect the approval of a stock offering that includes the minimum and maximum allowed amounts of funding, and the approximate terms of the offering.

3. *Have a due diligence officer review all SEC filings:* The SEC can and will reject any registration documents if information is not presented exactly in accordance with its instructions. To avoid the delay of an SEC procedures review the filings prior to submission.

4. *Verify contents of security authorization letter:* The next step process is to authorize the issuance of shares to new investors. The treasurer authorizes the issuance of shares to new investors by completing a form letter to the company's stock transfer agent, itemizing the number of shares to be issued to each investor, and should match all information on the authorization letter to the supporting documentation prior to approving the letter.

9.7 ACTIVITY

1) What is equity?

2) Why is stock registration important?

3) Explain how stock registration is done.

4) Discuss the role of SEC in stock issues and registration.

5) What is the procedure of registering stocks with SEC?

6) Discuss what is meant by accounting for stock sales.

7) Discuss some equity related controls.

9.8 SUMMARY

The ability to issue high-quality registration documents to the SEC is the mark of a well-run company. It requires excellent knowledge of SEC regulations, as well as the support and direction of experienced company counsel. This is what this unit has tried to discuss. In addition, it should be remembered that the treasurer is heavily involved in stock sales, and so should be familiar with all forms of registration and related exemptions, as well as the controls and procedures needed to complete stock sales under each one.

The next unit will discuss Investment management.

UNIT 10

INVESTMENT MANAGEMENT

10.1 INTRODUCTION

Welcome to unit 10! This unit will discuss investment management.

10.2 AIM

The aim of this unit is to discuss investment management.

10.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss investment criteria;
2) Discuss investment options;
3) Evaluate the various investment strategies;
4) Discuss risk-reduction strategies;
5) Acquire knowledge of how to account for investments; 6) Discuss investment management controls.

10.4 TIME REQUIRED

You are advised to at least spend not less than 3 (three) hours studying this unit.

10.5 REFLECTION

Attempt to define investment. Write your answer in the space provided below.

……
……
……
……

10.6.1 INVESTMENT CRITERIA

Surplus funds not needed for either operating purposes or compensating bank balances are available for investment. Prudent use of these funds can add to income, though the treasurer must consider a range of investment criteria, types of investments, and investment strategies before selecting the appropriate investment vehicle.

When considering various forms of cash investment, the treasurer should first consider the **safety of the principal** being invested. A company policy should limit investments to a specific set of low-risk investment types. Some consideration should also be given to the **maturity** and **marketability** of an investment. It is best to only make investments where there is a robust market available for their immediate resale.

The final consideration when making an investment is its **yield**-and this is truly the last consideration after the previous items have already been reviewed. Within the boundaries of appropriate levels of risk, maturity, and marketability, the treasurer can then pick the investment with the highest yield. Since these criteria tend to limit one to very low-risk investments, the yield will also likely be quite low.

The investment criteria for a company that finds itself in a rapid growth situation are more circumscribed. It typically burns through its cash reserves quite rapidly, so the liquidity of its investments must be extremely high in order to allow rapid access to it. Unfortunately, high liquidity is commonly associated with low investment returns, so the treasurer is forced to invest in low-yield investments, because it is critically important to keep cash available to feed the company's growth engine. Since risk is also associated with return, the treasurer must, once again, favour low-yield investments for minimal risk.

10.6.2 INVESTMENT OPTIONS

Some of the most common options that have low risk levels, short maturity dates, and high levels of marketability include the following:

a) **Bankers' acceptances:** Banks sometimes guarantee (or accept) corporate debt, usually when they issue a loan to a corporate customer, and then sell the debt to investors. Because of the bank guarantee, they are viewed as obligations of the bank.

b) **Bonds near maturity dates:** A corporate bond may not mature for many years, but one can always purchase a bond that is close to its maturity date. There tends to be a minimal risk of loss (or gain) on the principal amount of this investment, since there is low risk that interest rates will change so much in the short time period left before the maturity date of the bond that it will impact its value. A variation on this type of investment is the municipal bond, for which there is no tax on the interest income; however, in consideration of this reduced liability, its yield also tends to be somewhat lower than on other types of bonds.

c) **Certificate of deposit (CD):** These certificates are essentially term bank deposits, typically having durations of up to two years. They usually pay a fixed interest rate upon maturity, though some variable-rate CDs are available. There is a perception that they are more secure than commercial paper, since CDs are issued by banks, which are more closely regulated than companies.

d) **Commercial paper:** Larger corporations issue short-term notes that carry higher yields than on government debt issuances. There is also an active secondary market for them, so there is usually no problem with liquidity. Commercial paper is generally not secured; however, staying with the commercial paper issued by "blue chip" organizations minimizes the risk of default. Most commercial paper matures in 30 days or less, and rarely matures in greater than 270 days, in order to avoid the registration requirements of the Securities and Exchange Commission (SEC). Commercial paper is issued at a discount, with the face value being paid at maturity.

e) **Money market fund:** This is a package of government instruments, usually composed of Treasury bills, notes, and bonds, that is assembled by a fund management company. The investment is highly liquid, with many investors putting in funds for as little as a day. There are varying levels of risk between different money market funds, since some funds are more active in trying to outperform the market (with an attendant increase in risk).

f) **Repurchase agreement:** This is a package of securities (frequently government debt) that an investor buys from a financial institution, under the agreement that the institution will buy it back at a specific date. It is most commonly used for the

overnight investment of excess cash from a company's cash concentration account, which can be automatically handled by the company's primary bank. The typical interest rate earned on this investment is equal to or less than the money market rate, since the financial institution takes a transaction fee that cuts into the rate earned.

g) **Treasury bills issuances:** Most countries' governments including Zambia issue treasury bills and invite the general public to invest in them. Normally, treasury bills are issued by the central bank on behalf of government as a way of controlling money supply in the economy. They are short term debt instruments that are considered as risk free investments. They are issued at a discount to their face value. They have lower yields than most other forms of investments though.

10.6.3 INVESTMENT STRATEGIES

The treasurer should develop a standard methodology for investing funds.

At the most minimal level of investment strategy, the treasurer can do nothing and leave idle balances in the corporate bank accounts. This is essentially an **earnings credit strategy,** since the bank uses earnings from these idle balances, then this is not an entirely bad strategy-the earnings credit can be equivalent of the modest rate of return, and if there is not enough cash to plan for more substantive investments, leaving the cash alone is a reasonable alternative.

A matching strategy simply matches the maturity date of an investment to the cash flow available dates listed on the cash forecast.

A laddering strategy involves creating a set of investments that have a series of consecutive maturity dates.

A tranched cash flow strategy requires the treasurer to determine what cash is available for short, medium, and long-term investment, and to then adopt different investment criteria for each of these investment tranches.

The exact investment criteria will vary based on a company's individual needs, but here is a sample of how the tranches might be arranged:

- The short-term tranche is treated as cash that may be needed for operational requirements on a moment's notice. This means that cash flows into and out of this tranche can be strongly positive or negative. Thus, return on investment is

not a key criterion-instead the treasurer focuses on very high levels of liquidity. The return should be the lowest of the three tranches, but should also be relatively steady.

- The medium-term tranche includes cash that may be required for use within the next 3 to 12 months and usually only for highly predictable events, such as periodic tax or dividend payments, or capital expenditures that can be planned well in advance. Given the much higher level of predictability in this tranche, the treasurer can accept longer-term maturities with moderate levels of volatility that have somewhat higher returns on investment.
- The long-term tranche includes for which there is no planned operational use, and which the treasurer feels can be safely invested for at least one year. The priority for this tranche shifts more in favour of a higher return on investment, with an attendant potential for higher levels of volatility and perhaps short-term capital loss, with a reduction in the level of liquidity.

To engage in the tranched cash flow strategy, cash flow strategy, the treasurer should regularly review the cash forecast, and adjust the amounts of cash needed in each of the three tranches. Inattention to these adjustments could result in an unanticipated cash requirement when the cash in the company's long-term tranche is tied up in excessively long-term, illiquid investment investments.

The preceding strategies were mechanical; the treasurer analyzes cash flows and engages in investments based on cash availability. The next two strategies are more speculative, since the treasurer is guessing at the possible direction of future yields.

Riding the yield curve is a strategy of buying longer-term securities and selling them prior to their maturity dates. This strategy works when interest rates on short-term securities are lower than the rates of longer-term securities, which is normally the case.

Under a **credit-rating strategy,** the treasurer buys the debt of a company that may be on the verge of having its credit rating upgraded. By doing so, the investment ends up earning a higher interest rate than would other investments with a comparable credit rating. This strategy works best when the treasurer is very familiar with the debt issuer and has some confidence in his credit assessment. However, the company's stated investment policy may prevent the treasurer from buying debt below a fairly high credit rating, which eliminates this strategy from

consideration. Also, it is difficult to time a possible credit-rating upgrade to be within the term of an investment. And finally, delving into lower-grade debt increases the risk of an outright default on payments by the debtor.

For all of the investment strategies noted here, the treasurer must closely monitor the credit rating of the debt issuer. A credit downgrade can result in a substantially lower return to the company if the treasurer needs to sell the debt prior to its maturity date.

The strategies outlined here involve a broad range of ongoing investment activities; the more active ones call for additional staffing that a treasury department may not have available. If so, outsourcing investment management, as noted in the next section, may be a viable alternative.

10.6.4 OUTSOURCED INVESTMENT MANAGEMENT

A treasurer may conclude that investment management is not a core competency, or have little funding for a professional in-house investment staff. If so, it is possible to shift cash into a separate account that is managed by outside investment advisors. Under this arrangement, the outside firm invests the cash under the terms of a customized investment agreement with the company. The company can choose from the variety of possible investment strategies, as well as restrict investments to certain classes of assets.

This approach gives the company access to an experienced group of investment managers that presumably uses strong systems of control when initiating and tracking transactions. The fees a company incurs through an outsourcing arrangement can be quite competitive in comparison to the cost of maintaining a similarly experienced in-house staff.

10.6.5 RISK-REDUCTION STRATEGIES

A simple risk-reduction strategy is to avoid investments in the securities of any single entity, in favour of investments solely in one or more money market funds. These funds provide instant diversification across a multitude of issuers, with the attendant risk being constantly reviewed by a staff of risk management professionals. The use of money market funds is especially cost-effective for smaller treasury departments that cannot afford the services of an in-house investment manager.

10.6.6 ACCOUNTING FOR INVESTMENTS

A company will normally invest in **marketable securities**, so that it can more easily liquidate its investments. Marketable securities are investments that can be easily liquidated through an organized exchange, such as the New York Stock Exchange and the Lusaka Stock Exchange (LUSE). For accounting purposes, marketable securities must be grouped into one of the following three categories at the time of purchase and re-evaluated periodically to see if they still belong in the designed categories:

1. ***Available for sale:*** This category includes both debt and equity securities. It contains those securities that do not readily fall into either of the following categories. These are securities are reported on the statement of financial position at their fair value, while unrealized gains and losses are charged to an equity account and reported in other comprehensive income in the current period. The balance in the equity account is only eliminated upon sale of the underlying securities. If a permanent reduction in the value of an individual security occurs, the unrealized loss is charged against earnings, resulting in a new and lower cost basis in the remaining investment above the new cost basis cannot be formally recognized in earnings until the related security is sold, and so the interim gains will be temporarily "parked" in the unrealized gains account in the equity section of the balance sheet.

   All interest, realized gains or losses, and debt amortization are recognized within the continuing operations section of the income statement. The listing of these securities on the balance sheet under either current or long-term assets is dependent upon their ability to be liquidated in the short term and to be available for disposition within that time frame, unencumbered by any obligations.
2. ***Held to maturity:*** The category includes only debt securities for which the company has both the intent and ability to hold them until their time of maturity. Their amortized cost is recorded on the balance sheet. These securities are likely to be listed on the balance sheet as long-term likely to be listed on the balance sheet as long-term assets.

   If marketable securities are shifted into the held-to-maturity category from debt securities in the available-for-sale-category, their unrealized holding gain or loss should continue to be stored in the equity section, while being gradually amortized down to zero over the remaining life of each security.
3. ***Trading securities:*** This category includes both debt and equity securities that the company intends to sell in the short term for a profit. They are recorded on the statement

of financial position at their fair value. This type marketable security is always positioned in the statement of financial position as a current asset.

No matter how an investment is categorized, a material decline in its fair value subsequent to the balance sheet date but prior to the release of the financial statements should be disclosed. Further, clear evidence of permanent impairment in the value of available-for-sale securities prior to the release date of the financial statements is grounds for re-statement to recognize permanent impairment of the investment.

a) Transfers between Available-for-Sale and Trading Investments

An investment designated as a trading security can be shifted into the available for sale portfolio of investments with no recognition of a gain or loss on the value of the investment, since this type of investment should have been adjusted to its fair value in each reporting period already. If a gain or loss has arisen since the last adjustment to fair value, this amount should be recognized at the time of the designation change. If an investment designed as an available for sale security is shifted into the trading portfolio of investments, any gain or loss required to immediately adjust its value to fair value should be made at once. This entry should include an adjustment from any prior write-down in value that may have occurred when securities were classified as available for sale.

b) Accounting for Investments in Debt Securities

A debt security can be classified as either held for trading or available for sale (as previously defined for equity securities), or as held to maturity. The held-to-maturity portfolio is intended for any debt securities for which a company has the intent and ability to retain the security for its full term until maturity is reached. An investment held in the held-to-maturity portfolio is recorded at its historical cost, which is not changed at any time during the holding period, unless it is shifted into a different investment portfolio. The only exceptions to this rule are (1) the periodic amortization of any discount or premium from the face value of a debt instrument, depending on the initial purchase price; and (2) clear evidence of a permanent reduction in the value of the investment.

c) Transfers of Debt Securities among Portfolios

The accounting for transfers between debt securities portfolios varies based on the portfolio from which the accounts are being shifted, with the basic principle being that transfers are recorded at the fair market value of the security on the date of the transfer.

d) **Recognition of Deferred Tax Effects on Changes in Investment Valuations** A deferred tax benefit or tax liability should be recognized alongside the recognition of any change in the fair value of an investment listed in either a trading or availablefor-sale portfolio or of a permanent decline in the value of a debt security being held to maturity. The tax impact varies by investment type and is noted as follows:

- ***Gains or losses on the trading portfolio:*** The deferred tax effect is recognized in the income statement. If there is a loss in value, then debit the deferred tax benefit account and credit the provision for income taxes account. If there is a gain in value, then debit the provision for income taxes account and credit the deferred tax liability account.
- ***Gains or losses on the available-for sale portfolio:*** The same treatment as noted for gains or losses on the trading portfolio, except that taxes are noted in the other comprehensive income section of the income statement.
- ***Gains or losses on the held-to maturity portfolio:*** There is no tax recognition if changes in value are considered to be temporary in nature. If there is a permanent reduction in value, the treatment is identical to the treatment of losses in the trading portfolio, as just noted.

10.6.7 INVESTMENT REPORTING

It is useful to have a summary-level reporting itemizing the investment, current market value and return on investment for each investment made. An example is shown below, where all investments re in equity. This approach yields a quick view of a company's overall level of investment and results thereof.

If the investment portfolio includes debt, then the report could be expanded to include either individual or average maturity dates.

Security	Number of Shares	Market Value	Purchase Price	Rate of Return	Total Dividends Y-T-D

1) Airtel	500	K37,000	K31,000	5.2%	K800
2) Bata PLC	100	K2,400	K2,400	6.3%	K75
3) Zambia Sugar PLC	1,000	K30,000	K31,000	6.5%	K1,000
4) Puma Energy PLC	1,000	K65,500	K64,000	7.8%	K2,000
5) Lafarge PLC	100	K1,900	K1,875	7.5%	K70
6) National Breweries PLC	500	K42,000	K38,000	5.3%	K1,000
Total or Average		K178,800	K168,275	6.5%	K4,945

10.6.8 INVESTMENT MANAGEMENT PROCEDURES

The process of issuing funds for an investment is unique in that every step in the process is a control point. Without regard to controls, the only step required to make an investment is for an authorized person to create and sign an investment authorization form (itself a control point) and deliver it to the bank, which invests the company's funds in the designated investment. There are a number of additional steps, all designed to ensure that there is an appropriate level of control over the size and duration of the investment, and that the earnings from the investment vehicle are maximized.

a) ***Create and approve a cash forecast:*** There must be some basis for both the size and duration of an investment. Otherwise, a mismatch can develop between the need for cash and its availability, resulting in liquidity problems or an excessive amount of underutilized cash. By requiring that a cash forecast be completed and approved by an authorized person, there is less risk of these problems occurring.

b) ***Record proposed investment and duration on the cash forecast:*** Though the cash forecast alone should be a sufficient control over the determination of the correct size and duration of an investment, it helps to also formally write this information directly on the cash forecast, so there is no question about the details of the proposed investment.

c) ***Obtain approval of investment recommendation:*** A manager should sign off on the proposed investment. By placing the approval signature line directly on the cash forecast, the approver can review the accuracy of the forecast as well as the resulting investment recommendation, giving sufficient information for the approver to determine if the recommendation is correct.

d) ***Obtain and document-quote for each investment:*** An investment officer may have a favourite bank and will continue to invest with it, even if its rates are not competitive. It is also common for the investment staff to not want to go to the effort of obtaining multiple quotes on a regular basis. By requiring them to complete a quotation sheet, this control ensures that the best investment rate is obtained.

e) ***Issue a signed investment authorization form to the issuer:*** Banks will not invest funds without a signed investment authorization form from the company. From the company's perspective, a signed authorization also ensures that the appropriate level of management has approved the investment.

f) ***Match authorization form to a transaction report:*** The bank may unintentionally invest funds incorrectly or neglect to invest at all. By matching the signed authorization form to any investment transaction report issued by the bank, the company can verify what action the bank took as result of the authorization.

g) ***Forward records to accounting for storage:*** There is some risk that a person in the investment department will alter investment authorization documents after the fact to hide evidence or inappropriate investments. To reduce this risk, require them to immediately forward a set of supporting documents to the accounting department for storage in a locked location. The accounting staff should stamp the receipt date on each set of documents received, which the internal auditors can use to determine if any documents were inappropriately delayed.

In addition to the basic process above, the following controls are also useful ensuring that documented investments were actually obtained and stored:

- ***Periodically match the approved cash forecast, quote sheets, and investment authorization to actual investments completed:*** Though all the supporting paper work may be in order, unauthorized form of investment. To detect such transactions, the internal audit department should periodically match all supporting documents to the actual investments made, as reported by the issuing entities.
- ***Assign securities custody to independent party:*** It is not common for a company to physically control securities; instead, they are more commonly stored by a third-party custodian, which represents a higher level of control. If some securities must be stored on-site, then ensure that the person responsible for their physical security is not also responsible for recording the securities in the accounting records.

The controls noted here assume that a company is investing in simpler investments, such as CDs, that are handled through a bank. It will be necessary to expand these controls for more complex investments that are not routed through banks.

10.6.9 INVESTMENT MANAGEMENT CONTROLS

An investment policy is used to define the level of risk that a company is willing to tolerate and defines the exact types of investment vehicles to be used (or not used). Such a policy should cover the level of allowable liquidity. For example, the policy may state that all investments must be capable of total liquidation upon notification or that some proportion of investments must be in this class of liquidity. Thus, the policy could state that 75% of all investments must be capable of immediate liquidation (which rules out real estate holdings), or that any investments over a base level of K50 million can be invested in less liquid instruments. Generally speaking, the policy should severely restrict the use of any investments that cannot be liquidated within a short period of time, since this gives a company maximum use of the money in case of special opportunities (such as acquisition) or emergencies (such as natural disaster destroying a facility). Such careful delineation of investment liquidity will leave a small number of investments that a treasurer can safely use.

The second policy criterion is risk. Many companies have decided that they are not in the business of making investments, and so they avoid all income by putting all excess cash in government securities. Other companies take the opposite tack and attempt to derive a significant proportion of their profits from investment income. No matter which direction a company takes, it is necessary to delineate which kinds of investments are to be used, thereby keeping the treasurer focused on a specific set of investment options.

The final policy criterion is return on investment. For the purposes of a company's short-term investments, this should be the least of the three investment criteria; it is much more important for a company to achieve high levels of liquidity and risk minimization than it is to achieve a high rate of return. If a company is willing to rebalance the three criteria in favour of a greater return, then it should at least consider making return on investment a lesser factor for its short-term investments, since they have less time to recover from the vagaries of the financial market.

The investment policy should be closely aligned with the investment strategy. For example, if the treasurer elects to divide available cash into short-term, medium-term, and long-term tranches, then the policy should allow for long-term tranche. Similarly, if the treasurer plans to aggressively anticipate interest rates, then the policy should allow for some degree of speculation. Conversely, if the board of directors specifically does *not* allow certain activities, then this limits the treasurer's available strategies.

There are a multitude of possible investment policies that a treasury can adopt; some are designed to restrict its choices of possible investments, while others are needed to account for investments. The following policy possibilities are sorted into the categories for funds investment, designations of types of investments, transfers of investments among different portfolio designations and investment accounting.

a) Funds Investment

- *At least K___shall be invested in overnight investments and in negotiable marketable obligations of major Zambian issuers.* This policy forces the treasury staff to attain a minimum level of liquidity. The fixed Kwacha amount used in the policy should be regularly reviewed to match upcoming budgeted working capital requirements.
- *No more than _____ percent of the total portfolio shall be invested in time deposits or other investments with a lack of liquidity.* This policy is similar to the preceding one, except that it ignores a fixed liquidity requirement, focusing instead on a maximum proportion of the total portfolio that must be retained for short-term requirements. This policy tends to require less periodic updating.
- *The average maturity of the investment portfolio shall be limited to ____years.* This policy is designed to keep a company from investing in excessively long maturities. The policy can be broken down into more specific maturity limitations for different types of investments such as 5 years for any Zambian government bonds, 1 year for bank fixed deposits, and 270 days say for commercial paper.
- *Investments in foreign commercial paper shall be limited to those unconditionally guaranteed by a prime Zambian issuer and fully hedged.* This policy is designed to lower the risk of default on investments in securities issued by foreign entities.
- *Investment in commercial paper shall be limited to those of companies having long-term senior debt rating of A or better.* This policy is designed to limit the risk of default on commercial paper investments by focusing investments on only the highest-grade commercial paper. The policy can be expanded to require the sale of any commercial paper when the debt rating is adjusted downward below the minimum rating level.

- *Investments shall only be made in investment backed by the Zambian government's debt obligations.* This policy can be used in place of the preceding ones that specify allowable investments in non-governmental investments. This policy tends to be used by highly risk-averse companies who place less emphasis on the return generated from their investments.
- *Securities physically held by the company shall be stored with an accredited third party.* This policy improves the physical control over securities.
- *If an employee is responsible for the physical security of securities held by the company, then this person cannot also be responsible for recording the securities in the accounting records.* This policy prevents an employee from removing securities and then eliminating evidence of the securities from the accounting records.

b) Investment in debt securities accounting

- *The unrecognized amount of gains or losses on held-to maturity securities shall be regularly reported to the board of directors.* Management may designate poor-performing debt securities as held-to maturity, in which case any changes in their fair value will not be recognized. This policy is designed to reveal any gains or losses that would be recognized if these securities were to have any other portfolio designation, so the Board is aware of any

  "hanging" gains or losses.
- *Debt securities shall not be classified as held to maturity unless sufficient investments are already on hand to cover all budgeted short-term cash requirements.* Generally accepted accounting principles (GAAP) already requires that debt securities not be classified as held to maturity if a company does not have the ability to hold the securities for the required time period-this policy is more specific in stating that all anticipated cash flows be fully covered by other investments before any debt securities receive the held-tomaturity classification. The policy makes it more likely that a company will not be forced to prematurely liquidate its held-to-maturity debt portfolio.

c) Investment portfolios, transfer between

- *The board of directors shall be notified of the reasons for any significant shift in the designation of securities among the held-to-maturity, available-for-sale, and trading portfolios, and the approximate impact on different categories of*

income. This policy is designed to require management to justify its actions in shifting securities between portfolios, which is likely to reduce the amount of shifting, while also keeping the board informed of any likely movements of gains or losses between the operating income and other comprehensive income parts of the income statement.

d) Investment portfolios, transfer of debt securities among

- *The board of directors must authorize all shits in investment designation out of the held-to-maturity portfolio.* There are specific accounting instances where the transfer of securities out of the held-to-maturity portfolio will preclude a company's subsequent use of the held-to-maturity portfolio will preclude a company's subsequent use of the held-to-maturity portfolio. Accordingly, the board of directors should be notified of the reasons for such a designation and give its formal approval before the designation change can be made.

e) Marketable equity securities accounting

- *All securities purchases shall be designed as trading securities at the time of purchase.* This policy is intended to avoid the designation of an investment as "available for sale", which would allow management to avoid recording shortterm changes in the fair value of the investment in reported earnings. The policy removes the ability of management to alter financial results by shifting the designation of an investment.
- *All losses on securities designated as available for sale shall be considered permanent.* Accounting rules allow one to avoid recognizing losses on available-for-sale securities by assuming that the losses are temporary. By using this policy to require an immediate write-down on all losses, management no longer has the ability to manipulate earnings by making assumptions that losses are temporary in nature.
- *Available-for-sale securities shall not be sold solely to recognize related gains in their fair market value.* Accounting rules do not allow on-going recognition of gains in the value of available-for-sale securities in earnings until they have been sold, so there is a natural temptation to manage earnings by timing their sale. This policy is designed to set an ethical standard for management to prevent such actions from taking place. In reality, this is a difficult policy to

enforce, since management can find reasonable excuses for selling securities when their unrecognized gains are needed for bookkeeping purposes.

A company may find that its investment policy, while appearing to be reasonable on paper, does not match the realities of the marketplace, resulting in a suboptimal investment portfolio. To bring the investment policy into better alignment with the market, use *backtesting* with the company's historical investment records to see what various changes to the policy would have done to the company's investment policy, especially if performed at regular intervals to account for on-going changes in the market place.

10.7 ACTIVITY

1) When a treasurer is considering an investment option, what factors must he/she consider?
2) Discuss some investment options available for a treasurer.
3) Evaluate the various investment strategies that are available for a treasurer.
4) Discuss the various ways of accounting for investment.
5) What are some of the investment management controls available to a treasurer?
6) Discuss some of the common investment management controls a treasurer can use during investment.

10.8 SUMMARY

Investing cash is a key role of the treasurer, and this unit has provided a great deal of guidance for doing so. In sorting through the large number of strategies, controls, policies, and accounting issues related to investment, the treasurer should particularly keep in mind the following three items:

1. *Investment policy is critical.* If there is a large quantity of cash to be invested, then there is a corresponding large risk of losing it through inappropriate investments. To mitigate this risk, absolutely have a detailed investment policy that is regularly updated to match the changing needs of the company and the investment environment.

2. *Return is least important.* Do *not* let return on investment become the key criterion for an investment strategy, or else the attendant increase in risk could result in a significant loss of principal.
3. *Use all controls.* Because of the large volumes of cash involved, there is a risk of investment fraud, as well as of inadvertently making incorrect investments. Strong controls mitigate this risk.

In short, the treasurer's primary goal in investment management is to guard the principal; a respectable return is nice, but not as critical as ensuring that no cash is lost. The investment strategy should be built around this key underlying goal.

Having discussed Investment management, the next unit will now discuss Clearing and settlement systems.

UNIT 11

CLEARING AND SETTLEMENT SYSTEMS

11.1 INTRODUCTION

Welcome to unit 11! This unit will discuss the characteristics of clearing and settlement system.

11.2 AIM

The aim of this unit is to discuss clearing and settlement systems.

11.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss the characteristics of the clearing and settlement system; 2) Discuss the FED wire, and the Automated Clearing House (ACH) system; 3) Discuss other clearing systems such as CHIPS, ZECH,etc.
4) Discuss the continuous link settlement (CLS) system

11.4 TIME REQUIRED

You are advised to spend at least 3 (three0 hours studying this unit.

11.5 REFLECTION

Attempt to define cheque clearing?

……

……

11.6 OVERVIEW OF THE CLEARING AND SETTLEMENT PROCESS

Clearing is all of the steps involved in transferring funds ownership from one party to another except for the final step which is settlement.

Settlement involves the finalization of a payment, so that a new party takes possession of transferred funds. The treasurer should be aware of these processes in order to understand the timing of payment transfers.

In clearing and settlement systems, the banks for the payer and beneficiary exchange information regarding monetary transfers; the result of this exchange is payments between the banks.

The general concept of clearing and settlement is for the banks of the paying party (the payer) and the receiving party (the beneficiary) to exchange information regarding monetary transfers, resulting in the transfer of funds between the two banks. The banks, in turn, debit the account of the beneficiary. Given the massive volume of such transactions, formal clearing and settlement systems have been installed to streamline the process.

Clearing and settlement systems are generally organized around individual countries or economic regions. Banks located within these areas can have an account with the local clearing and settlement institution; settlement takes place between the accounts of the banks held at the clearing institution. Banks located outside of these areas do not have such an account, and so must use a local bank as a correspondent bank that handles payment instructions on their behalf.

11.6.1 Settlement types

Payment can be on a **gross basis,** where each bank pays the total amount owed. Payments handled through a gross settlement system are more likely to have a requirement for immediate execution, where payment instructions are processed separately for each individual transaction. The cost of gross settlement transactions is high, so individual transactions running through these systems tend to involve larger amounts of funds or be very time sensitive.

Payments can also be on a **net basis,** where a large number of transactions are accumulated and offset against each other, with only the net differential being transferred between banks. Payments handled through a net settlement system usually wait until the end of the day, when all transactions between the banks are summarized and offset against each other by clearing institution then sends the net transfer information to the settlement institution, which executes the transfer of funds between banks.

The clearing institution normally completes its daily summarization process and transmits net transfer information to the settlement institution after the cutoff time of settlement institution. This means that the transfer of funds to the account of the beneficiary bank will be delayed by one business day. A few clearing institutions compile net transfer information to settlement institutions not only before their cutoff times, but several times per day, which allows for settlement speeds similar to those of gross settlement systems. The cost of net settlement transactions is low, so lower-value transactions are usually settled through these systems.

Banks prefer to use net settlement systems because payments processed through them require greatly reduced funds transfers (and therefore considerably less liquidity) than gross settlement systems.

11.6.2 Transaction types

Each clearing and settlement system is primarily designed to handle a certain type of transaction. One such type is the ***high-value payment***, which must be executed immediately. Because of the time-constraint, high value payments cannot wait for end-of-day netting, and so are settled on a gross basis, with an immediate cash transfer between banks. Another payment type is the ***low-value payment,*** which does not require immediate execution and tends to be for smaller amounts. Because of the reduced need for immediate execution, these payments are handled through a net settlement system.

11.6.3 US CLEARING AND SETTLEMENT SYSTEMS

11.6.3.1 FEDWIRE

The Fedwire system is a gross settlement system that is operated by the U.S. Federal reserve and processes large-value items with same day, real-time settlement. Nearly all U.S. banks and the agencies of foreign banks participate in the Fedwire system. If a bank is sending funds through the Fedwire system on behalf of a client company, then the deadline for initiating such transfers is 6 p.m. eastern time. The fee for a Fedwire payment is relatively inexpensive.

11.6.3.2 AUTOMATED CLEARING HOUSE (ACH) SYSTEM

The Automated Clearing House (ACH) system is the net settlement system used for electronic payments in the United States, and is used by most banks in the country. The ACH system is used for large-volume, low-value payments, such as payroll direct deposits, business-to-business payments, dividends, tax payments and social security payments. The transfer of funds from the payer to the beneficiary can take several days, depending on the payer's payment instructions. This system is significantly more complex than the Fedwire system, since it comprises a network of bank associations and privately owned processing entities. The cost of an ACH payment is quite low, usually just a few cents per transaction.

The ACH process flow is for a company to submit an electronic file to its bank (also known as the *originating depository financial institution*), containing payment information; the bank lets these submissions accumulate until the end of the day. At that time, it directly pays any of the authorized items with a book transfer if the recipient has an account with the bank. If not, the bank batches and forwards the remaining payment authorizations to its designated ACH operator (usually a regional branch of the Federal Reserve, or the Electronic Payments Network). The ACH operator collects the ACH submissions from all of the banks in its region and calculates the net settlement amounts that they must pay to each other. The ACH operator then aggregates the remaining transactions involving banks outside of its region, subdivides them by ACH region, and transmits them to the ACH operators responsible for conducting similar processing for their regions. Payments are made to the beneficiary's banks (also known as *receiving depository financial institutions*), which in turn pay the beneficiaries. Payments between banks associated with different ACH operators are settled on a gross basis.

Settlement timing is based on the payment date specified by the payer in the ACH file submitted to its bank. A company can deliver ACH debit instructions that are no earlier than one banking day prior to settlement date, while ACH credits can be delivered no earlier than two banking days prior to the settlement date. If the ACH file is delivered too late to settle on the specified date, then the ACH operator will use the next business day as the settlement date.

11.6.3.3 CLEARING HOUSE INTERBANK PAYMENTS SYSTEM (CHIPS)

CHIPS is a net settlement system that is operated by the Clearing House in New York, which in turn is owned by a group of banks. It settles primarily foreign exchange and Eurodollar trades, with total daily volume exceeding $1 trillion. CHIPS accumulates payment instructions

throughout the day, and then calculates the net payment to be paid by each bank in the system. The net payment is then made through the Fedwire system. The CHIPs system is designed for high-value payments, and can also transmit remittance information along with payment instructions.

The CHIPS process flow is for a payer's bank to send payment instructions to CHIPS, which calculates net positions at the end of the day and sends this information to the Fed. The Fed then debits the paying bank's account at the Fed, and credits the account of the beneficiary's bank at the Fed. A separate instruction from CHIPS to the beneficiary's bank results in the beneficiary's bank crediting the account of the beneficiary.

There is some risk for a bank using CHIPS because it is not being paid until the end of the day, but could make funds available to its customers prior to the end of the day; if a bank owing it money were to fail, the bank would then have no means of recovering the funds already paid to its customers. Banks control this risk by imposing intraday credit limit on their transactions with each other.

11.6.3.4 CHECK CLEARING

A check issued from a payer to a beneficiary can be cleared and settled through one of the several routings. They include:

a) ***Same bank settlement:*** The bank on which a check is drawn may be the same bank used by the beneficiary. If so, the bank simply makes a book entry to shift the inhouse funds from the account of the payer to the account of the beneficiary.
b) ***Correspondent bank settlement:*** A company may receive a check from an entity located outside of its country. If so, the company's bank sends the check to a correspondent bank in the originating country; clearing and settlement occurs in that country, and then the correspondent bank uses an electronic transfer to shift back to the beneficiary's bank.
c) ***Check-clearing institution:*** Checks are physically processed through the Federal Reserve System. The process flow is for the beneficiary (check recipient) to send the check to its bank, which in turn sends the check-clearing institution (CCI), which is frequently the Fed. The CCI aggregates the information on all checks received and sends a payment instruction to the Fed. The Fed then debits the account of the payer's bank at the Fed and credits the account of the beneficiary's bank at the Fed.

Meanwhile, the CCI will notify the banks of both the payer and beneficiary, who debits the payer's bank account and credits the beneficiary's account, respectively.

d) ***Electronic check presentment:*** Either the recipient of a check or its bank can scan a check to create an electronic image, and then send the image to the CCI. Since no transport of actual checks occurs, the clearing process can be several days quicker than traditional paper check clearing.

11.6.3.5 THE CONTINUOUS LINK SETTLEMENT (CLS) SYSTEM

Foreign exchange settlement presents a risk of one party's defaulting before a transaction has been completed because settlement takes place through accounts in the correspondent banks in the countries where the relevant currencies are issued. Because the various national payment systems are located in different time zones around the world, one side of a foreign exchange transaction will likely be settled before the other side of the transaction. For example, dollar payments are settled later than euro payments, which in turn are settled the euro side of the payment before receiving any dollars. If the counterparty were to fail in the midst of this transaction, the transaction initiator would have paid dollars but lost the offsetting euros. This risk is called ***settlement risk.***

To avoid this settlement risk while also speeding up the settlement process, a number of major banks banded together to create the CLS system. The system is operated by CLS Bank International, of which the founding banks are shareholders. Other banks can submit their foreign exchange transactions through these member banks. The following currencies can be settled in the CLS system:

Australian dollar	Israeli shekel	Singapore dollar
British pound	Japanese yen	South African rand
Canadian dollar	Korean won	Swedish krona
Danish krone	Mexican peso	Swiss franc
Euro	New Zealand dollar	U.S dollar
Hong Kong dollar	Norwegian krone	

The CLS system has proven to be a popular one, with system settlements averaging more than $2 trillion per day.

CLS maintains an account with the central bank controlling each of the above currencies. Also, each member bank of CLS has its own account with CLS, which is subdivided into a sub-

account for each currency. The member banks submit their foreign exchange transactions to CLS, which uses a gross settlement system to debit the account of a particular currency, CLS requires that it have sufficient balances in its other subaccounts (less a small margin to account for possible fluctuations in exchange rates during the day) to act as collateral for the debit position. If a member bank's debit position exceeds a present limit, then that bank has to replenish its subaccount in the currency having the debit position.

The CLS settlement process flow is for member banks to send their foreign exchange transaction information to CLS during the day, after which CLS creates a schedule of net payments that the member banks must pay to CLS. CLS then processes both sides of each individual foreign exchange transactions, so that the account of one member bank is debited, while the account of another member bank is credited. CLS processes these transactions on a first-in, first-out basis. If, during the processing sequence, a member bank's cash position with CLS becomes too low, CLS will shunt aside and postpone its remaining transactions until additional funds are provided by the member bank.

After CLS has completed this process, it transfers the updated balances of the settlements back to the accounts that the member banks hold at the central banks in their home countries. Since these payments are the result of the aggregation of a multitude of smaller transactions, they are on a net basis. This processing period must be completed during a five-hour period that covers the overlapping business hours of the participating national settlement systems.

How does CLS impact the corporation? It gives the treasurer exact information about when settlements will occur in various currencies, which previously had been difficult to predict with precision. With better foreign exchange settlement information, the treasury staff can now optimize its short-term investment strategy.

11.6.3.6 ELECTRONIC CLEARING SYSTEM IN ZAMBIA

The Bank of Zambia (BoZ) and Banker Association of Zambia (BAZ) have introduced a special variant of Electronic Funds Transfer (EFT) in the form of electronic clearing system (ECS), which is an up-gradation of the paper-based payment system. Due to the absence of infrastructure, processing is not entirely based on the principles of EFT. It combines:

a) ***Electronic credit clearing:*** It was launched by the Zambia Electronic Clearing House (ZECH) in 2002, to provide an alternative to bulk payment transactions passing through clearing houses which has greatly obviated the need for issuing a large number of paper-based instruments. This system facilitates transactions which have single debit and multiple credits. Details of the transactions are furnished on magnetic media for processing, and the banks are getting credit data on magnetic media duly encrypted.

b) ***Electronic debit clearing:*** This payment system works on the principle of multiple debits and single credits. Consumers willing to participate in the scheme give a mandate to debit their accounts on receipt of a debit clearing advice from the debit clearing system, and the subscriber has an option to indicate a maximum limit for each debit entry. In 2002, the ZECH introduced the ECS-RAPID (Receipt and Payment of Instrument/Document) in Lusaka for payment of service charges to the service provider from consumers. In this system, a MICR (magnetic ink character recognition) reader captures details of payments from a specially designed bill and the schedules are prepared on magnetic media for onward processing.

c) ***Floppy input clearing:*** This is another mode of electronic clearing introduced by ZECH in 2002, for the clearing house managed by it. Participating banks submit their inter-bank claim statement in electronic format on floppies. At the clearing house, this data is consolidated and settlement statements are generated. This is a much more accurate and fast settlement process.

11.6.3.7 NATIONAL ELECTRONIC FUND TRANSFER (NEFT) SYSTEM

With the availability of integrated technology consisting of computers and communication facilities, distances no longer remain a constraint in providing better customer service and expediting the funds transfer mechanism. EFT facilities quick movement of funds through the electronic media. EFT mechanisms involving inter-bank funds settlement at a national level has come up only recently as an aftermath of the recommendations of the BAZ committee on technology issues in the payment and settlement system constituted by ZECH in 2001. The basic infrastructure in various branches/offices, at present, however, is not capable of supporting such a system, which is dependent on sophisticated communication systems integrated with computers. Due to these factors, the EFT system introduced by ZECH is largely built around the existing infrastructure for cheque clearing. The BAZ committee had suggested a hybrid system, both paper and electronic media for message transmission. It had suggested that high value institutional funds transfer should be batched every hour. The batch system has since been implemented in 2002. The ZECH acts as the service provider as well as the system regulator.

The Bank of Zambia has introduced a system called 'The Bank of Zambia National Electronic Funds Transfer System' which may be referred to as 'NEFT system' and shall include the set of procedural guidelines detailed hereunder, for the participating banks and institutions with the required computer system and communication network through which funds transfer operations would take place.

a) **Coverage**

   To facilitate quick transfer of NEFT messages, it is essential that only networked branches of banks are part of the systems. Banks' own network could be used for inter-branch communication.

b) **NEFT characteristics**

   NEFT transactions may be made for amounts inclusive of ngwee component. There is no upper value limit for putting through an individual NEFT transaction.

c) **NEFT-Process flow**

   The parties to a fund transfer under the NEFT system are the sending bank, the sending NEFT service centre, the NEFT clearing centre, the receiving service centre and the beneficiary branch.

d) **Role of customer**

A bank customer (i.e. or originator) willing to avail of the remittance facilities offered by a sending bank shall submit a NEFT application from authorizing the sending bank to debit the sender's account and transfer funds to the beneficiary specified in the NEFT application form.

e) Role of Sending Bank

The sending bank shall prominently display at its premises the cut-off time schedule up to which it shall receive NEFT application forms from its customers. The sending bank branch would prepare the 'Structured Financial Messaging System' (SFMS) message to sending service centre, for processing/data upload.

The sending service centre shall transmit the NEFT SFMS message to the NEFT clearing centre by using the communication network designed by the Bank of Zambia. The SFMS messages would be transmitted to the national clearing centre.

f) Role of Clearing Centre

The National Clearing Cell (NCC), of the ZECH at COMESA centre, COMESA point, Lusaka, will be the data processing NEFT clearing centre. SFMS messages generated for banks will be transmitted to the receiving service centre of each of the receiving banks using SFMS after data validation at the receiving NEFT Clearing Centre.

g) Role of Receiving Bank

On receipt of the NCC SFMS message, the receiving NEFT service centre shall process these files and forward them to the destination branches using SFMS.

Alternatively, the receiving NEFT service centre may use the 'Straight Through processing Interface' (STPI) available in SFMS and upload these SFMS messages to their internal banking solutions directly, to give the beneficiaries instantly on the same day by crediting the specified account of the beneficiary or otherwise placing funds at the disposal of the beneficiary.

h) Acknowledgement for NEFT

No acknowledgments are envisaged under the NEFT scheme. A message that is not returned unaffected before the next settlement day is treated to have been completed and credit afforded to the beneficiary's account by the beneficiary branch. It is, therefore, vital that unaffected credits are re-transmitted back as fresh NEFT transactions immediately.

i) Sender to be advised in case of refund

If the beneficiary specified in the sender's payment order fails to get payment through the NEFT system for some valid reasons, the sender shall be informed immediately

after the sending bank gets the returned NEFT. The sending bank shall also arrange to make payment to the sender by crediting the account of the sender, or otherwise placing funds at the disposal of the sender.

j) Beneficiary bank to advise the beneficiary of the payment

After crediting the account of the beneficiary, the beneficiary bank shall advise the beneficiary of the payments made. The statement of account, pass book entry or any online messaging system shall indicate briefly the source of funds as well. In case of a holiday at the beneficiary branch, they have to effect credit as on the same day, or latest at the commencement of business on the next working day.

h) Settlements for NEFT transactions

Every participating bank shall open and maintain in the NEFT centre, Lusaka, a settlement account for settlement of payment obligations arising under funds transfer executed under the NEFT system.

i) Advantages of the NEFT

(1) NEFT facilitates an efficient, secure, economical, reliable and expeditious system of funds transfer and clearing in the banking sector throughout Zambia, and

(2) NEFT relieves the stress on the existing paper based funds transfer and clearing system.

11.6.3.8 REAL-TIME GROSS SETTLEMENT (RTGS) SYSTEM

RTGS is an electronic payment environment where payment instructions processed on a 'continuous' or 'REAL-TIME' basis; and settled on a GROSS or individual basis without netting the debits against credit payments so effected are 'final', and 'irrevocable' settlement is done in the books of the central bank-the ultimate liquidity depository of the country. RTGS uses the SFMS.

- Each bank has a single gateway interface called Participant Interface of PI for the RTGS system. The payment message/enquiry/clearing settlement originates from the participant's host system (branch/treasury).
- The message is passed on by the participant and clearing system interface (PI) to inter-bank funds transfer processor (IFTP), which acts as a broker. Communication between the PI and RTGS systems is through IFTP only and not directly.
- IFTP stores the message and in case of payment message constructs, the settlement message containing a core subset of the information required for settlement (settlement message) is routed to the RTGS System at the Bank of Zambia.

- On receipt of the subset, the Bank of Zambia checks whether the sending bank has sufficient covering funds in its account and informs the IFTP of the status of transfer, queued or settled.
- The settlement message is processed and fate is advised to IFTP.
- Based on response, IFTP enriches the message received from the RTGS system by adding back the corporate details and sends settlement advice to both the originating and beneficiary participant in case of successful settlement or failure advice to the originating participant in case of failed settlement.
- Thus, business information, which is exchanged between the sending and receiving bank, is not known by the settlement agent. The ZECH therefore, acts only as a settlement agent. **a) RTGS Business Day**

  The RTGS business day is divided into four phases:

  1) *RTGS Open Phase:* All normal transactions are accepted and processed.
  2) *IDL Shut Phase:* All transactions are processed as in RTGS open phase but no new IDL will be requested.
  3) *IDL Close Phase:* Restriction on fresh IDL continues and no transaction which debits a member with an outstanding IDL will be settled.
  4) *RTGS Close Phase:* Transactions reversing outstanding IDL will only be permitted. Outstanding multilateral net settlement batch (MNSB) transactions for settlement (provided the member is designated as a clearing entity) and MNSB return transactions are also permitted.

  The existing clearing system despite RTGS's full operations has also continued.

b) Future perspectives

There are more than 250 bank branches in Zambia, of which over 70% are in semiurban and urban areas (Simataa, 2010). All the branches have attained hundred per cent computerization. RTGS is available in all these branches and the national electronic funds transfer through individual banks' head offices. This would require up-scaling of the RTGS' operations in terms of capacity and speed, for which efficient and effective continued use of the system is sustained.

11.6.3.9 CHEQUE TRANCATION SYSTEM (CTS)

In 2013, Bank of Zambia in conjunction with the Zambia Electronic Clearing House (ZECH), introduced a clearing system called **Cheque Truncation System (CTS).**

Zambia Electronic Clearing House Limited (ZECHL) is a joint venture between the Bank of Zambia (BOZ) and the Bankers Association of Zambia (BAZ). ZECHL manages the interbank clearing system on behalf of the banking industry.

The reason for the introduction of the CTS was part of the two stakeholders' ongoing reform programme of the national payment system with regard to cheque processing and electronic funds transfer processes. ZECHL together with the banking industry embarked on the implementation of CTS project, which included enhancements to the Direct Debit And Credit Clearing (DDACC) stream.

Cheque truncation System (CTS) is a system that involves the process of clearing cheques using cheque images between banks as opposed to using physical cheques presented for payment in a bank by individuals or corporate bodies. (ZECHL, 2013).

a) **General benefits of cheque truncation**
   1) The clearing period has been shortened and standardized across the country to one clearing day (T+1) unlike the previous (T+3) days for local cheques and more days for up-country cheques (cheques drawn on other towns). This means that value to cheques is now given a day after the date the cheque is deposited and thus facilitates early access to funds by the customers and that enhances service delivery.
   2) It has improved the velocity of cash flow between transacting parties in the economy owing to reduced clearing periods.

3) It has reduced risks associated with manual handling and physical movement of cheques between banks.

b) How to identify a CTS compliant cheque

Apart from the visible standard features of the cheque such as the name of the bank/branch on which the cheque is drawn, date, amount, bank sort-code, name of drawer, crossing, MICR Code line; one should look out for the following distinct features:

1. **Watermark:** This is a translucent image in the cheque that is seen when the cheque is held against a light source.
2. **Check Digit Value:** These are the last 2 digits on the MICR code line after the space.

Cheques without these two features to be accepted for deposit after the cut-over date was scheduled for 1st February, 2013. However, non-CTS compliant cheques issued prior to cut-over date of 1st February, 2013 were not going to be accepted for clearing up to 31st July, 2013, after which non-CTS complaint cheques would not be completely phased out of the system. However, due to some teething issues in its implementation delayed and full implementation only occurred in 2014.

c) Advantages of using CTS compliant cheques

1. CTS compliant cheques are processed faster than non-CTS compliant cheques.
2. Customers who deposit CTS compliant cheques receive value for their money faster than those who deposit non-CTS complaint cheques.

d) Customer responsibilities

As a customer, you have a role to play to ensure the system services you efficiently and effectively.

1) Handle the cheques meant for deposit carefully to avoid damage. **Do not** fold, or handle cheques with greasy or soiled hands or store them in a humid place.
2) Check that all cheque details such as date, amount in words and figures are valid before depositing.
3) Fill in separate deposit slips i.e. CTS compliant cheques alone on one deposit slip and non-CTS compliant cheques alone on another deposit slip.
4) When writing out your cheque, use **Black ink** only. Do not use any other colours as they tend to be faint during image capture.
5) Do not write anything in the MICR code line space.

6) Do not mask/cover any part of the cheque with masking or cello tape because doing so may obscure the essential details of the cheque thereby delay the processing of the cheque and inconvenience the payee.
7) Keep the cheque leaves and books in a safe place at all times to prevent theft which may lead to fraudulent use of the cheques.
8) Do not issue cheques on an insufficiently funded account, as it is a criminal offence under the National Payments Act of 2007.

11.7 ACTIVITY

1) Discuss how cheque clearing works?
2) Discuss the FED wire and ACH system of clearing?
3) Compare the CHIPS and ZECH clearing systems.
4) What is Continuous Link Settlement (CLS) system?
5) What is RTGS?
6) Explain how the Cheque Truncation System (CTS) works.

11.8 SUMMARY

With the exception of the CLS system, this unit has described only the clearing and settlement systems operating within the United States and Zambia. Similar systems operate in other countries and economic regions, and also fall into the general categories of high-value, gross settlement systems or low-value, net settlement systems. If a treasurer's company transacts significant business in region outside of the United States, it may be worthwhile to research the process flows of the clearing and settlement systems used in those regions.

The next unit will discuss treasury systems.

UNIT 12

TREASURY SYSTEMS

12.1 INTRODUCTION

Welcome to Unit 12! This unit has been designed to discuss Treasury systems.

12.2 AIM

The main aim of this unit is to discuss different treasury systems available for a treasurer.

12.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss the treasurer's technological needs;
2) Discuss the treasury management system;
3) Demonstrate how the SWIFT connectivity works.

12.4 TIME REQUIRED

You are advised to spend at least 3 (three) hours studying through this unit.

12.5 REFLECTION

What technology do you think is used by the treasurer?

.....

.....

……...

12.6 TREASURY SYSTEMS- AN INTRODUCTION

The treasurer requires information that is not normally available through a company's standard accounting system, or even from its enterprise resources planning (ERP) systems. Even though an ERP system is designed to aggregate all of the information from a variety of external sources regarding investments, foreign exchange positions, interest rates, and so on. Consequently, treasury systems are needed so as to integrate information from a variety of sources, yielding real-time information that the treasury staff can use to efficiently perform their tasks. This unit therefore will discuss the treasurer's technology needs, and whether to install a treasury management system or not.

12.6.1 TREASURER'S TECHNOLOGY NEEDS

The treasurer is in a difficult position of requiring information from many sources, most of which are not required by any other company manager. Since treasury is relatively a small department, it may not command the resources that larger departments command. Consequently, the treasurer must frequently prioritize information needs. While priorities may vary by company, the following list establishes a reasonable set of priorities, in declining order:

1. **Cash position:** The treasurer's overriding obligation is to ensure that the company has adequate cash to fund its operations. Thus, the following technology needs are critical:
    - Cash book balance tracking.
    - Multibank reporting of balance information.
    - Cash pooling management.
    - Cash forecasting and reconciliation of actual to projected cash flows.
    - Funds transfer capability.
    - Investment management, including money market dealing.
    - Interest income calculation.
    - Debt management.
    - Interest expense calculation.
    - Payment processing.
    - Intercompany transaction settlement with notional accounts.
2. **Foreign exchange transactions:** Some companies have so little foreign business that foreign exchange transactions are negligible, but larger firms with established treasury

operations will likely need the ongoing purchase and sale of multiple currencies. Thus, the following technology needs arise:

- ❖ Rate feeds from Bloomberg or Reuters.
- ❖ Intercompany netting capability.
- ❖ Foreign exchange forecasting and position analysis.
- ❖ Foreign exchange bid summarization.
- ❖ Foreign exchange deal-making capability.
- ❖ Foreign exchange confirmation processing.

3. Hedging: For those companies that elect to hedge their interest rates or foreign exchange positions, consider the following technology requirements:

- ❖ Rate feeds from Bloomberg or Reuters.
- ❖ Exposure modeling capability.
- ❖ Hedge deal-making capability.
- ❖ Hedge confirmation processing.
- ❖ Hedge documentation capability for Statement of Financial Accounting Standards (SFAS) 133 requirement.

In all cases, transactions generated by the treasury system should automatically create accounting entries that are interfaced directly into the corporate general ledger.

In addition to these three core areas, the treasurer also needs a reporting system that reveals the global cash position, investment portfolio, debt portfolio, cash forecast, and foreign exchange transactions. The system should also provide mark-to-market valuations, scenario analysis, and counterparty risk summaries. This information can be presented through a customized online dashboard that is updated in real time.

The treasurer's technology needs also to extend to the efficiency of and control over treasury activities. Thus, the following requirements should be considered:

a) Efficiency Issues

- *Minimize data entry:* The system should never require the manual entry of a transaction into the system more than once, and preferably should involve automated data collection and posting, so that no manual entry is required at all.
- *Work flow processing:* If supervisory approval is required, the system should electronically route the pertinent transaction to the correct supervisor for approval.

b) Control issues

- *Audit trail:* All treasury transactions should result in a clearly defined audit trail that identifies who made a transaction and the date, amount and accounts impacted by the transaction.
- *Segregation of duties:* The system should limit access to certain modules and require approval of key transactions.
- *Warning indicators:* The system should automatically notify users if transaction confirmations have not been received, if hedging policies are being violated, if there are negative cash balances, and so on.

12.6.2 TREASURY MANAGEMENT SYSTEM

All of the technology requirements noted in the last section are available in treasury management systems today. However, these systems are expensive.

Why should treasurers spend so much money in installation of treasury technologies? This is because it automates so many of the rote treasury tasks. For example, if an employee buys an investment and runs the transaction through this system, it will create a transaction for the settlement, one for the maturity, and another for the interest. It will then alter the cash forecast with this information, and create a wire transfer to send the money to the financial intermediary that is handling the purchase.

A treasury management system will be less expensive and easier to install if the treasurer reduces the number of outside banking relations that the company has, so that there are fewer customized interfaces to construct. Since it can take multiple months to unwind a banking relationship, this issue should be dealt with well before the system's scheduled installation date.

A key issue with a treasury management system is its complexity, which can be substantial. There is a significant risk that the system will not be fully utilized, since employees will gain expertise only in those specific functions that they were performing prior to the system installation. To avoid this issue, schedule thorough training for all affected personnel in *all* functions of the system, and also schedule an audit of system usage several months after installation, to determine which features are not being used. The likely result of this review will be targeted training for underutilized functionality.

What if a company cannot find a cost-beneficial way to acquire a treasury management system? One solution is to examine the cost of a failed manual system, such as a spreadsheet error. If such an error is discovered, its cost may be justification for acquiring a system. Another option is to enlist the internal auditors, who can argue that the risk of a transactional error may result in a Sarbanes-Oxley control breach that would have to be revealed in the company's public filings. A third alternative is to explore a third-party hosting solution, which substitutes the large up-front capital cost of a treasury management system for an ongoing monthly fee.

If none of these alternatives work, the treasurer should at least attempt to install those portions of a treasury management system that reduce the amount of data entry and repetitive work by the treasury staff, thereby giving them more time to work on such value-added activities as risk management, fund raising, and hedge analysis. As the size of the company gradually increases, it may then be possible to add to the functionality of the existing system.

12.6.3 SWIFT CONNECTIVITY

The Society for Worldwide Interbank Financial Telecommunication (SWIFT) operates a worldwide network that banks use to exchange standardized electronic messages that are known as SWIFT MT codes. Most banks around the world are part of the SWIFT system this includes Zambian banks as well. The SWIFT network is highly secure and is designed strictly to transport messages between participants-it does not provide a clearing or settlement service.

Companies are now able to access the SWIFT network by any one of the four methods as follows:

a) **Standardized corporate environment (SCORE):** Under this approach, a company can communicate with all member banks in a closed user group. Companies allowed to use this method must be listed on selected stock exchanges in specific countries. SWIFT invoices companies directly for their message traffic. This is the most efficient method, because users have direct access to nearly all banks.
b) **Member-administered closed user group (MA-CUG):** A company can join a separate MA-CUG for each bank with which it wishes to communicate. Each MACUG is administered by a bank, rather than SWIFT. The bank running each CUG will invoice member companies for their message traffic. This approach may call for membership in multiple MA-CUGs, which is less convenient to all types and sizes of companies.

c) **Alliance lite:** SWIFT has made this method available to smaller companies having low transaction volumes. It allows them to use either a manual browser-based payment entry system or to integrate directly into their treasury management systems.

d) **SWIFT Bureaus:** Third-party providers have set up their own access to the SWIFT network and allow companies access through their systems for a per-transaction fee. This approach avoids the need for any in-house systems maintenance, but connectivity to any in-house treasury management systems is likely to be limited.

In none of the preceding access methodologies is a company allowed to deal directly with another corporation; it can only send messages through bank intermediaries.

Access to the SWIFT network is important for larger companies, because they can link their treasury management systems directly into the SWIFT network. By doing so, they avoid having to establish individual interfaces with the reporting systems of all the banks with which they do business, and instead can rely on a single standard messaging format to initiate transactions with and acquire information from bank accounts all over the world.

Each SWIFT MT code used to send messages within the SWIFT network contains a standard set of information fields. Thus, a different SWIFT MT code is used for each type of transaction. For example, a company can issue an MT 101 to move funds, an MT 104 to debit a debtor's account, an MT 300 for a foreign exchange confirmation, an MT 320 for a loan confirmation, and an MT 940 to request bank account information. Given the high degree of standardization, these messages can be automatically generated by a company's treasury management system and transmitted through SWIFT, while all incoming messages can also be dealt with by the treasury management system in a highly automated manner.

In summary, there are multiple ways available for a company to gain access to the SWIFT system, which it can then integrate into its treasury management system. Doing so streamlines a number of treasury transactions, which makes the entire system more cost-effective to operate.

12.7 ACTIVITY

1. Discuss the treasurer's technological needs.
2. Demonstrate how the SWIFT connectivity works.

12.8 SUMMARY

This unit has been able to outline some of the technologies the treasurer can use. It should be remembered that a company can subsist on spread sheetbased systems if it engages in a trifling number of treasury transactions. However, it will soon find with increased volume that the volume that the amount of rote data entry labour and outright errors associated with such systems will eventually call for the implementation of formal treasury systems. These systems operate best if the treasurer insists on a high level of staff training, as well as some reduction in the number of external banking relationships. Also, the increased availability of the SWIFT network to corporations makes it possible to engage in a high degree of transactional automation, thereby giving the treasury staff more time for tasks that better utilize their skills.

The next unit will discuss cash transfer methods.

UNIT 13

CASH TRANSFER METHODS

13.1 INTRODUCTION

Welcome to unit 13. The unit is designed to discuss cash transfer methods.

13.2 AIM

The aim of this unit is to discuss cash transfer methods.

13.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss cheque payments;
2) Discuss wire transfers;
3) Discuss procurement cards;
4) Discuss cash payments; 5) Discuss cash transfer controls;
6) Discuss cash transfer procedures.

13.4 TIME REQUIRED

You are advised to spend at least 3 (three) hours reading through this unit.

13.5 REFLECTION

List some of the ways in which cash can be transferred.

……
……
……

13.6 CASH TRANSFER METHODS-AN INTRODUCTION

A treasurer should understand the implications of different methods of transferring cash to or from a company since there are significant differences in the costs and cash flow speed of each one. Further, the level of manual processing and related controls is significantly different for each kind of finance and accounting functions. This unit intends to give the treasurer a thorough understanding of each form of cash transfer; when it should be used; what it will cost; and any required policies, procedures, and controls.

13.6.1 CHEQUE PAYMENTS

A cheque payment is made on a paper document, which has traditionally been physically routed from the payer to the payee, to the payee's bank, and then back to the payer's bank. The number of routings and the need for physical handling of the check results in significant delays in the transfer of cash between the principal parties.

The vast majority of cheques are issued directly by companies. However, they may also be issued directly by a bank, which is called *a bank cheque or bank draft.* A bank cheque is a payment on behalf of the payer, which is guaranteed by the bank (and therefore of value to the payee). The bank can safely issue this guarantee because it immediately debits the payer's account for the amount of the cheque, and therefore has no risk. Not only is this a safe transaction for the bank, it is also beneficial to the bank, since the bank has ownership of the funds from the time when it debits the payer's account to when the money is eventually paid to the payee (which could be several weeks, depending on when the payer elects to send the cheque to the payee).

When a company has written a large volume of cheques that have not yet cleared, the available cash balance shown by the company's bank will be larger than the company's ledger balance. If the treasurer can reliably predict how long it will take for the cheques to clear, it is then possible to invest some of the cash that is available due to un-cleared cheques.

If a company has significant cash holdings, then it may be worthwhile to spend time investing in float-related funds. However, maintaining an abnormally small cash balance

requires active float monitoring on a daily basis. If there is a gap of even a single day in float monitoring, then the company will very likely not have sufficient funds for all presented cheques, and will incur expensive account overage fees.

a) Remote deposit capture

Remote deposit capture allows a company to avoid the physical movement of received cheques to its bank. Instead, one can use a special scanner and scanning software to create an electronic image of each cheque, which it then transmits to the bank. The bank accepts the online image, posts it to the company's account, and assigns funds availability based on a predetermined schedule.

The key benefit of remote deposit capture is the complete elimination of the transportation float that arises when shifting cheques from the company to the bankwhich can be a considerable delay. If the delivery person misses the bank's cutoff time, then an entire extra day is added to this float. However, since remote deposit capture typically has extended processing hours, there is far less risk of incurring this extra delay.

Another benefit is that a company no longer needs a bank that is physically located near the company location. Instead, it can consolidate its banking relationships and use just a single provider, who may be located anywhere in the country.

The company using such a system should be aware that it is financially liable for the accuracy of the information they enter into the system.

13.6.2 WIRE TRANSFERS

A wire transfer sends funds to the recipient's bank account more rapidly than any other form of payment, and is the standard form of international payment. To initiate a wire, the treasury staff must send the following information to its bank:

- Sending company's name, address, and account number;
- Recipient's name and account number;
- Recipient bank's name and bank identification number; Amount of the payment.

To send a wire transfer, a company can access a bank's web site to enter the wiring instructions, fax the information to the bank, or call in the information. No matter what method is used, the wiring information must be received by the bank prior to its *cutoff time.*

Any wiring instructions received after this time will be delayed until the beginning of the next business day.

A wire transfer between countries may not be credited to the account of the recipient for several days. Part of the problem is a bank-imposed delay in the value date. In addition, if the initiating bank does not have a direct correspondent relationship with the receiving bank, then it must route the payment through a third bank that does have a correspondent relationship with the receiving bank, which takes time. Also, some banks still manually review incoming wires and apply them to beneficiary accounts which take more time. Finally, the time of day when a wire is initiated can be a factor; if it begins near the close of business, then an extra day may be added to the transaction.

There are several ways to improve the situation. One is to use the same bank at both ends of a transaction. This makes the transfer a simple *book transfer*, which passes immediate value dating to the recipient. The receiving bank may be making its profits on delayed value dating; if so, specify a different receiving bank.

When a company sends funds internationally via a wire transfer, the recipient will likely be charged a stiff foreign exchange conversion fee by the receiving bank. To avoid the fees, the paying company can offer the recipient to pay in the recipient's currency in exchange for a lower price. The payer can then shop among several foreign exchange providers for the best exchange rate. At a minimum, a company should have its international customers quote in both the company's home currency and the supplier's local currency, so that the company can see if it can achieve a better exchange rate through its foreign currency provider.

Both the issuing and receiving banks in a wire transfer transaction charge high transaction fees. In particular, the payment recipient is charged a *lifting fee*, which its bank imposes for handling the transaction. This fee is deducted from the amount of the funds being transferred, which means that the amount ultimately received is always somewhat less than the amount expected, which interferes with account reconciliation.

If a company engages in a large number of wire transfer transactions, these fees can add up to a significant expense. To avoid this, some countries like Zambia have created local electronic payment facilities such as DDACC and RTGS (discussed in detail in unit 11 of this module).

13.6.2.1 Letters of Credit

When dealing with counterparties in other countries, it is difficult to evaluate their financial condition, and the legal systems in their countries make it more difficult to collect overdue receivables. For these reasons, exporters are more concerned with using cash transfer methods that have a high probability of payment.

A well- established method for doing so is the *letter of credit.* This is an arrangement where the importer's bank (the *issuing bank*) formally authorizes an obligation to pay the exporter's bank during a specific period of time, assuming that several documented conditions have been met. The documents that must be presented to the issuing bank include an invoice and proof of delivery. In addition, a certificate of insurance may be required, as well as a quality certificate. The issuing bank creates the letter of credit, and therefore has control over its terms.

When all terms of the letter of credit have been completed, the exporter presents all required documents to its bank, which pays the exporter the amount noted in the letter of credit. If the exporter's bank is un-willing to make payment, then it is called the *advising bank*, and merely passes along the documentation to the issuing bank, which is now designated as the *nominated bank*, and makes the payment. If the exporter is uncertain of the reliability of the nominated bank, it may ask its own bank to confirm the letter of credit. If the bank agrees, then it is designated as the *confirming bank*, and enters into an agreement to pay the exporter immediately following receipt of the required documents. Upon completion of this step, the letter of credit is said to be a *confirmed* letter of credit. A bank will charge a fee in exchange for being the confirming bank. This fee can be quite high if a banker feels that there is a significant risk of the issuing bank's failing to pay, and a banker may refuse confirmation entirely if the risk is perceived to be excessively high.

The key party in making a letter of credit work is the issuing bank, which is guaranteeing the credit. In order to do so, this bank may block out a portion of its credit to the importer, which it will not release until the letter of credit transaction has been completed and the bank has been paid by the importer.

The letter of credit is useful for the importer, who receives proof of title to the goods, as well as evidence that goods have been shipped. However, the real benefit goes to the exporter,

who obtains a guarantee of payment from a bank, which is a distinct improvement over a guarantee of payment from the importer.

The letter of credit requires significant administrative time and manual effort by the importer, exporter, and their banks. There is also a significant risk that a bank will refuse to pay under the terms of a letter of credit if there are any discrepancies in the submitted documents. For these reasons, there has been ongoing pressure for a number of years to use other, more streamlined ways to transfer cash.

13.6.3 PROCUREMENT CARDS

A company can make smaller-scale payments with a procurement card program. This can involve the use of *debit cards,* which deduct cash directly from a company's bank account, but more commonly employ *credit cards*. Under this system, the bank managing the procurement card program will bill the payer on a monthly basis for all charges made during the month, while remitting funds to the payee within a few days of each charge. If the payer pays the monthly bill late, then the bank charges interest on the open balance. A procurement card program is an excellent tool for any company making payments to its suppliers, since it circumvents the lengthy and expensive process of issuing purchase orders, matching receiving documents to supplier invoices, and making cheque payments.

If a company accepts payments via credit cards, then it must sign a credit card agreement with a credit card processor, under which it agrees to be charged a processing fee in exchange for being in the program. The fee charged varies considerably by type of credit card, as well as by the amount of card volume that a company transacts. Typically, the company is significantly lower for debit card transactions.

Because of the fee charged to the recipient of funds under a credit card program, the payee is not usually willing to accept large payments-it would seriously impact profits. Instead, procurement cards are normally used for smaller, high-volume transactions, especially in a retail environment.

Procurement cards are less commonly used for international payments, because the payer is also charged a fee for any conversions of foreign currencies back into the credit card processor's home currency.

13.6.4 CASH PAYMENTS

Inbound cash payments tend to be for very small transactions, though possibly in very high volume, especially in retail situations. However, business-to-business cash payments are not common.

If a company receives large volumes of cash, it is usually more secure to hire a professional money transporter to move the cash to the bank on behalf of the company. To do so, the company locks the cash into a cash box, which the transporter takes to the bank. The receiving bank also has a key for the cash box; the bank staff opens the cash box and then counts and deposits the cash for the company.

Cash is bulky, requires significant controls to maintain on-site, and does not earn interest income until deposited. Given the extra cost of counting it at the bank, it is also expensive to deposit. Consequently, companies have a strong incentive to avoid both paying with or accepting cash.

13.6.5 CASH TRANSFER CONTROLS

Cash transfers can subject a company to a considerable risk of loss, and so require a broad array of controls, which vary considerably by transfer method.

13.6.5.1 Controls for cheque payments

The control system for cheque payments assumes that a fully documented packet of payables information has already been created, which contains receiving, purchase order, and supplier invoice information. With this information in hand, the following controls should be used to create and monitor cheque payments:

- *Remove cheque stock from locked storage:* Unused cheque stock should always be kept in a locked storage cabinet. In addition, the range of cheque numbers used should be stored in a separate location and cross-checked against the cheque numbers on the stored cheques to verify that no cheques have been removed from the locked location.
- *Restrict access to cheque-signing equipment:* If a company uses any form of computerized cheque-printing equipment, it may be necessary to lock down all access to it. This can include any printers in which cheque stock is maintained, signature plates, and signature stamps.

- *Require a manual signature on cheques exceeding a predetermined amount:* This control is useful when signature plates are used for smaller cheque amounts. When signature plates are used, there is no longer a final review of payments before they are mailed. Therefore, requiring a "real" signature for large cheques adds a final review point to the payment process.

- *Check signer compares voucher package to cheque:* The cheque signer must compare the backup information attached to each cheque to the cheque itself, verifying the payee name, amount to be paid, and the due date. This review is intended to spot unauthorized purchases, payments to the wrong parties, or payments being made either too early or too late. This is a major control point for companies not using purchase orders, since the cheque signer represents the only supervisorylevel review of purchases.
- *Implement positive pay:* A strong control that virtually eliminates the risk of an unauthorized cheque's being cashed is "*positive pay".* Under this approach, a company sends a list of all cheques issued to its bank, which clears only cheques in this list, rejecting all others. However, this approach also calls for consistent use of positive pay concept, since any manual cheques issued that are not included on the daily payments list to the bank will be rejected by the bank.
- *Use electronic payments:* There are several types of fraud that employees can use when a company pays with cheques, while outside parties can also modify issued cheques or attempt to duplicate them. This problem disappears when electronic payments are made instead. In addition, the accounts payable staff no longer has to follow up with suppliers on uncashed cheques, or be concerned about remitting payments to state governments under local escheat laws since there are no cheques.
- *Reconcile the cheque account everyday:* This approach ensures that any fraudulently modified cheques or cheques not processed through the standard accounting system will be spotted as soon as they clear the bank and are posted on the bank's website

13.6.5.2 Controls for remote deposit capture

Remote deposit capture eliminates the need to physically transfer cheque payments to the local bank for deposit. However, the replacement process of converting the cheques to an electronic format still requires some controls. These include:

- *Verify receipt by the bank:* When processing cheques through a remote deposit capture system, always print out a verification statement that the image was successfully sent to and received by the bank, and attach this verification to copies of the cheques for storage. This ensures that the deposit has taken place.
- *Train for handling of foreign cheques:* It is important that staff are trained for handling foreign cheques.

13.6.5.3 Controls for electronic payments

Electronic payments can involve large amounts of money, and so require a stringent set of controls to mitigate the risk of loss. Controls include:

- *Restrict access to master vendor file:* For those electronic payments being made automatically by the accounting software, it is important to keep tight control over changes to the vendor master file, since someone could access the file and alter the bank account information to which payments are being sent.
- *Require signed approval document for manually initiated electronic payments:* In a high-volume payment environment, nearly all electronic payments are routed through the accounting software, which handles the payments automatically. However, since a manually initiated payment falls outside the controls already imposed on the regular accounts payable process, the addition of an approval document is mandatory, preferably requiring multiple approval signatures.
- *Require password access to payment software:* It is necessary not only to enforce tightly limited access to the software used to initiate electronic payments, but also to ensure that passwords are replaced on a frequent basis. This is a critical control and should be rigorously enforced.
- *Require additional approvals:* Additional approvals are useful for larger payments. Another approval should be required whenever a new supplier is set up for electronic payment, since this is an excellent spot to detect the initiation of payments to a shell company. The additional approval could be linked to the generation of a credit report on the supplier, to verify its existence as a valid business entity. The highest level of control over electronic payments would be to require dual approvals for *all* such payments, though this may prove too onerous for ongoing business operations.
- *Require an end-of-day payments review:* A standard detection control should be to have a third party who is unrelated to the electronic payments process review all

payments made at the end of each day. This review should encompass a comparison of authorizing documents to the actual amounts paid, as well as verification that payments are made to the correct supplier accounts.

The preceding set of controls relate only to the process of either issuing an electronic payment to a supplier or of allowing a supplier to debit the company's bank account, and not address problems such as unauthorized account debits and the sheer size of potentially incorrect or fraudulent payments. The following controls address the additional issues:

- *Impose on outright debit block on all company accounts:* If the company does not wish to incur any risk of having a third party initiate a debit transaction against one of its bank accounts, it can impose a blanket debit block on those accounts, thereby preventing debit transactions from posting to a company account. A result of this control is that all electronic payments be initiated solely by the company, not by its trading partners.
- *Request a daily cumulative limit for authorized trading partners' debits:* Even if a company has installed a system like a DDACC system where debit filters authorize only certain suppliers to initiate a DDACC debit, there is still a risk that employees of an authorized supplier could fraudulently initiate a very large DDACC debit. To mitigate this risk, see if the company's bank can impose a daily cumulative limit on those suppliers who are allowed to initiate account debits.
- *Request notification of duplicate debits*: If a supplier initiates a debit transaction that is identical to one posted in the past day or two, there is an increased risk that this could be a duplicate charge. To reduce the risk of this problem going undetected, have the bank notify the company whenever a duplicate debit is posted or (even better) prior to posting.
- *Use a separate bank account as a source of electronic payments:* Because there is a risk of making extremely large fraudulent electronic payments, a useful control is to use a separate bank account as the source of electronic payments, with cash levels kept only high enough to fund those electronic payments made during the normal course of business, based on historical patterns. If an extremely large electronic payment is due to be made, this should initiate additional perusal of the transaction before additional cash is shifted to the account from which the payment will be made. To achieve a greater level of control, the person responsible for shifting funds into the electronic

payments account should not be the same person who initiates or approves electronic payments.

13.6.5.4 Controls for Letters of Credit

The greatest risk with a letter of credit is not being paid. The documents that a payee must present in order to be paid are usually prepared and controlled by other parties, and the terms of those documents may be very strictly defined, thereby delaying payment. All controls related to letters of credit are designed to improve the odds of being paid. The following controls should be implemented prior to finalizing a letter of credit:

- *Avoid approval by the buyer:* There should not be any documentation requirement under the letter of credit where the document must be signed by someone who is under the control of the buyer, since the buyer, since the buyer can use this requirement to effectively block payment.
- *Avoid tight dating:* The letter of credit contains dates by which goods must be shipped and documents presented. The seller should add a large margin of error to these dates to ensure that they can be met. If they cannot be met, then the seller should avoid shipping anything until the dates have been revised and agreed to in writing by all parties. In addition, the expiry date of the letter of credit should be sufficiently far off to allow for several rounds of document resubmissions.
- *Minimize presentment documents:* If there are fewer documentation requirements under the letter of credit, then it will be easier for the payee to obtain payment. This must be negotiated with the buyer prior to issuance of the letter of credit.
- *Require a financially strong issuing bank:* The issuing bank assumes the risk of the buyer's insolvency, so the payee insists that a bank with a strong credit rating be the issuing bank.
- *Require confirmation by a confirming bank:* The issuing bank is in suspect financial condition, or if the political stability of the issuing bank's country is questionable, then a confirming bank can confirm the payment obligation. If the confirmation cost is too high or cannot be obtained at all, then try to have the letter of credit issued outside of the political area that is causing the problem.
- *Require payment prior to buyer possession:* The letter of credit should require that the seller is paid before the buyer takes possession of the goods. Otherwise, if there are discrepancies in the presented documents, a buyer that is already in possession of the

goods will be less likely to waive the discrepancies, thereby delaying payment to the seller.

The following control involves the timing of shipments under a letter of credit to ensure that the buyer will pay in the event of an amendment to the letter of credit:

- *Complete amendments prior to shipment:* If there are issues with the letter of credit that require amendment by the buyer and seller (and agreement by the issuing and confirming banks), then verify that these changes have been completed and approved in writing *prior to* shipping the goods to the buyer. Otherwise, it is possible that the bank will not allow the amendments and refuse to pay, even though the goods have been shipped.

The following control involves the creation of the seller's invoice to the buyer, to ensure that it will be acceptable to the bank.

- *Match the description of goods on the invoice to the letter of credit:* A bank will not pay under the terms of a letter of credit unless the description of goods on the invoice exactly match those on the letter of credit. The invoice preparer should be well trained on this issue, and it may also be useful to have a second person verify the invoice information.

13.6.5.5 Controls for procurement cards

- *Enter receipt in procurement card transaction log:* When employees use procurement cards, there is a danger that they will purchase a multitude of items and not remember all of them when it comes time to approve the monthly purchases statement. By maintaining a log of purchases, the card user can tell which statement line items should be rejected.
- *Reconcile transaction log with monthly card statement:* Each cardholder must review his or her monthly purchases, as itemized by the card issuer on the monthly card statement.
- *Fill out missing receipt form:* Each card user should attach original receipts to the statement of account, in order to verify that they have made every purchase noted on the statement. If they do not have a receipt, they should fill out a missing receipt form, which itemizes each line item on the statement of account for which there is no receipt. The department on the statement of account for which there is no receipt. The

department manager must review and approve this document, thereby ensuring that all purchases made are appropriate.

- *Fill out line item rejection form:* There must be an organized mechanism for cardholders to reject line items on the statement of account. A good approach is to use a procurement card line item rejection form, which users can send directly to the card issuer.
- *Complete reconciliation checklist:* The statement of account reconciliation process requires multiple steps, some of which cardholders are likely to inadvertently skip from time to time. Accordingly, a standard reconciliation checklist that they must sign is a useful way to ensure that the procedure is followed.
- *Supervisory review for inappropriate or split purchases:* There must be a thirdparty review of all purchases made with procurement cards. An effective control is to hand this task to the person having budgetary responsibility for the department in which the cardholder works. By doing so, the reviewer is more likely to conduct a detailed review of purchases that will be charged against his or her budget.

The preceding set of controls relate only to the process of tracking receipts and reconciling them against the monthly card statement. There are several additional controls that can be used to keep excessive or inappropriate purchases from being made; they include the following:

- *Restrict purchasing levels:* A major control over the use of procurement cards is the restriction of amounts that can be purchased. This may be a maximum amount of daily purchases, a limitation on the total purchased over a month, or restriction to purchases only from suppliers having certain Standard Industry Classification (SIC) codes. This approach is extremely useful for ensuring that cardholders do not amok with their card purchases, while ensuring that losses are restricted if cards are stolen and used by a third party to make purchases.
- *Require supervisory approval of changes in spending limits:* The initial spending limitation on a procurement card is intended to meet the purchasing needs of the user. Consequently, any request for an increase in the spending limit may mean that the user fraudulently intends to purchase beyond the budgeted spending level. To ensure that spending limits do not result in a significant level of overspending, all spending limit changes should be closely monitored and approved by a supervisor.

- *Verify that purchases are made through an approved supplier:* The purchasing staff may have negotiated special volume pricing deals with selected suppliers, so it may be necessary to review statements of suppliers. This control can be made more robust by issuing an approved supplier "yellow pages" to all cardholders, so they know where they are supposed to make purchases.

13.7 ACTIVITY

1) Discuss the importance of a cheque.
2) Explain how cheque payments through a lock box works.
3) What is a wire transfer?
4) Discuss the procedure of how a letter of credit is processed.
5) What are procurement cards?
6) Discuss cash transfer controls available for the following:
   a) Cheque payments;
   b) Remote deposit capture;
   c) Electronic payments;
   d) Letters of credit;
   e) Procurement cards.

13.8 SUMMARY

This unit has been able to discuss cheque payments, wire transfers, electronic transfers, letters of credit, procurement cards and why the treasurer should have good knowledge about these cash transfer methods.

We should remember that a company the wants to create an efficient cash transfer system should stay away from cash, cheques and even letters of credit because these methods simply involve too many controls, process steps and administrative procedures. The company should try by all means use wire transfers, electronic transfers such as DDACC and procurement cards because these involve fewer problems.

While a company normally deals with all forms of cash transfers noted in this unit, it is advisable that the treasurer pushes for a larger proportion of electronic transfers and procurement cards wherever possible.

The next unit will discuss Bank of Zambia's treasury role in the Zambian economy.

UNIT 14

BANK OF ZAMBIA-TREASURY ROLE

14.1 INTRODUCTION

Welcome to unit 14. This unit is designed to highlight Bank of Zambia (Zambia's central bank)'s treasury role.

14.2 AIM

The aim of this unit is to discuss Bank of Zambia's role in treasury management.

14.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss Bank of Zambia's role in Treasury Management.
2) Discuss the role of the Financial Markets Department in managing the treasury function of the bank.
3) Discuss the role of the Economics Department.

14.4 TIME REQUIRED

You are advised to spend at least 3 (three hours) reading through this unit.

14.5 REFLECTION

Pause for a moment and try to think of which department at Bank of Zambia has the role of treasury management. Attempt to write your answer in the space provided below.

……

14.6 BANK OF ZAMBIA-OVERVIEW

The Bank of Zambia was established by an Act of Parliament on 7th August, 1964. The act has since been replaced by ‘’The Bank of Zambia Act No. 43 of 1996. The Zambian government is the sole subscriber of the bank’s capital. The Bank of Zambia formulates and implements monetary and supervisory policies. The powers of the bank are vested in the Board of Directors who is responsible for policy formulation and general administration of the bank. The president appoints the Governor and two Deputy Governors for a contract period of five years. The Minister of Finance and National Planning appoints members of the board.

The mission statement for Bank of Zambia reads as follows:

“The principle purpose of the Bank of Zambia is to achieve and maintain price and financial system stability for balanced macro-economic development”.

Bank of Zambia’ vision is “to be a dynamic and credible central bank that adds value to the economic development of Zambia”.

The main objectives of the BOZ are contained in the BOZ Act, section 4. It reads: The bank shall formulate and implement monetary policy and supervisory policies that will ensure the maintenance of price and financial systems stability so as to promote balanced macroeconomic development. (www.boz.zm).

Functions of the bank, without prejudice to the generality of subsection (1) and subject to the other provisions of this Act, the bank shall:

a) license, supervise and regulate the activities of banks and financial institutions so as to promote safe, sound and efficient operations and development of the financial system;
b) promote efficient payment mechanisms;
c) issue notes and coins to be legal tender in the Republic and regulate all matters relating to the currency of the Republic;
d) act as banker and fiscal agent to the Republic;
e) Support the efficient operation of the exchange system;
f) Act as advisor to the government on matters relating to economic and monetary management;
g) Act as banker to other banks.

The Bank of Zambia also has the following roles to play:

- To develop the financial system in Zambia.
- To maintain the stability of the external value of the national currency-the Zambian Kwacha.
- To determine the exchange rate policy in conjunction with the Ministry of Finance and National Planning.
- To manage the country's foreign exchange reserves.

Bank of Zambia has many departments which include; Financial Markets department, Economics department, Banking Division department and Non-Bank Financial institutions Supervision department.

The departments that are involved in treasury management are the **Financial Markets department and the Economics department.**

14.6.1 FINACIAL MARKTES DEPARTMENT

The purpose of the Financial Markets department is to implement **monetary policy of the Bank of Zambia (BoZ) and act as a fiscal agent of the government.** In this regard, the department has a critical role in contributing to the Bank of Zambia's overall objective of achieving price stability. Price stability is an essential element in financial markets as it creates greater opportunities for investment by allowing the business community to plan well.

14.6.1.1 Key functions of the financial markets department

The following are the key functions of the department:

- Implementing monetary policy;
- Acting as fiscal agent of the government;
- Administering the Overnight Lending Facility;
- Managing foreign exchange reserves;
- Undertaking foreign exchange dealings.

14.6.1.2 Implementing Monetary Policy

On 2nd April, 2012, the Bank of Zambia introduced the BoZ Policy Rate as the key interest rate to signal its monetary policy stance. A committee was appointed called the **Monetary Policy Committee (MPC)** which meets every quarter to discuss the policy rate. Currently (January, 2016), the BoZ policy rate is 15.5%. However, adjustments to the BoZ Policy Rate are guided by economic indicators such as output, expected inflation and the exchange rate. Thus, the

mandate of the department is to regulate liquidity in the banking system in a manner that would influence the overnight interbank rate to be in line with the BoZ Policy Rate on a daily basis. For this purpose, the Bank has established a corridor that defines the band within which the overnight interbank rate will be allowed to fluctuate. At present this is set at plus or minus 200 basis points of the Bank of Zambia Policy Rate.

In this regard, the Bank's stance on whether to buy or sell financial assets is determined by the level of short-term interest rates particularly in the overnight market. The aim is to steer the overnight interbank rate towards the BoZ Policy Rate. This is achieved by the department undertaking **Open Market Operations (OMO)** to either supply or withdraw liquidity using various financial instruments at its disposal depending on whether the overnight rate is above or below the corridor, respectively. Through these operations, a stable liquidity position is maintained in the financial system in line with the desired level of interest rates. It is envisaged that influencing short-term interest rates at appropriate levels will assist in achieving price and financial stability.

14.6.1.3 Administering the Overnight Lending Facility

The department also operates an Overnight Lending Facility (OLF), a liquidity financing facility, which commercial banks can access when in need of liquidity from Bank of Zambia at the end of the day. In this manner, the OLF complements other refinancing facilities such as Treasury bill Rediscount Facility and the Intraday Credit Facility. The primary purpose of the OLF is to enhance the operations of the money markets by assuring commercial banks liquidity assistance from the Bank of Zambia. Currently, lending takes place at 10% above the policy rate and the Bank has the option to revise it from time to time depending on market conditions. It is in this way that the facility serves as another avenue through which monetary policy signals can be conveyed to the economy.

14.6.1.4 Fiscal Agent of the Government

The fiscal agency role is executed by the department through the issuance and redemption of Government securities. In this regard, the department conducts primary auctions of Government securities, secondary trading of securities, redemptions and interest payments as well as maintaining the Government securities databases. The department also advises and

provides domestic debt projections and estimations of domestic debt service to the Ministry of Finance for the annual national budget purposes.

In terms of auctions, Treasury bills are now sold on a fortnightly basis while Government bonds auctions are done on a quarterly basis.

14.6.1.5 Managing Foreign Exchange Reserves

The department also manages international reserves. International reserves are assets denominated in foreign currency which are held by a monetary authority. These assets generally consist of cash, financial instruments, gold and Special Drawing Rights (SDRs). International reserves are considered as a major public asset. The BoZ's international reserves portfolio currently comprises cash, deposits, bonds and SDRs held at the IMF. Although, gold is a permissible reserves assets under the law, BoZ currently does not hold gold.

Like other central banks, BoZ holds international reserves for a variety of reasons, which include the following:

- Intervening in the foreign exchange market;
- Execution of payments for the exchange market;
- Servicing government's foreign currency liabilities and debt;
- Providing funds during emergencies or national disasters; and
- Support investor confidence that the country can meet its external obligations.

14.6.1.6 Understanding Foreign Exchange Dealings

Additionally, the department undertakes foreign exchange dealings in a manner that enables the Bank to respond to its core mandate of ensuring price stability and promoting financial system stability. Zambia has a flexible exchange rate system, which means that the price of foreign money is set by the market forces of demand and supply. Therefore, the exchange rate policy management function is aimed at ensuring relative stability in the exchange rate is executed by monitoring the foreign exchange market and interventions by either selling or purchasing foreign exchange.

The Financial Markets Department is operationally divided into three divisions: Market & Risk Analysis, Foreign Markets Operations and Domestic Market Operations.

By performing the above functions, the Department contributes to achieving the BoZ overall objective of price and financial systems stability.

14.6.2 THE ECONOMICS DEPARTMENT

The Economics Department generates monetary policy relevant information through the four divisions that make up the Department. These are:

a) **Macroeconomic Analysis:** This division is responsible for preparation of analytical and policy related reports on monetary, fiscal and other economic developments.
b) **Balance of Payments and External Debt:** The division is responsible for compilation and analysis of balance of payments and external debt development. The division is also responsible for the preparation and monitoring of the macroeconomic programme.
c) **Information and Statistics:** This division is responsible for compilation of data on economic developments and dissemination of information to the general public.
d) **Research:** This division undertakes and co-ordinates research activities related to monetary and other economic issues so as to foster new ways of doing things better.

Further, the Bank, through the Economics Department, works closely with Economic Ministries and acts as focal point in the area of international co-operation with institutions such as IMF, World Bank, UNECA, MEFMI and COMESA.

14.6.2.1 The role of the Economics Department

The Economics Department plays a significant role in meeting the Bank of Zambia's mission statement. The Department provides economic and financial information and advisory services to the Bank of Zambia Management which facilitates appropriate monetary policy formulation and implementation.

14.6.2.2 Departmental Objectives

1) Collection, compilation and reporting of accurate data on financial/monetary systems, balance of payments and other economic statistics.
2) Monitoring and analyzing monetary developments, and compliance of set monetary targets.
3) Monitoring and analyzing the balance of payments and other macroeconomic developments.
4) Conducting research on policy related and relevant monetary and macroeconomic issues.
5) To advise BoZ management on the appropriate monetary policy formulation and implementation.
6) Production of Bank of Zambia publications.

14.6.2.3 Bank of Zambia's role in economic diversification

The role of the Bank of Zambia in economic diversification is derived from the Bank's mission statement. The Bank's mission statement specifies that the primary function of the Bank of Zambia is to formulate and implement monetary and supervisory policies that will ensure price and financial system stability, in order to promote balanced macroeconomic development.

The achievement of long run macro-economic stability especially price stability is critical to the creation of an enabling environment for economic diversification.

14.6.2.3.1 Price Stability

Price stability is both important and desirable because a rising price level imposes substantial economic costs on society. Among the major costs of high inflation are:

a) *Increased costs of investment*-High interest rates are normally associated with high inflation. High interest rates inhibit borrowing for investment thereby slowing down economic growth and the development process. In addition, high inflation adversely affects savings mobilization in the financial markets because the saving public opts to hedge against inflation. This leads to a reduction in the availability of loanable funds and ultimately investment.
b) *Eroded public and investor confidence, and contributes to exchange rate depreciation*- This leads to currency substitution or dollarization, which in turn, reduces the effectiveness of monetary policy.
c) *Eroded external competitiveness,* which may lead to a fall in export earnings, worsening of the balance of payments and constrained growth in the country's gross domestic product (GDP).
d) *Entrenched poverty* through the erosion of people's incomes.

1) ***Supervisory policies and financial system stability***

Through involvement in financial regulation and supervision, as well as in the oversight of payments systems operations, the BoZ plays a fundamental role in preserving and enhancing the efficiency, safety and soundness of the banking and financial system. An efficient, safe and sound financial system attracts domestic savings, foreign investment capital and lowers transaction costs. Clearly, economic diversification can only flourish in a safe, sound and stable financial environment. *2)* ***Initiatives by BoZ***

In 2002, BoZ decided to convert the Kwacha Foreign Exchange Statutory Reserves into US dollars. The conversion was expected to eliminate some of the distortions in the financial sector. Specifically, the objectives of the conversion were to contribute to efforts aimed at lowering the cost of funds for the commercial banks arising from the movement in the exchange rate, which had tended towards depreciation, on the non-remunerated foreign currency statutory reserve requirements and increasing the funds available for lending.

The Bank of Zambia's efforts at economic diversification also witnessed the introduction of a **voluntary** scheme to lower the effective statutory reserve ratio for participating commercial banks making new loans, advances or leases to the agricultural sector from the total deposit liabilities base used when calculating required statutory reserves ratio on Kwacha deposit liabilities **for participating Commercial Banks.** Thus, lowering the **effective** statutory reserve ratio, on Kwachs deposit liabilities faced by commercial banks' lending to the agricultural sector. By this measure, the BoZ expects lending base rates, on qualifying Kwacha loans and advances, to decline substantially.

To further improve the efficiency of the foreign exchange market, the Bank of Zambia has after consultations with key stakeholders, **introduced a broad based inter-bank foreign exchange market** and has suspended the auction system at the Bank of Zambia dealing window. Institutional arrangements regarding remittances of foreign exchange will also be strengthened. However, to achieve long-term stability and competitiveness in the exchange rate, the country must improve its export earnings through economic diversification into sectors and activities that will not only increase production but also the quality of domestic goods and services.

Zambia continues to have a huge external debt burden whose reduction to sustainable levels remains a majority priority.

BoZ has and continue to play an important role in attaining macro-economic stability. To develop the financial sector in Zambia, Bank of Zambia prepared a Financial Sector Development Plan (FSDP). The objectives of FSDP were to:

1. To have a more competitive financial sector that provides credit to a wider private sector at affordable interest rates.
2. To define the role of government in the financial system.
3. To broaden the role of the financial system to support development.

4. Reduce the reliance of the financial system on Treasury bill income for their profitability.
5. To have an effective legal and judicial system to ensure timely disposal of cases involving the collection of bad loans.
6. To devise means of reducing operating costs by financial institutions.
7. To have more transparent procedures for liquidating banks by the Bank of Zambia i.e. to make BoZ a more independent central bank.
8. To have more financially viable public financial institutions.

All in all, the FSDP which is now in operation represents a comprehensive strategy that was formulated to address current weaknesses in the Zambian financial system as well as guide efforts aimed at realizing the vision of a monetary and financial system that is stable, sound and market-based and that supports efficient resource mobilization necessary for economic diversification and sustainable growth.

The vision for Bank of Zambia has always been (as mentioned in part above) to have a stable, financially sound and competitive financial system that performs the financial intermediation role efficiently and provides adequate financial services to all sectors of the economy and all parts of the country including rural areas as well as achieve a more developed capital market in order to mobilize sufficient savings to effectively contribute to balanced economic development.

14.7 ACTIVITY

1. Explain how Bank of Zambia (BoZ) uses OMO (Open Market Operations) to control money supply.
2. Explain how Bank of Zambia implements monetary policy.
3. What is the role of the Financial Markets Department?
4. What is the role of the Economics Department?

14.8 SUMMARY

This Unit has discussed the overview of Bank of Zambia, the Financial Markets Department and the Economics Departments that are actively involved in the country's treasury functions.

The next unit will discuss the Ministry of Finance and National Planning and its role in the country's treasury activities.

UNIT 15

MINISTRY OF FINANCE AND NATIONAL PLANNING-TREASURY ROLE

15.1 INTRODUCTION

Welcome to unit 15! This is the last unit in this module. The unit is intended to discuss the Ministry of Finance and National Planning's role in managing the treasury of Zambia.

15.2 AIM

The aim of this unit is to discuss the role of the Ministry of Finance (MOF) in managing the country's treasury activities.

15.3 OBJECTIVES

At the end of this unit you should be able to do the following:

1) Discuss the role of the Ministry of Finance and National Planning; 2) Discuss divisions and departments responsible for treasury management; 3) Discuss the role of the Secretary to the Treasury.

15.4 TIME REQUIRED

You are advised to spend at least 3 (three) hours studying this unit.

15.5 REFLECTION

Write down five (5) functions of the Ministry of Finance.

……

……

……

…

……

…

……

…

……

15.6 INTRODUCTION

The Ministry of Finance (MoF) derives its mandate from part X of the Constitution of Zambia. It is also guided by statutory and portfolio functions as stipulated in the Government Gazette Notice No. 183 of 23rd March, 2012. The Ministry is charged with economic, national development planning and budgeting, and financial management responsibilities. It is also the ministry that deals with the country's treasury functions.

The Ministry is headed by a Minister assisted by a deputy Minister. The administrative and technical team is headed by the Secretary to the Treasury assisted by two permanent secretaries, namely; Permanent Secretary-EMF (Economic Management and Finance) and Permanent Secretary-B (Budget).

MoF's **vision** is "to be a world class institution in the mobilization and management of public resources for sustainable national development".

The Ministry's **mission statement** is "to efficiently and effectively co-ordinate national planning and economic management, mobilize and manage public resources in a transparent and accountable manner for sustainable national development.

MoF's **value statement** is as follows: "the Ministry of Finance will uphold excellence, profession, team work, integrity, commitment, transparency, accountability and value for money".

The Ministry has **twelve (12) Directorates, namely;**

1. Human Resource and Administration;
2. Budget Office Department;
3. Government Stores;
4. Centralized Computer Services Department;
5. Office of the Accountant General;
6. Investment and Debt Management;
7. Central Statistical Office;
8. Office of the Controller of Internal Audit;
9. Economic Management Department;
10. National Planning Department;
11. Monitoring and Evaluation Department;
12. National Policy and Programme Implementation Department.

Statutory bodies and grant aided institutions that report to the Ministry include the following;

Bank of Zambia, Development Bank of Zambia (DBZ), Indo Zambia Bank, Financial Intelligence Centre (FIC), National Savings and Credit Bank (NATSAVE), National Road Fund Agency, Lotteries Control Board, Pension and Insurance Authority (PIA), Zambia National Commercial Bank (ZANACO), State Lotteries Board, Zambia National Building Society, Zambia Public Procurement Authority, Zambia Revenue Authority (ZRA), Lusaka Stock Exchange (LUSE), Securities and Exchange Commission (SEC), Zambia State Insurance Corporation (ZSIC), Zambia Institute of Chartered Accountants (ZICA), National Economic Advisory Council, Revenue Appeals Tribunal, Zambia Centre for Accountancy Studies (ZCAS) and Zambia Institute for Policy Analysis and Research.

15.6.1 DIVISIONS AND DEPARTMENTS

a) Financial Management and Administration

The Financial Management and Administration Division (FMA) is mandated to manage the administrative support services, public affairs and procurement in order to assist the Ministry meet its financial, planning and economic management goals. In this regard, the Division manages and develops human resources in order to enhance quality service delivery and productivity.

Some of the particular objectives of FMA include: the reduction of procurement costs on goods, services and public works for cost effectiveness; the improvement of financial management systems in order to ensure provision of timely and efficient

accounting services; the design and implementation of integrated information communication technology infrastructure and supportive information systems and efficient information delivery;

Human Resource and Administration, Office of the Accountant General, Internal Audit, Centralized Computer Services Department and Government Stores are all under this Division.

b) Planning and Economic Management Division

The planning and Economic Management Division (PEMD) is the national "thinktank" in terms of moving the country to greater heights of development. In this regard, it is responsible for coordinating the preparation of the national long-term development vision, integration of the sector and regional development programmes and strategies into the national development plans and programmes, linking the Budget to Macro-economic framework as well as monitoring and evaluating development programmes. The Division's mandate includes spearheading the implementation of the Fifth National Development Plan (FNDP).

Economic Management Department, National Planning Department, Monitoring and Evaluation Department, National Policy and Programme Implementation Department and Department of Census and Statistics all fall within this Division.

c) Budget and Economic Affairs Division

The Budget and Economic Affairs Division (BEA) facilitates preparation of the national annual budget and its implementation in order to provide the framework for timely implementation in order to provide the framework for timely allocation of financial resources for National Development. The Division is further charged with the responsibility to manage Government investments in order to maximize returns. In addition, BEA must also develop and implement effective debt management strategies and framework in order to reduce national debt stock.

Budget Office and the Investments and Debt Management Department are part of this Division.

15.6.2 THE SECRETARY TO THE TREASURY

The Secretary to the Treasury is a member of the Presidential cabinet. This person is the head of the Department of the Treasury, and deals with all financial and monetary matters directly relating to the government. The secretary is the principal economic adviser to the president

and plays a major role in formulating economic policy. This role is roughly equivalent to that of Finance Minister. The Secretary to the Treasury is also responsible for providing economic reviews and forecasts from time to time. Normally these reviews include; the country's Gross Domestic Product (GDP), inflation, the country's debt management, the balance of payment and performance of different sectors of the economy such as Energy, Mining, Tourism, Health, Transport, Agriculture, Education and so on. Usually the reviews include information on the performance of the global and domestic economy.

The Secretary to the Treasury in the Ministry of Finance, may in writing (according to the Bank of Zambia Act 1996 No 43) appoint a senior official in that ministry to be alternate directorto the Secretary to the Treasury and to attend any meeting of the Board which the Secretary to the Treasury is unable to attend, and when such alternate attends such meetings, the provisions of subsection (2) of the Act shall apply, *mutatis mutandis*, to such alternate.

The Secretary to the Treasury is responsible for formulating and recommending domestic and international financial, economic, and tax policy, participating in the formulation of broad fiscal policies that have general significance for the economy, and managing the public debt.

Essentially, the Secretary to the Treasury manages the country's finances.

The Secretary to the Treasury also does issue treasury information briefs on issues such as the budget performance, the country's current debt status, monetary policy, commodity prices (copper), and external sector development

Visit the Ministry of Finance's website www.mofnp.gov.zm for more information on treasury management.

15.7 ACTIVITY

1. Explain the role of the Ministry of Finance's:
   a) Financial Management and Administration Division
   b) Planning and Economic Management Division
   c) Budget and Economic Affairs Division.
2. Discuss the role of the Secretary to the Treasury.

15.8 SUMMARY

The treasury functions of Zambia are carried out by the Ministry of Finance and National Planning and the Secretary to the Treasury is responsible for treasury implementation.

Well done for reading this module this far. You have now come to the end of the module. I hope and trust that the module has provided you enough information on treasury management.

15.9 MODULE SUMMARY

This module has been able to tackle the following topics; introduction to treasury management, working capital/liquidity management, management of interest rate risk, currency risks, currency risk management, cash forecasting, cash concentration, debt management, equity management, investment management, clearing and settlement systems, treasury systems, cash transfer methods, Bank of Zambia's treasury role and finally Ministry of Finance and National Planning's treasury role.